Panasonic

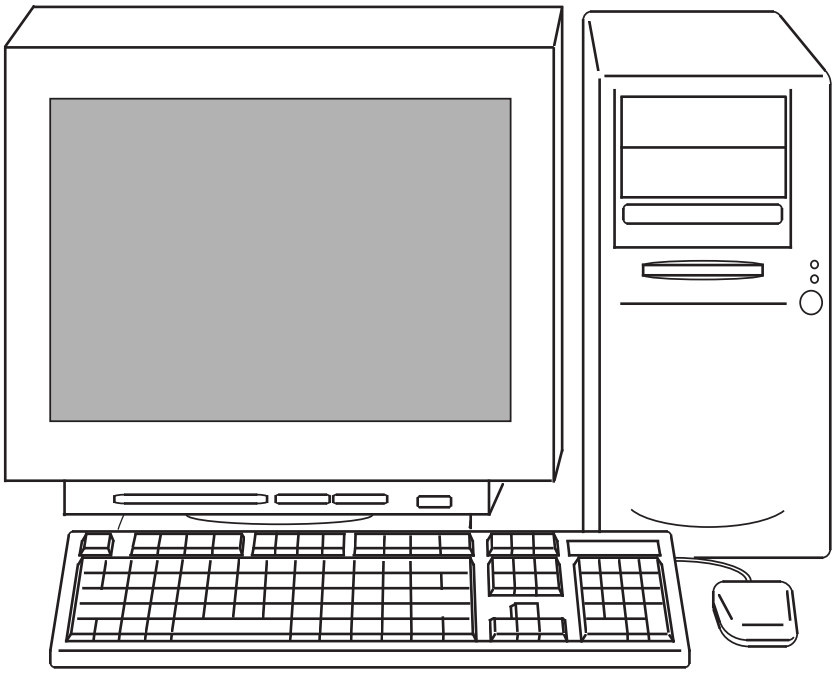
Operating Instructions

Production Analysis

MES for Electronics Assembly

PanaCIM-EE

Model No. **NM-EJS3B**



Thank you for purchasing the “Panasonic product.”
- Read the operating instructions carefully for proper and safe usage.
- Before placing the machine in service, be sure to read “Safety Precautions.”
- Store the operating instructions carefully.

N7201A444E14 (V08.08)
201801

Original Instructions

For using this system:
This system is designed with a purpose of visualization of work productivity, more efficiency and quality warranty in electronic component mounting process. Use this system solely for the purpose.

Do not use this system for other purposes.

Operating machines:
Any person who operates the machine should receive a special training.

Any operator who has not received the training is not allowed to operate.

For training, contact Panasonic Smart Factory Solutions Co., Ltd., Sales company or Sales agent.

⚠ Warning

Do not operate the machine without receiving special training



Risk of injury due to improper operation.

Prohibited

●About training, conduct in your company or contact us.

For machine training in Europe, you may contact Panasonic Industry Europe:

Panasonic Industry Europe GmbH FA Division
Robert-Koch-Strasse 100, 85521 Ottobrunn Germany
Phone number: 0049-(0)89-453541192

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About the Instruction

■ Operating instruction contents

The following operating instructions are included in the DVD-ROM attached to this machine. The instructions are divided according to work or function. Use one that is appropriate for the purchased module (application) or your purpose.

∗ BM series machine is included in the explanation of this manual, but its functions are not supported.

Type	Description
<General>	
Introduction	Overview of machine and basic setting/operation/analysis/utility.
Basic Operation (PanaCIM Web Application)	PanaCIM Web application's basic operation.
<Modules>	
Production Analysis	Collect and analyze machine's running information and display its statistics on real time basis to support productivity improvements (recognition/pickup error rate improvement, machine trouble stop time reduction) for mounting line.
Traceability	Combine PCB information read by a barcode reader on machine or conveyor and production record or material/part information of machine to manage history of PCBs for quality control and market support.
Material Verification	Verify material/part information and recipes while setting up material/parts in production setup/model change/splicing/part exhaust to prevent wrong material/part set up.
Product Changeover and Control	Auto download recipe and adjust lane width based on PCB ID for quicker model change.
Production Monitoring	Retrieve production information and operation information on real time basis and display line status on the monitor and PDA to provide operators with instructions for events such as error or part exhaust. This is to reduce mount line stop time or to prepare in advance for better productivity.
Enterprise Link	A function to output the data collected from machines during production (material verification data, trace data, operation performance data) in the standard format so external applications can reuse the data. Part information can be loaded also in the standard format.

Type	Description
Material Control	A function to provide the location and the number of parts or feeders stocked in the production or preparation area. MSD function controls MSD parts' expired date.
Component Supply Navigator	The Component Supply Navigator optimizes the supply sequence to provide a shortest route of an operator without component shortage. It can prevent short time breakdown in the production line occurred due to component shortage and contribute to manpower saving.
Maintenance	For the purpose of reducing time spent on maintenance planning / control operations, asset (machines, heads and feeders) profiles can be controlled on the floor, or the maintenance history of each asset can be recorded.

License CD names for modules are as follows.

Module Name	License CD Name
Production Analysis	Production Analysis
Traceability	Traceability
Material Verification	Material Verification
Product Changeover and Control	Product Changeover and Control
Production Monitoring	Production Monitoring & Dispatch
Enterprise Link	Enterprise Link
Material Control Base	Material Control Base
Material Control Allocation	Material Control Allocation
Material Control MSD	Material Control MSD
Component Supply Navigator	Component Supply Navigator
Maintenance	Maintenance

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1. OVERVIEW

This chapter explains the overview of PanaCIM-EE/Production Analysis.

1.1 Overview

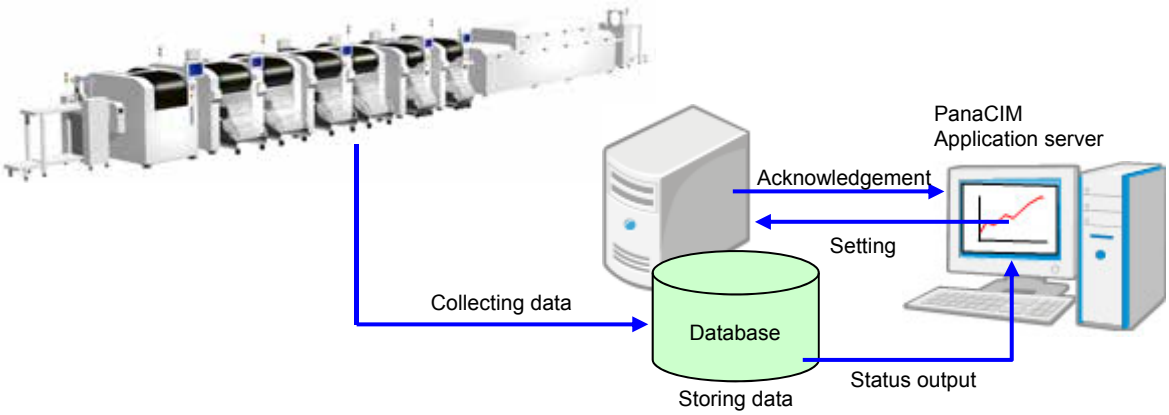
PanaCIM-EE (PanaCIM Enterprise Edition: hereinafter referred to as “PanaCIM”) offers the PanaCIM/Production Analysis, a web-based system for monitoring the conditions of the production line controlled by the PanaCIM.

The Production Analysis allows the users to collect the production information of the machines, to analyze the information, to display the statistical data and to show information in real time, thus helping to improve the productivity of the SMT line (to enhance pickup rate and to reduce mounting rate and downtime). Factory performance data can be viewed in real time on any computers (even those outside the factory) that can connect the PanaCIM application through Internet/Intranet.

Main functions of the PanaCIM/Production Analysis include the following:

- Comparing production performance and plans
- Creating charts of production performance and plans
- Analyzing production performance

The following shows how these functions work in the system.

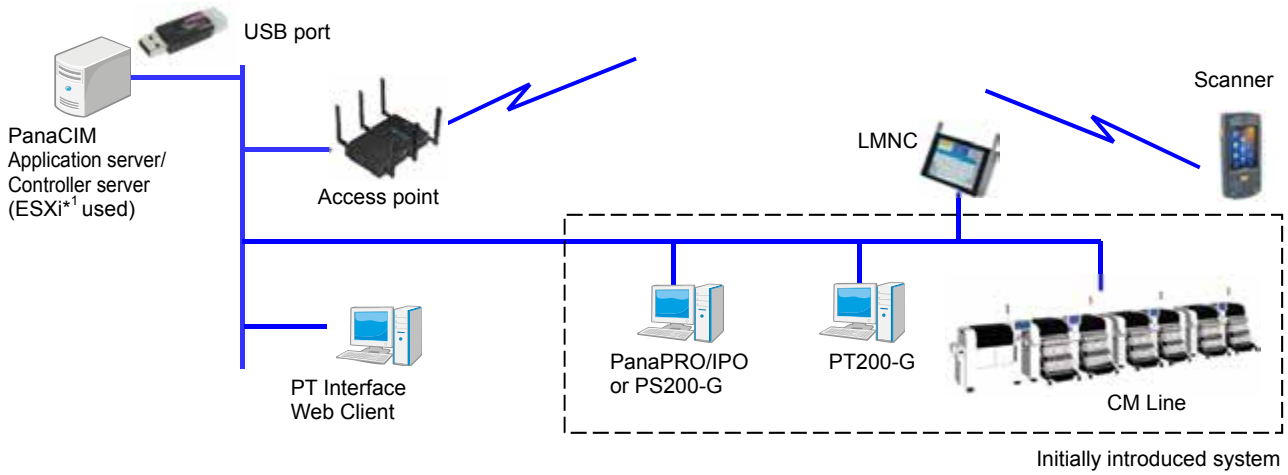


1.2 System Configuration

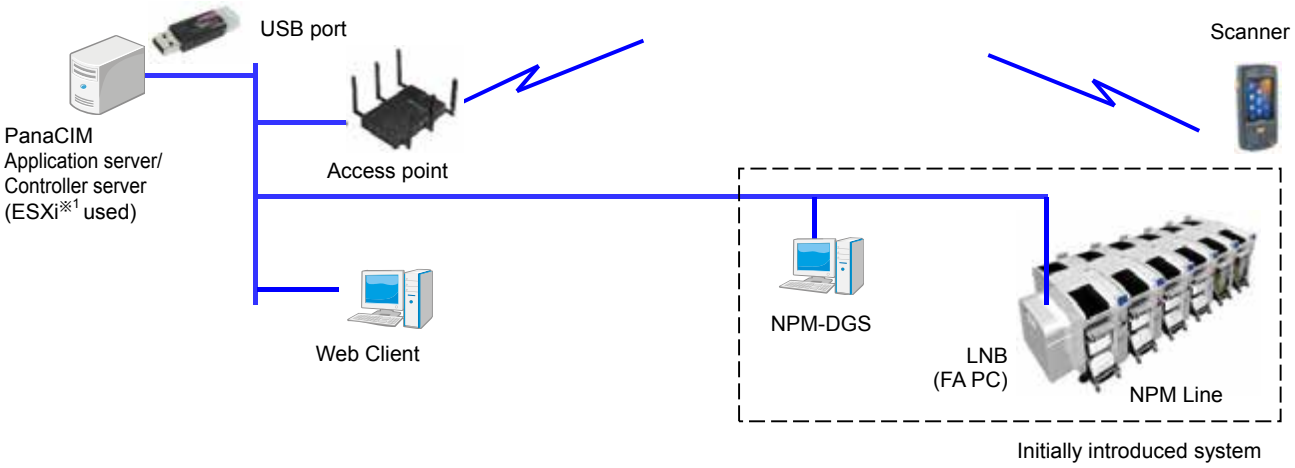
This section shows the schematics of systems with PanaCIM for small-scale and large-scale production.

■System with PanaCIM (for small-scale production)

CM Line



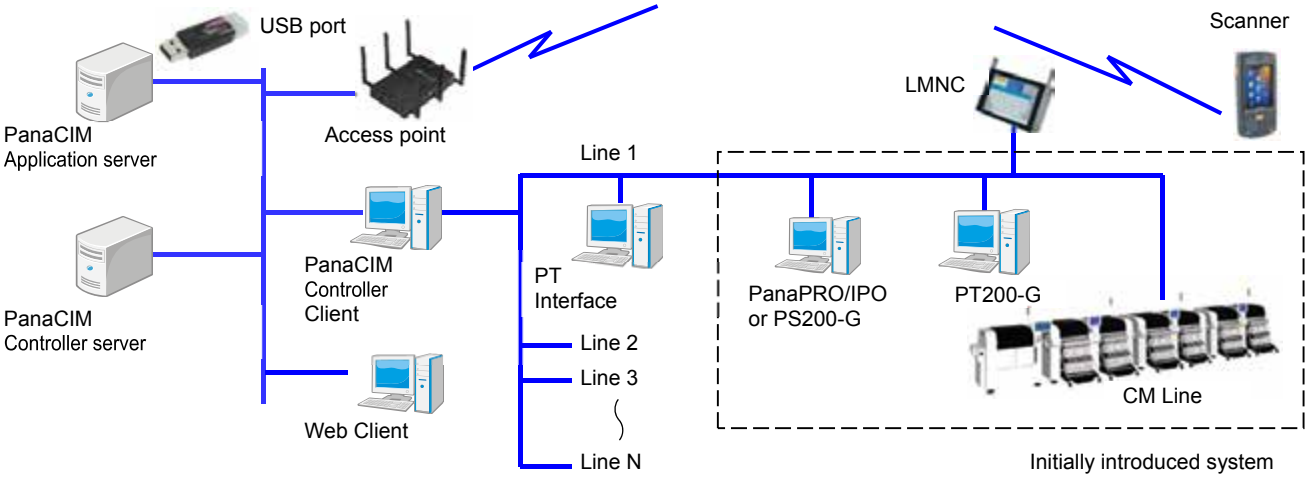
NPM Line*2



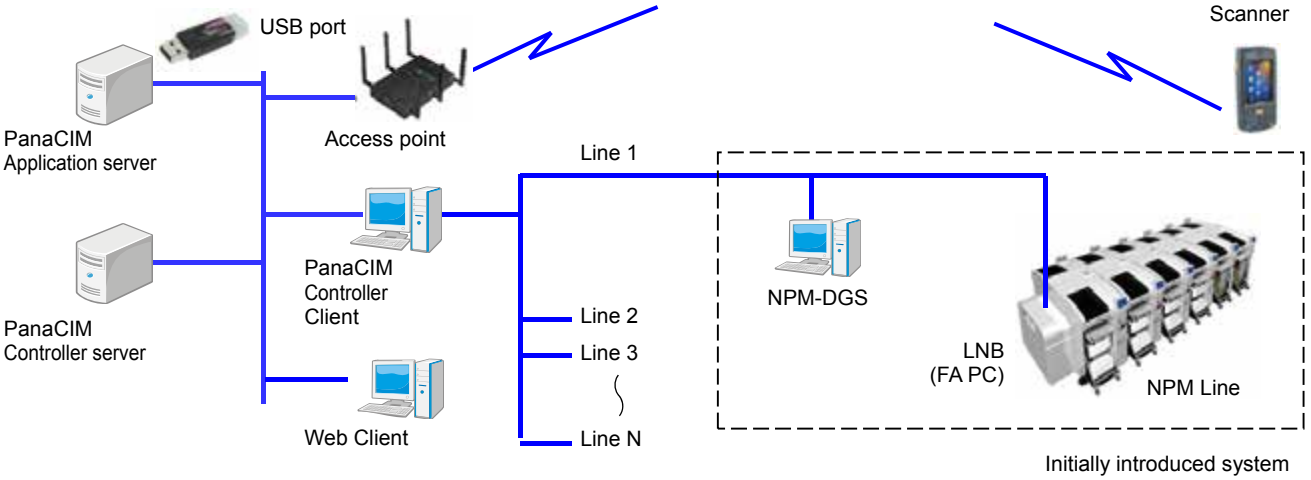
*1. ESXi: A virtualization tool that allows multiple applications to run on a single server.
*2. NPM line can also include the models such as AM100, SPD, SPG, SPV, and SP70 as NPM-related machines.

■ System with PanaCIM (for large-scale production)

CM Line



NPM Line[*1]



*1. NPM line can also include the models such as AM100, SPD, SPG, SPV and SP70 as NPM-related machines.

2. SETTINGS

This chapter explains Settings of a PanaCIM-EE Production Analysis.

2.1 Overview

■ List of Production Analysis module functions

Main functions	Detailed functions		Description
	Large category	Small category	
Data Collection	Data Type		Production management information output from the machines
	Max. Machines to Collect Data		100 stages (per PanaCIM controller server)
	Max. Lines to Collect Data		Lines equivalent to the above machines (per PanaCIM controller server)
	When to Collect Data		Each time a PCB is produced
Data Storage	Backup	Auto Backup	Backs up data automatically at the specified intervals.
		Store Outside	Stores the backed up data outside (on other media).
	Restore	Restore	Data can be restored from the backup data.
Charts and Reports	Statistical Charts	Type	Shows the production analysis chart of the next item (partly).
		Save	- Saves the parameters (attributes, range, etc.) required to create charts as chart configuration files. - Saves the created charts as image files (in JPEG format).
		Quick Run	Calls up the saved configuration files to display charts.
		Time to Show	Shows a chart of the finished setting in a short time after Generate or Quick Run is selected.
	Reports	Type	Shows the production analysis report of the item after next (partly).
		Save	Saves the parameters (attributes, range, etc.) required to create reports as report configuration files.
		Quick Run	Calls up the saved configuration files to display reports.
		Time to Show	Shows a report of the finished setting in a short time after Generate or Quick Run is selected.
	Real Time Charts	Auto Update	Updates all statistical charts at the specified intervals automatically.
		Update Interval	Updates the charts at the specified intervals in minutes.
		Save	Saves the updated charts as image files (in JPEG format) automatically.
Settings	Chart Options		Enables settings (range, etc.) required to create charts.
	Thresholds		Sets the thresholds for the attributes of charts.
	Show Exceeded Thresholds	Highlight	Visually notifies the user of any attributes that have exceeded the threshold on the chart.

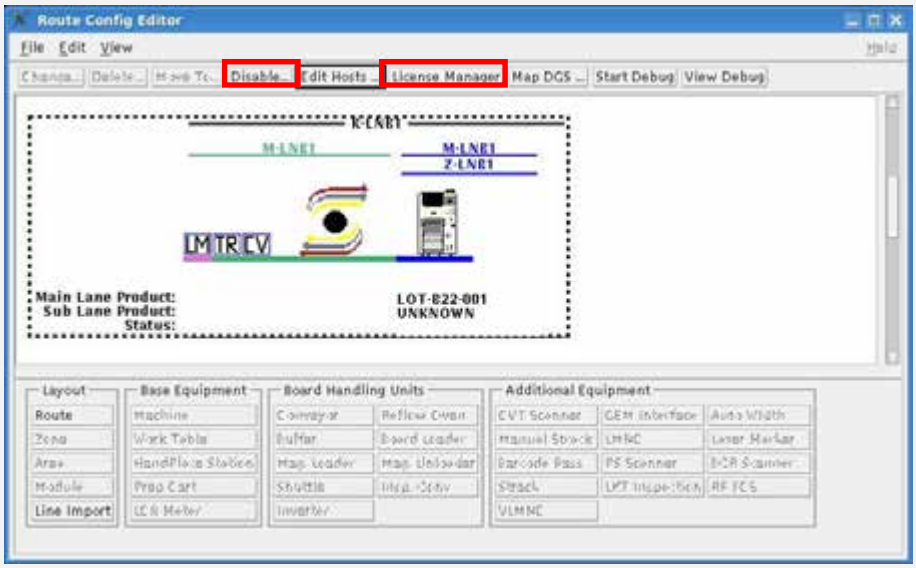
2.2 Settings

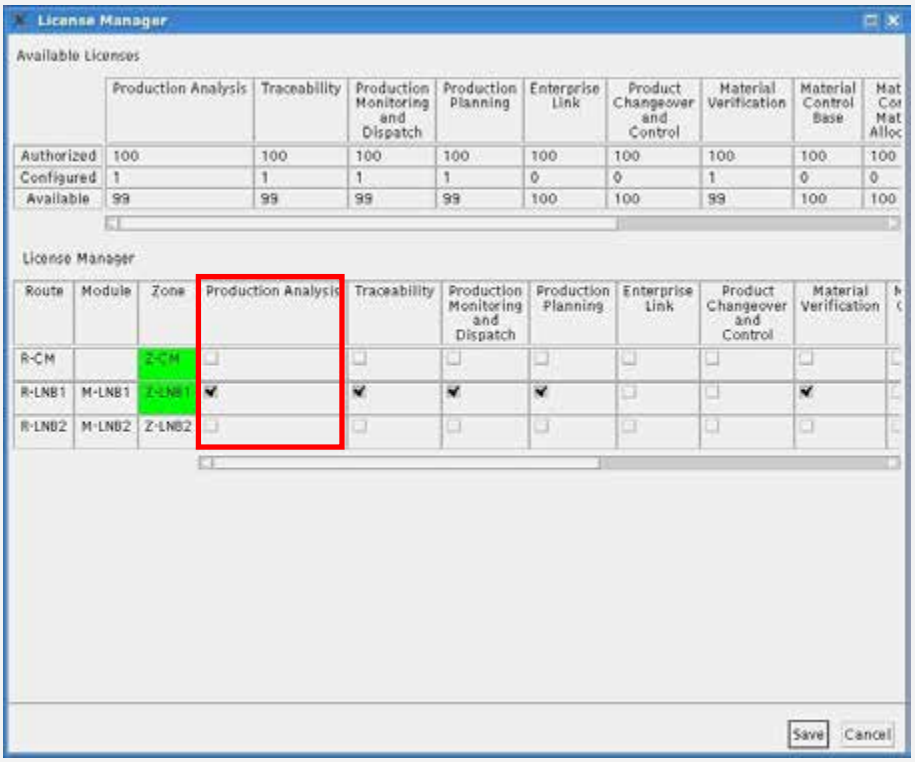
2.2.1 Settings of CIMC

■ Route Editor

- **Configure Licenses**

  Before using the PanaCIM/Production Analysis, the license needs to be enabled via the Route Editor of the PanaCIM controller server.

Step	Procedure
1	Select **Route Editor** icon on the application toolbar of the PanaCIM controller to open the **Route Editor** Application. 
2	Select the route to change the license configuration and click **Disable**. 
3	From the menu on the toolbar, select **License Manager** to launch the License Manager.

4	Under **Available licenses**, select the license for modules and zones. To use the PanaCIM/Production Analysis, place a check mark in the following checkboxes: • **Product Analysis:** Enables the selected license zone by zone.  * The lines available in the Route Editor appear in green. To change the configuration of the license, disable the lines by removing the check mark in the box.
5	After configuring the license, click **Save** and exit the License Manager.

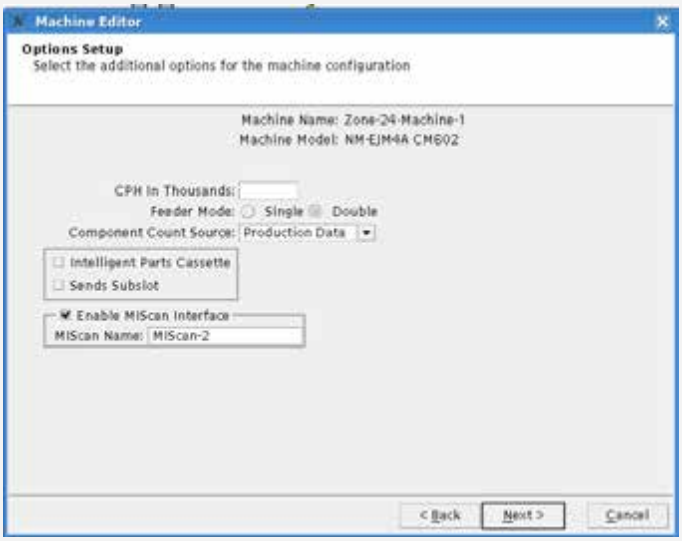
Machine Configuration

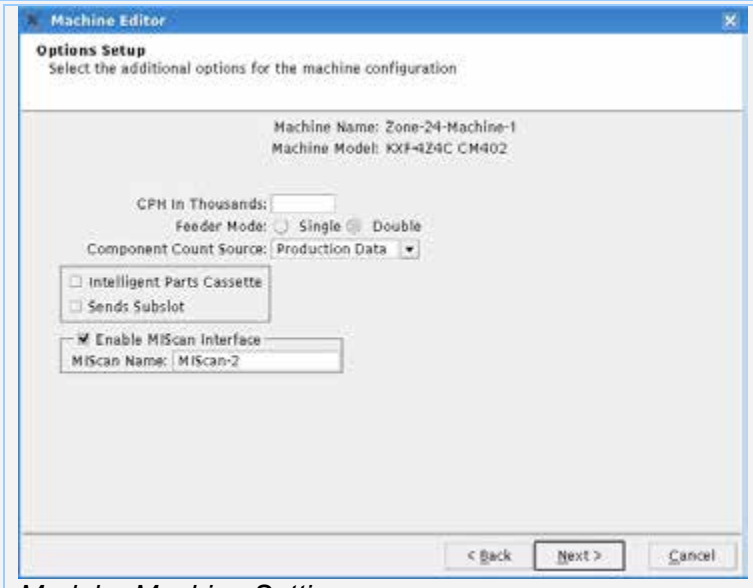
Refer to the action table below to configure Machines to collect management data using the Route Editor Application. Please refer to the PanaCIM Base documentation for creating and configuring a new route. The same setting method applies regardless of the machine type.

Notes:

- Management data collection is per equipment type. BM series machines use periodic polling (with defined interval) whereas Modular machines use board by board collection.

- PT200 Configuration: Ensure the PT200 is setup to output the Production files, the Machine Event files and the PT Event file output. To do this, select the Setup tab on the PT200 machine and then double-click on ProViewer. Ensure that the Production File Output, Machine Event File Output and PT Event File Output have the Yes radio button selected.

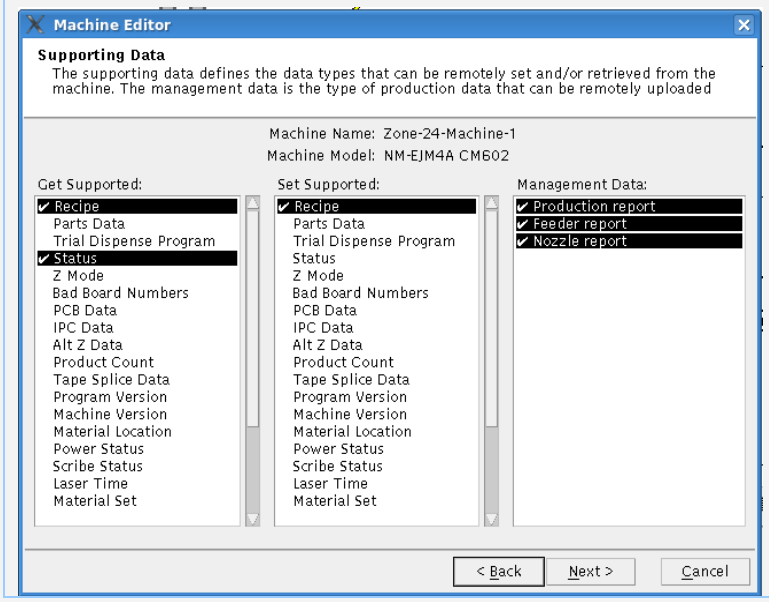
Step	Procedure
1	Open the **Route Editor** Application from the PanaCIM Main Menu. Select a **machine**.
2	Click **Next** until you reach the **Options Setup** page.
3	Set the following in the **Options Setup** page. • CPH in Thousands: Set the interval for the PanaCIM to get the management data. (min) Default: 60  *BM Series Machine Settings*



Modular Machine Settings

* Although it depends on the line configuration or the number of the machines, setting too small a value causes the load to the network to be increased. When changing this value, contact our service representative for details.

4	Click **Next** until you reach the **Supporting Data** page.
5	Select the Management Data types: **Production report**, **Feeder report**, and **Nozzle report**.
6	Click **Next** until you reach the last page that includes a **Save** option.
7	Click **Save** to save the machine configuration changes for Management Data collection.

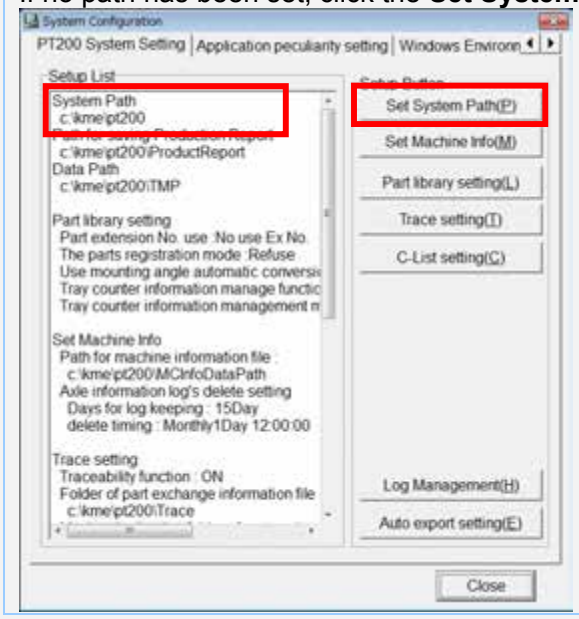
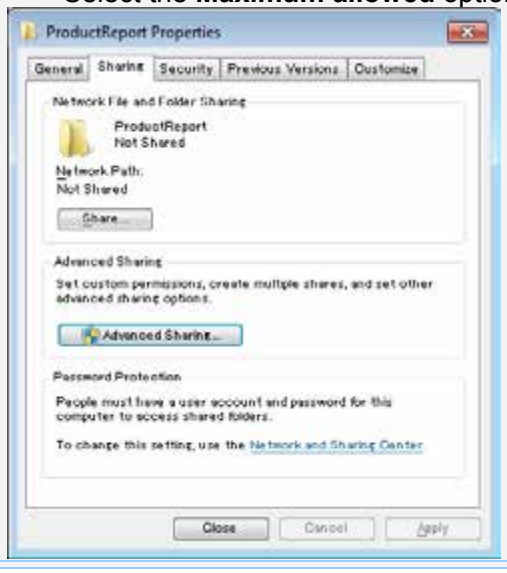


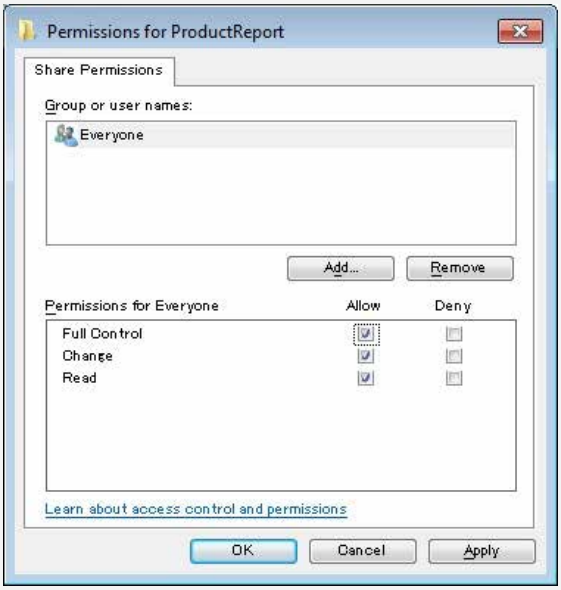
2.2.2 Settings of PT200

■ Setup

● Configure system

PanaCIM collects production information based on the production management information files saved in the PT200. Set the shared folder for accessing the PT200 from the PanaCIM.

Step	Procedure
1	From the **Setting** menu of the PT200, select the **Systems Setup** icon. 
2	On the **System configuration** dialog, check the Path to save production information under **List of settings**. If no path has been set, click the **Set System Path** button and set the desired folder. 
3	On the Windows Explorer, select **Sharing and Security** of the 'Path to save production information' folder.
4	Under the Sharing tab on the **ProductionReport Properties** dialog, set as follows: • Select the **Share this folder** option to share the 'ProductionReport' folder. • Select the **Maximum allowed** option. 

5	Click the **Permissions** button. Give Full Control to the **Everyone** user, and click **OK**. 

3. FUNCTION - [Reports]

This chapter explains function- [Reports].

3.1 Overview

To find the [Reports], access the PanaCIM EE Web Application and select **Charts and Reports** from the menu. Then, select **Reports** from the balloon-menu.



3.1.1 List of Reports

The table below is the list of selectable reports.

■ Production Analysis

Report	Description
Component Usage Report	Shows various statistics relating to component usage over the course of production cycle.
Component Usage Report (incl. costs)	Shows various statistics (including costs) relating to component usage over the course of production cycle.
PCB Production Report	Shows a summary of production runs on a machine/route during a selected period.
Work Order Usage Report	Shows details of Work Order runs on a machine/route during a selected period of time.
Event Trace Report	Shows details of all events occurring on machines in a route during a selected period.
Event Trace Summary Report	Shows the number of events occurring on machines in a route during a selected period.
Down Time Report	Shows the duration of how long the equipment was down, the reason for the downtime, and the operator who initiated it.
Down Time Summary Report	Shows the number of times the equipment was down during a selected period.
State Trace Report	Shows all information of states changed on a machine/route during a selected period.
State Trace Summary Report	Shows the number of times state changed on a machine/route per state during a selected period.
Part Supply Patrol Report	Shows the time spent for part supply and details of tasks.

■ Feeder Maintenance

Report	Description
Tagged Feeder Repair Report	Shows the list of feeders tagged for maintenance.
Feeder Repair Count Report	Shows the number of times that a particular feeder is repaired during the selected period of time.
Feeder Repair History Report	Shows the list of history of feeder maintenance during the selected period of time.
Feeder Report	Shows the characteristics to identify a feeder.

■ Production Monitoring Acknowledgment

Report	Description
PMD Event History Report	Shows all the events that occurred on the selected machine/route over a selected period.
PMD Event Summary Report	Shows the number of times of events occurred on the selected machine/route over a selected period.
Work in Progress Report	Show a list of work orders currently running.
Upcoming Work Orders Report	Showa list of work orders scheduled for production.
Line Down Time Report	Shows the Route Down Times for each route, along with the reason for the down time and the operator who initiated it.

■ Enterprise Link

Report	Description
Import History Report	Shows information about the data source and its related records.

3.2 Setting

3.2.1 Filter criteria

For filter criterial of a report, refer to Basic Operations (Web application 3.2.2 [Filter Criteria].

3.3 Operation

3.3.1 Operation Procedure

For how to output reports, refer to refer to Basic Operations (Web application) 3.3.1 [Operation Procedure].

3.4 Report Item

3.4.1 Report Item

The tables below are items displayed in [Report].

■ Base

● Product Setup Information Report

Item	Description
Machine File Name	Name of a program file associated with a particular placement machine. Placement machine uses this file for placing components on the PCB.
Model ID	This is machine file name of Product Setup. This corresponds to model string under Setup tab of Product Setup dialog.
Line Name	Name of the route on which products are manufactured.
Zone Name	Name of the production zone consisting of placement machine. Zones exist within a route.
Equipment Name	Name of the placement machine that places parts on the PCB.

● Mix Information Report

Item	Description
Line Name	Name of the route on which products are manufactured.
Mix Name	Name of the mix consisting of individual products (boards to be produced) and their associated manufacturing data in order to share single setup of manufacturing resources like placements machines, component feeders etc.
Product Name	Name of the product in the mix. An individual product corresponds to a single PCB on which different types of electronic components are placed by fetching components from feeders attached to placement machines.

● Alternate Model Barcodes

Item	Description
Line Name	Name of the route on which products are manufactured.
Product Name	Name of the individual product in the mix. An individual product corresponds to a single PCB on which different types of electronic components are placed by fetching components from feeders attached to placement machines.
Mix Name	Name of the mix consisting of individual products (boards to be produced) and their associated manufacturing data in order to share single setup of manufacturing resources like placements machines, component feeders etc.
Model String	A number associated with product which is matched to the model code on the panel. If the model code on the panel is different from Product model string, changeover is initiated.
Alt 1 to 50	Alternate name of the individual product in the mix.

● Feeder Allocation Report

Item	Description
Mix Name	Name of the mix consisting of individual products (boards to be produced) and their associated manufacturing data in order to share single setup of manufacturing resources like placements machines, component feeders etc.
Equipment Name	Name of the placement machine that is used for placing

Item	Description
	components on the PCB.
Product Name	Name of the individual product in the mix. An individual product corresponds to a single PCB on which different types of electronic components are placed by fetching components from feeders attached to placement machines.
Supply Method	Identifier for the way in which components are fed to placement machines. For example, supply could be in the form of reel, tray, stick etc.
Total Count	Total number of feeders of particular type whose identifier is mentioned under **Supply Method** column.

- Part Allocation Report

Item	Description
Part No.	Part number of a component. Each component placed on the PCB is uniquely identified by the part number.
Count (Panels)	The number of panels that enter a single stage of modular machine. Panel refers to the overall board that comprises individual patterns (products).
Total Count (Panels)	Total number of panels needed to manufacture a particular product. These are the number of panels that enter a single stage of modular machine over the course of production of a product.
Count (Patterns)	It refers to number of products that make up a panel. A single product can be repeated over a panel thereby producing multiple copies of same product on a single panel which enhances productivity and overall throughput.
Total Count (Patterns)	Total number of boards of a particular product that will be produced at the end of production run
Current Quantity	Total number of boards of a particular product that have been produced so far from production start time till the time when the report was opened.

- Work Order History Report

Item	Description
Work Order Name	Name of the work order under which production started. A work order contains important information related to production schedule of a product like start date / end date of production, quantity of boards produced on that date etc.
Mix Name	Name of the mix consisting of individual products (boards to be produced) and their associated manufacturing data in order to share single setup of manufacturing resources like placements machines, component feeders etc.
Product Name	Name of the individual product in the mix. An individual product corresponds to a single PCB on which different types of electronic components are placed by fetching components from feeders attached to placement machines.
Start Time	Time when the production started for a product.
End Time	Time when the production ended for a product.
Duration(DD:HH:MM:SS)	Total time for which production of a product lasted in Days, Hours, Minutes and Seconds. It is calculated as difference of Start Time and End Time.
Equipment Name	Name of the Placement machine used in the production.
Start Operator	Name of the operator in the shift when the production started.
End Operator	Name of the operator in the shift when the production ended.

Item	Description
Closing Type	This denotes how the work order was closed. It can be Auto or Manual. Manual is when user sees that it would take a long time before the work order gets closed automatically. Hence the user closes it manually.
Reason	The reason for closing the work order, usually fed when user closes the work order manually
Scheduled Quantity	The number of planned boards for work order.
Actual Quantity	The number of boards actually produced.

- SP Changeover Report

Item	Description
Equipment Name	Name of the Placement machine used in the production.
Product Name	Name of the individual product in the mix. An individual product corresponds to a single PCB on which different types of electronic components are placed by fetching components from feeders attached to placement machines.
Printer Common File Name	Name of the Mapped product in the mix.

- Changeover Report

Item	Description
Line Name	Name of the route on which a product needs to be manufactured after changeover is initiated based upon the model string.
Zone Name	Name of the production zone consisting of placement machine. Zones exist within a route.
Equipment Name	Name of the placement machine that will place parts on the PCB after changeover is initiated.
Stage No.	Stage number of a modular machine. Modular machines consisting of two tables will have a single stage while modular machines consisting of four tables have two stages. For example in CM602, Table 1 and 2 comprise stage 1 while Table 3 and 4 comprise stage 2. For placement machine consisting of two stages, some components will be placed on board in stage 1 while rest of the components will be placed on the board in stage 2.
Mix Name	Name of the mix for which production will be started after changeover.
Product Name	Name of the individual product in the mix.
Side	Side (Top or Bottom) on which parts will be placed as part of manufacturing the product.
Mix File Name	Name of the mix file generated by PanaPRO and imported to PanaCIM via PEM.
Output File Name	Name of the program file which contains all the data required for placing parts on the PCB.
Machine File Name	Name of the program file which contains all the data required for placing parts on the PCB.
Model String	A number associated with product which is matched to the model code on the panel. If the model code on the panel is different from Product model string, changeover is initiated.
Z/PU #	Feeder location at which feeder needs to be mounted during changeover. For example, Z/PU value of 20014 indicates that feeder needs to be mounted to supply parts to Slot 14 Table 2 of placement machine.
Slot	Slot to which parts are supplied after a feeder is mounted.
SubSlot	Indicates whether a part is supplied by left-side or right-side of

Item	Description
	double feeder, thin-type single feeder or auto loading feeder (spliceless feeder). It can either be L (left subslot) or R (right subslot).
Part No	Part number of the component supplied by the feeder
Supply Method	Supply code indicating the characteristics of the feeder. For example, supply method used by the feeder could be in the form of tape reel, tray, stick etc.
Master Z	Master Z No.

- Supplier Part Number Mapping Report

Item	Description
Part No	Part number of the component.
Supplier Part No	Part Number defined by supplier.

■ Material Verification-Base

● Component Trace Report

Item	Description
Line Name	Name of the route on which production occurred.
Zone Name	Name of the production zone consisting of placement machine.
Equipment Name	Name of the placement machine that will place parts on the PCB.
Stage No.	Stage number of a modular machine. Modular machines consisting of two tables will have a single stage while modular machines consisting of four tables have two stages. For example in CM602, Table 1 and 2 comprise stage 1 while Table 3 and 4 comprise stage 2. For placement machine consisting of two stages, some components will be placed on board in stage 1 while rest of the components will be placed on the board in stage 2.
Z/PU #	Machine location from which parts are picked for placement. For example Z/PU value of 20014 indicates that parts are picked from a feeder located at Table 2 Slot 14 of placement machine.
Slot	Machine slot number from which parts are picked for placement.
SubSlot	In case of double feeder, thin-type single feeder or auto loading feeder (spliceless feeder), when components are supplied from left side and right side, subslot corresponds to a single side (left or right) from which parts are picked. Generally Subslot value of 0 corresponds to left side while subslot value of 1 corresponds to right side. In this report, L corresponds to left subslot while R corresponds to right subslot.
Material Barcode	Reel Barcode of the reel consisting of electronic parts that are placed on the PCB during the manufacturing process.
Part No.	Part number of a component. Each component placed on the PCB is uniquely identified by the part number.
Expected Part	Part number expected at a feeder location based upon the product setup information downloaded to the scanner
Lot No	Entity that is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing.
Vendor	Identifier for the part manufacturer.
Feeder barcode	Barcode label that uniquely identifies a feeder.
Quantity	Initial quantity of parts existing in the feeder at the time of mounting it to placement machine.
Time On	Date and Time when the part was mounted on the machine.
Time Off	Date and Time when the part was unmounted from the machine.
Mount Operator	Name of the operator who mounted the part reel.
UserData1	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Unmount Operator	Name of the operator who unmounted the part reel.
Spliced Operator	Name of the operator who spliced.
Part Class	It refers to part shape that has a corresponding vision program used by the machine to recognize component shape. Some of the part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	An MCID is an identification alias, barcoded on each reel, which maps to a part number.
UserData2 to 5	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.

- Current Setup Report

Item	Description
Line Name	Name of the route on which products are manufactured.
Zone Name	Name of the production zone consisting of placement machine. Zones exist within a route.
Equipment Name	Name of the placement machine that places parts on the PCB
Lane	Lane number of the placement machine. Placement machine can have one lane or two lanes. In each lane, single product will be manufactured.
Stage No.	Stage number of a modular machine. Modular machines consisting of two tables will have a single stage while modular machines consisting of four tables have two stages. For example in CM602, Table 1 and 2 comprise stage 1 while Table 3 and 4 comprise stage 2. For placement machine consisting of two stages, some components will be placed on board in stage 1 while rest of the components will be placed on the board in stage 2.
Z/PU #	Machine location from which parts are picked for placement. For example Z/PU value of 20014 indicates that parts are picked from a feeder located at Table 2 Slot 14 of placement machine.
Slot	Machine slot number from which parts are picked for placement.
SubSlot	In case of double feeder, thin-type single feeder or auto loading feeder (spliceless feeder), when components are supplied from left side and right side, subslot corresponds to a single side (left or right) from which parts are picked. Generally Subslot value of 0 corresponds to left side while subslot value of 1 corresponds to right side. In this report, L corresponds to left subslot while R corresponds to right subslot.
Material Barcode	Reel Barcode of the reel consisting of electronic parts that are placed on the PCB during the manufacturing process.
Part No.	Part number of a component. Each component placed on the PCB is uniquely identified by the part number.
Expected Part	Part number expected at a feeder location based upon the product setup information downloaded to the scanner.
Lot No	Entity that is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing
Vendor	Identifier for the part manufacturer
UserData1	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Feeder barcode	Barcode label that uniquely identifies a feeder
Initial quantity	Quantity of parts existing in the feeder at the time of mounting it to placement machine.
Mount Time	Date and Time at which feeder was mounted to placement Machines.
Operator	Barcode of the operator who mounted the part reel.
Splicing	Specifies whether a reel was spliced to an existing reel.
Main Reel	In case a reel was spliced to an existing reel, it contains the reel barcode of the existing reel.
Splicing Reel	In case a reel was spliced to an existing reel, it contains the barcode of the spliced reel.
Splicing Reel 2 to 3	For stacking stick feeder.
Spliced Operator	Barcode of the operator who spliced.

Item	Description
Part Class	It refers to part shape that has a corresponding vision program used by the machine to recognize component shape. Some of the part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	An MCID is an identification alias, barcoded on each reel, which maps to a part number.
Audit Status	Status of the Setup Audit. Setup Audit is performed for a Setup to ensure that the operator placed all feeders in the correct slots. The Setup Audit operation will verify all part numbers or reel barcodes by comparing feeder arrangement and CVT data to ensure correct feeder placement. If an incorrect part number or reel barcode is scanned two times, and error message will appear until the correct part is scanned. It is to be noted that this feature of Setup Audit does not apply to non-modular machines.
Audit Failures	Number of Setup Audit failures. For example, a feeder is present at an incorrect location which does not exist in the setup.
UserData2 to 5	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.

- Work History Report

Item	Description
Line Name	Name of the route on which production occurred.
Zone Name	Name of the production zone consisting of placement machine.
Equipment Name	Name of the placement machine that will place parts on the PCB.
Stage No.	Stage number of a modular machine. Modular machines consisting of two tables will have a single stage while modular machines consisting of four tables have two stages. For example in CM602, Table 1 and 2 comprise stage 1 while Table 3 and 4 comprise stage 2. For placement machine consisting of two stages, some components will be placed on board in stage 1 while rest of the components will be placed on the board in stage 2.
Z/PU #	Machine location from which parts are picked for placement. For example Z/PU value of 20014 indicates that parts are picked from a feeder located at Table 2 Slot 14 of placement machine.
Slot	Machine slot number from which parts are picked for placement.
SubSlot	In case of double feeder, thin-type single feeder or auto loading feeder (spliceless feeder), when components are supplied from left side and right side, subslot corresponds to a single side (left or right) from which parts are picked. Generally Subslot value of 0 corresponds to left side while subslot value of 1 corresponds to right side. In this report, L corresponds to left subslot while R corresponds to right subslot.
Material Barcode	Reel Barcode of the reel consisting of electronic parts that are placed on the PCB during the manufacturing process.
Part No.	Part number of a component. Each component placed on the PCB is uniquely identified by the part number.
Expected Part	Part number expected at a feeder location based upon the product setup information downloaded to the scanner
Lot No	Entity that is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing.
Vendor	Identifier for the part manufacturer.
Feeder barcode	Barcode label that uniquely identifies a feeder.

Item	Description
Quantity	Initial quantity of parts existing in the feeder at the time of mounting it to placement machine.
Mount Time	Date and Time when the operation was initiated.
Unmount Time	Date and Time when the operation was completed.
Operator	Barcode of the operator who installed a part reel and completed the specified operation.
Splicing Operator	Name of the operator who spliced.
UserData1	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Operation	Type of the operation performed.
Part Class	It refers to part shape that has a corresponding vision program used by the machine to recognize component shape. Some of the part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	An MCID is an identification alias, barcoded on each reel, which maps to a part number.
Part Attribute Comparison	Comparison status (e.g.: No Comparison, Compared).
Old Part Attribute	Old part name.
Part Attribute Additional Information	It is an attribute for comparison between new and old. (e.g.: Vendor, Lot and User information).
UserData2 to 5	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.

- Transaction Trace Report

Item	Description
Line Name	Name of the production line.
Zone Name	Name of the production zone consisting of placement machine.
Equipment Name	Name of the placement machine that will place parts on the PCB.
Stage No.	Stage number of a modular machine. Modular machines consisting of two tables will have a single stage while modular machines consisting of four tables have two stages. For example in CM602, Table 1 and 2 comprise stage 1 while Table 3 and 4 comprise stage 2. For placement machine consisting of two stages, some components will be placed on board in stage 1 while rest of the components will be placed on the board in stage 2.
Z/PU #	Machine location from which parts are picked for placement. For example Z/PU value of 20014 indicates that parts are picked from a feeder located at Table 2 Slot 14 of placement machine.
Slot	Machine slot number from which parts are picked for placement
SubSlot	In case of double feeder, thin-type single feeder or auto loading feeder (spliceless feeder), when components are supplied from left side and right side, subslot corresponds to a single side (left or right) from which parts are picked. Generally Subslot value of 0 corresponds to left side while subslot value of 1 corresponds to right side. In this report, L corresponds to left subslot while R corresponds to right subslot.
Part No.	Part number of a component. Each component placed on the PCB is uniquely identified by the part number.
Expected Part	Major part name existing in a particular feeder location based on the product setup information downloaded to the machine.
Lot No	Entity that is used to keep track of parts that are placed at different

Item	Description
	points in time in order to know when a particular group of parts was used in the manufacturing.
Vendor	Identifier for the part manufacturer.
Feeder barcode	Barcode label that uniquely identifies a feeder.
Initial quantity	Quantity of parts existing in the feeder at the time of mounting it to placement machine.
Material Mount Time	Date and Time when the part reel was mounted.
Material Unmount Time	Date and Time when the part reel was unmounted.
Operation Type	Type of the operation performed at a particular feeder location.
Result	Result of the operation in 17 above, and if it failed, the reason for the failure.
Operator Name	Name of the operator who performed a particular CVT operation.
Uninstalled Operator	Name of the operator who removed a part from the machine.
Splicing Operator	Name of the operator who performed a splicing operation.
UserData1	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Material Barcode	Barcode of the reel that is existing at a particular feeder location.
Main Reel	Barcode of the reel that is mounted at the placement machine.
Splicing Reel	Material barcode of the reel that is spliced to the main reel.
Splicing Reel 2 to 3	For stacking stick feeder.
Stick Count	For stacking stick feeder.
Part Class	It refers to part shape that has a corresponding vision program used by the machine to recognize component shape. Some of the part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	An MCID is an identification alias, barcoded on each reel, which maps to a part number.
Comparison ID	It refers to the reel list barcode. If this feature is enabled in the **Mobile Scanner** tab of the **System Config** dialog, scanner prompts to scan a valid Reel List ID before scanning the Part Number during production and Presetup operations. String compare of the comparison ID data to the Part Number data is done and the operation continues further only if a valid string match is found.
Overwrite Reason	The reason for overwriting comparison of part attribute.
Part Attribute Comparison	Comparison status (e.g.: No Comparison, Compared).
Old Part Attribute	Old part name.
Part Attribute Additional Information	It is an attribute for comparison between new and old. (Ex.: Vendor, Lot and User information).
UserData2 to 5	Entity that is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing.
Lane 1 (Lot Name)	A lot name downloaded to lane 1 during transaction.
Lane 1 (Mixed Name)	A mixed name of lane 1.
Lane 2(Lot Name)	A lot name downloaded to lane 2 during transaction.
Lane 2 (Mixed Name)	A mixed name of lane 2.
Data Source	Source of the data import.

- Production Order Trace Report

Item	Description
Work Order Name	Name of the work order. It defines production schedule of a particular product in terms of number of PCBs that need to be manufactured, start time and end time of the production.
Equipment Name	Name of the placement machine that is used for placing components on the PCB.
Mix Name	Name of the mix consisting of individual products (boards to be produced) and their associated manufacturing data in order to share single setup of manufacturing resources like placements machines, component feeders etc.
Product Name	Name of the individual product in the mix. An individual product corresponds to a single PCB on which different types of electronic components are placed by fetching components from feeders attached to placement machines.
Start Time	Time when the production actually started corresponding to a work order.
End Time	Time when the production actually ended corresponding to a work order.
Material Barcode	Barcode of the reel used for manufacturing a product.
Part No.	Part number of a component. Each component placed on the PCB is uniquely identified by the part number.
Expected part	Part number expected at a feeder location based upon the program downloaded to the machine.
Vendor	Identifier for the part manufacturer.
Lot No	Entity that is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing.
UserData1	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Initial quantity	Quantity of parts existing in the feeder at the time of mounting it to placement machine
Part Class	It refers to part shape that has a corresponding vision program used by the machine to recognize component shape. Some of the part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	An MCID is an identification alias created by a user, barcoded on each reel, which maps to a part number.
UserData2 to 5	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Operator Name	Name of the operator who performed the specified operation.

- Loaded Feeder Report

Item	Description
Equipment Name	Name of the placement machine that is used for placing components on the PCB.
Feeder barcode	Barcode label that uniquely identifies a feeder.
Material Barcode	Barcode of the reel used for manufacturing a product.
Part No	Part number of a component whose component reel is loaded onto feeder.
Initial Quantity	Number of components existing in the reel at the time

Item	Description
	when this reel was loaded onto the feeder
Vendor	Identifier for the part manufacturer.
Lot No	Entity that is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing.
UserData1	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Registered Operator	Identifier of the operator who loaded the feeder with a component reel. Operator ID is usually could be in the form of barcode / operator name.
Machine Entry Time	Date and Time when a feeder was loaded with a component reel
Part Class	It refers to part shape that has a corresponding vision program used by the machine to recognize component shape. Some of the part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	An MCID is an identification alias, barcoded on each reel, which maps to a part number.
UserData2 to 5	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.

- Feeder Routine Check History

Item	Description
Equipment Name	Name of the placement machine that is used for placing components on the PCB.
Setup Audit Start Time	Time when the routine check started.
Setup Audit End Time	Time when the routine check ended.
Machine Side	Location of a feeder to be carried out routine audit. (Front or Rear)
Position	Feeder position consisting of slot or subslot on the particular table that is carried out routine audit.
Scanned Data	Part attribute scanned as part of carrying out routine audit. This is usually reel barcode/ part number
Part No	Part number of a component whose component reel is loaded onto feeder
Scanned Lot No	Scanned lot number.
Timestamp	Date and Time when the routine audit is carried out.
Operator Name	Name of the operator who carries out the routine audit.
Routine Audit Type	Type of the routine audit. There are three type of routine audit namely: **Manual Audit:** This is an on demand type of auditing and does not require scheduling as a prerequisite for performing an audit. **Program change audit:** Routine Audit is initiated when a program change has been detected at either the position or the machine. **Timed Audit:** Routine Audit is initiated at a preconfigured time interval or a shift schedule.
Feeder Routine Audit Status	Indicates the status of routine audit corresponding to a feeder location for the product about to be currently manufactured. It could be one of the following: **Complete**: The operator scanned the part in this slot and it matches the expected part. **Incomplete**: The operator has not been prompted to scan this part yet.

Item	Description
	Interlock: The operator could not complete the audit by scheduled end time and the machine is locked. **Wrong**: The operator scanned the part at a particular machine location and it does not match the expected part. **Skipped**: The operator chose to skip a slot when prompted to scan it.
Routine Audit Failures	Number of times routine audit failed for a particular feeder location. It corresponds to Setup Audit status "**Wrong**".
Current Board	Product whose manufacturing is currently scheduled and for which setup audit is initiated in order to avoid loading wrong feeders.

- Production Lock History

Item	Description
Equipment Name	Name of the placement machine that is used for placing components on the PCB.
Board Name	Name of the product that was being manufactured at the time machine got locked.
Event	Event that triggered the machine lock. For example, event could be Setup Audit that was not completed at the scheduled time.
Lock Time	Time when the machine got locked.
Unlock Time	Time when the machine was unlocked.
Unlock Method	Action performed to unlock the machine. For example, operator completing the setup audit of all feeder locations to unlock the machine.
Operator Name	Name of the operator during whose shift machine got locked.
Details	Additional information related to machine locking.
Material Barcode	Barcode of the reel used for manufacturing a product.

- Remaining Shift Time

Item	Description
Product Name	Name of the individual product in the mix. An individual product corresponds to a single PCB on which different types of electronic components are placed by fetching components from feeders attached to placement machines.
Part No	Part number of a component whose component reel is loaded onto feeder.
Vendor	Identifier for the part manufacturer.
Lot No	Entity that is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing.
UserData1	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Timestamp	Time to verify.

- LCR Verification History (Out of support)

Item	Description
Equipment Name	Name of the placement machine that is used for placing components on the PCB.
Material Barcode	Barcode of a reel verified for production of productions.
Part No	Part number of a component whose component reel is loaded

Item	Description
	onto feeder.
Run Time	-
Verification Result	Verification result. If failed, reasons for the failure.
Operator Name	Name of the operator who performs verification.
Part Attribute	-
Measured Value	-
Predicted Value	-
Predicted Tolerance	-
Manual Input	-
Remaining	-

- Routine Check History

Item	Description
Equipment Name	Name of the placement machine that is used for placing components on the PCB.
Operation	Type of the operation that the routine audit is carried out.
Work Area	Work area where the routine audit is carried out.
Position	Feeder position consisting of slot or subslot on the particular table that is carried out routine audit.
Scanned Data	Part attribute scanned as part of carrying out routine audit. This is usually reel barcode/ part number
Part No	Part number of a component whose component reel is loaded onto feeder
Scanned Lot No	Scanned lot number.
Scanned Quantity	Scanned quantity.
Running Time	Running time
Operator Name	Name of the operator who carries out the routine audit.
Routine Audit Type	Type of the routine audit. There are three type of routine audit namely: **Manual Audit:** This is an on demand type of auditing and does not require scheduling as a prerequisite for performing an audit. **Program change audit:** Routine Audit is initiated when a program change has been detected at either the position or the machine. **Timed Audit:** Routine Audit is initiated at a preconfigured time interval or a shift schedule.
Routine Audit Result	Results of routine audit.
Routine Audit Failures	Number of times routine audit failed for a particular feeder location. It corresponds to Setup Audit status “**Wrong**”.
Current Board	Product whose manufacturing is currently scheduled and for which setup audit is initiated in order to avoid loading wrong feeders.

■ Material Verification-Presetup

● Presetup Detail Report

Item	Description
Presetup Name	Identifier for the presetup that is mounted at a particular machine location.
Presetup Location	Name of the production zone where presetup cart is located.
Target Scanner	Scanner name associated with the machine.
Target Zone	Production zone for which presetup is being performed. This zone contains the machine name where the presetup will be mounted.
Carriage Position	Machine table at which Presetup cart will be mounted
Mix Name	Mix that is being used for performing presetup. This name is a combination of product job name, panel (board) name and serial number. Mainly used to identify boards to be produced if the resources like placements machines, component feeders, etc. are set to be shared with multiple panels (boards) in the product data.
Board Name	Name of the panel (board) or lot of the product program for the product being used for presetup.
Z/PU #	Machine location at which a part is scanned while performing presetup. For example, Z/PU value of 20014 indicates that part that is scanned is located at Table 2 Slot 14 of placement machine.
Slot	Slot at which part scanned during presetup is located.
SubSlot	Subslot at which part scanned during presetup is located.
Feeder barcode	Barcode label of the feeder used while performing presetup
Part number	Part number of the component that is scanned during presetup。
Supply Method	Specifies the way in which components are fed to placement machines. For example, supply could be in the form of reel, tray, stick, etc.
Expected Part No	Part number expected at a feeder location expressed as slot or subslot number, based upon the product program downloaded to the scanner
Material Barcode	Reel barcode of the reel used during presetup. This column will be populated if presetup is performed in material control mode.
Vendor	Identifier for the part manufacturer
Lot No.	Lot No. which component scanned during presetup. Lot No. is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing.
Initial Quantity	Number of components existing in the reel at the time the reel was mounted for presetup.
UserData1	First data customized as part information of the feeder loaded on the presetup cart. This customized part data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Presetup Operator	Name / barcode of the operator who performed the presetup.
Presetup Load Time	Time when the presetup was mounted to the machine.
Estimated Quantity	This is the quantity in the reel entered during setup, or the expected current quantity from which the number of components consumed for production after setup has been subtracted.
Part Class	Characterizes the shape of the part used for the presetup. This part shape has a corresponding vision program which is used by the machine to recognize it. Some of the part class values are 1

Item	Description
	(Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	Material component ID of the component used in the Presetup. It is an identification alias, barcoded on each reel, which maps to a part number.
UserData2 to 5	Data customized as part information of the feeder loaded on the presetup cart. This customized part data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.

- Presetup Summary Report

Item	Description
Presetup Name	Identifier for the presetup that is mounted at a particular machine location.
Presetup Location	Name of the production zone where presetup cart is located.
Target Scanner	Scanner name associated with the machine.
Target Zone	Production zone for which presetup is being performed. This zone contains the machine name where the presetup will be mounted.
Carriage Position	Machine table at which Presetup cart will be mounted.
Board Name	Name of the panel (board) or lot of the product program for the product being used for presetup.
Setup Complete	Date and Time when the presetup was last performed. It does not necessarily indicate that presetup was completed at this point in time.
Status	Presetup state when it was last performed. For example, presetup could be completed or it was halted midway due to which it would be in an incomplete state.

- Modular Presetup Cart Mount History Report

Item	Description
Cart ID	Unique identifier of the presetup cart.
Equipment Name	Name of the modular machine on which the presetup was mounted.
Table No	Table number at which presetup cart was mounted. It would take a value between 1 and 4 corresponding to table 1, table 2, table 3 and table 4 respectively.
Time ON	Date and Time when the presetup cart was mounted on a modular machine.
Time Off	Date and Time when the presetup cart was unmounted from the modular machine. If this column contains the text “**Still on machine”**, it indicates that presetup cart is still mounted.
Operator Name	Name/barcode of the operator who mounted / unmounted the presetup cart.

- Modular Presetup Cart Current Contents Report

Item	Description
Cart ID	Unique identifier of the presetup cart.
Location	Name of the production zone where presetup cart is located.
Z/PU #	Machine location at which a part is scanned while performing presetup. For example, Z/PU value of 20014 indicates that part that is scanned is located at Table 2 Slot 14 of placement machine.
Slot No	Slot to which parts will be fed by the presetup cart after being

Item	Description
	Mounted.
SubSlot	In case of double feeder, thin-type single feeder or auto loading feeder (spliceless feeder), when components are supplied from left side and right side, subslot corresponds to a single side (left or right) from which parts are picked. Generally Subslot value of 0 corresponds to left side while subslot value of 1 corresponds to right side. In this report, L corresponds to left subslot while R corresponds to right subslot.
Feeder Barcode	Barcode label of the feeder loaded to the presetup cart. From this feeder, parts are fed to corresponding slot and subslot of the machine after presetup cart is mounted to the machine table.
Material Barcode	Barcode of the reel loaded to the feeder
Part No	Part number of the component that is scanned during presetup.
Expected Part No	Part number expected at a feeder location expressed as slot or subslot number, based upon the product program downloaded to the scanner.
Vendor	Identifier for the manufacturer of the part existing in the feeder loaded to the presetup cart.
Lot No	Lot number of the part existing in the feeder loaded to the presetup cart. Lot number helps to know when a particular group of parts was used in manufacturing a PCB.
UserData1	First data customized as part information of the feeder loaded on the presetup cart. This customized part data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Timestamp	Time when the feeder loaded with part reel was mounted to the presetup cart.
Presetup Operator Name	Name / barcode of the operator who performed the presetup.
Estimated Quantity	This is the quantity in the reel entered during setup, or the expected current quantity from which the number of components consumed for production after setup has been subtracted.
Initial Quantity	Number of components existing in the reel at the time the reel was mounted for presetup.
Part Class	Indicates characteristics of the shape on the presetup cart. This part shape has a corresponding vision program which is used by the machine to recognize it. Some of the part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	Material component ID of the component used in the Presetup. It is an identification alias, barcoded on each reel, which maps to a part number.
UserData2 to 5	Data customized as part information of the feeder loaded on the presetup cart. This customized part data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.

- Modular Presetup Cart Feeder History Report

Item	Description
Cart ID	Unique identifier of presetup cart.
Equipment Name	Name of the placement machine that uses parts existing on the presetup cart for placement on the PCB.
Board Name	Name of the board (PCB) or lot of the product program for the product being used for presetup.
Z/PU #	Machine location at which a part is scanned while performing presetup. For example, Z/PU value of 20014 indicates that part

Item	Description
	that is scanned is located at Table 2 Slot 14 of placement machine.
Slot No	Machine slot to which parts are supplied by the feeder mounted on the presetup cart
SubSlot	In case of double feeder, thin-type single feeder or auto loading feeder (spliceless feeder), when components are supplied from left side and right side, subslot corresponds to a single side (left or right) from which parts are picked. Generally Subslot value of 0 corresponds to left side while subslot value of 1 corresponds to right side. In this report, L corresponds to left subslot while R corresponds to right subslot.
Feeder barcode	Barcode label of the feeder loaded to the presetup cart. From this feeder, parts are fed to corresponding slot and subslot of the machine after presetup cart is mounted to the machine table.
Material Barcode	Barcode of the reel loaded to the feeder
Part No	Part number of the component that is scanned during presetup.
Expected Part No	Part number expected at a feeder location expressed as slot or subslot number, based upon the product program downloaded to the scanner
Vendor	Identifier for the manufacturer of the part existing in the feeder loaded to the presetup cart.
Lot No	Lot number of the part existing in the feeder loaded to the presetup cart. Lot number helps to know when a particular group of parts was used in manufacturing a PCB.
UserData1	First data customized as part information of the feeder loaded on the presetup cart. This customized part data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Time On	Time when the feeder loaded with part reel was mounted to the presetup cart
Presetup Operator	Name / barcode of the operator who performed the presetup.
Time Off	Time when a particular feeder (identified by feeder barcode) was unmounted from the presetup cart.
Operator Name	Name of the operator who mounted / unmounted the feeder with respect to presetup cart.
Estimated Quantity	This is the quantity in the reel entered during setup, or the expected current quantity from which the number of components consumed for production after setup has been subtracted.
Initial Quantity	Number of components existing in the reel at the time the reel was mounted for presetup.
Part Class	Indicates characteristics of the shape on the presetup cart. This part shape has a corresponding vision program which is used by the machine to recognize it. Some of the part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	Material component ID of the component used in the Presetup. It is an identification alias, barcoded on each reel, which maps to a part number.
Comparison ID	It refers to the reel list barcode. If this feature is enabled in the **Mobile Scanner** tab of the **System Config** dialog, scanner prompts to scan a valid Reel List ID before scanning the Part Number during production and Presetup operations. String compare of the comparison ID data to the Part Number data is done and the operation continues further only if a valid string match is found.
Overwrite Reason	“MaterialResuseOverride” appears only when the following

Item	Description
	options are checked in the **Mobile Scanner** tab of the **System Config** dialog. - Disable the use of loaded/spliced parts: Checked - Confirm overwrite: Specified any option other than **None**
UserData2 to 5	Data customized as part information of the feeder loaded on the presetup cart. This customized part data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.

- Modular Presetup Cart Mount Status Report

Item	Description
Cart ID	Unique identifier of the presetup cart.
Position	Area where presetup cart is located.
Timestamp	Date and Time when the presetup cart was mounted to a placement machine table.
Status	Status of the presetup cart Unmounted: Cart is removed from the machine. Mounted: Cart is mounted to the machine.
Table No	Table number of the placement machine to which presetup cart is mounted.

- Modular Presetup Cart Check Report

Item	Description
Cart ID	Unique identifier of the presetup cart.
Line Name	Line Name that prodiction is performed
Equipment Name	Name of the modular machine on which the presetup was Mounted.
Board Name	Name of the board (PCB) or lot of the product program for the product being used for presetup
Mix Name	Mix that is being used for performing presetup. This name is a combination of product job name, panel (board) name and serial number. Mainly used to identify boards to be produced if the resources like placements machines, component feeders, etc. are set to be shared with multiple panels (boards) in the product data.
Table No	Table number of the placement machine to which presetup cart is mounted
Parts on Cart	Number of components existing on the presetup cart.
Parts on Product	Number of parts that need to be placed on the PCB.
Verification Count	Number of parts existing on the presetup cart that are also existing in the product.

- Modular Presetup Production Data Check

Item	Description
Equipment Name	Name of the modular machine on which the presetup was Mounted.
Table No	Table number of the placement machine to which presetup cart is mounted.
Cart ID	Unique identifier of the presetup cart.
Parts on Cart	Number of components existing on the presetup cart.
Parts on Product	Number of parts that need to be placed on the PCB.
Verification Count	Number of parts existing on the presetup cart that are also existing in the product.

- Modular Presetup Cart Routine Check

Item	Description
Cart ID	Unique identifier of the presetup cart.
Routine Check Start Time	Time when the routine check started.
Routine Check End Time	Time when the routine check ended.
Position	Feeder position consisting of slot or subslot on the particular table that is carried out routine audit.
Scanned Data	Part attribute scanned as part of carrying out routine audit. This is usually reel barcode/ part number
Part No	Part number of a component whose component reel is loaded onto feeder
Timestamp	Date and Time when the routine audit is carried out.
Operator Name	Name of the operator who carries out the routine audit.
Routine Audit Type	Type of the routine audit. There are three type of routine audit namely: **Manual Audit:** This is an on demand type of auditing and does not require scheduling as a prerequisite for performing an audit. **Program change audit:** Routine Audit is initiated when a program change has been detected at either the position or the machine. **Timed Audit:** Routine Audit is initiated at a preconfigured time interval or a shift schedule.
Feeder Routine Audit Status	Indicates the status of routine audit corresponding to a feeder location for the product about to be currently manufactured. It could be one of the following: **Complete**: The operator scanned the part in this slot and it matches the expected part. **Incomplete**: The operator has not been prompted to scan this part yet. **Interlock**: The operator could not complete the audit by scheduled end time and the machine is locked. **Wrong**: The operator scanned the part at a particular machine location and it does not match the expected part. **Skipped**: The operator chose to skip a slot when prompted to scan it.
Routine Audit Failures	Number of times routine audit failed for a particular feeder location. It corresponds to Setup Audit status “**Wrong**”.

- Modular Presetup Cart ID List

Item	Description
Cart ID	Unique identifier of the presetup cart.
Cart Serial No	Unique identifier of assigning to a cart

■ Material Control -Base

● Material Quantity Report

Item	Description
Part No	Part number of the component. Each component placed on the PCB is uniquely identified by the part number
Material Barcode	Barcode of the reel that could be mounted on the machine or existing in the offline storage area
Location	Identifies the location of part reel. If the reel is present in the offline storage, it could refer to the name of material area / storage unit. If the component is mounted on the machine, name of the machine would be listed.
Z/PU #	Specifies the feeder location at which reel is mounted. If reel is stored in an offline storage area, it will be blank. Z number is for Panasert machine while PU number is used for CM machines.
Vendor	Unique identifier for the part manufacturer.
Lot No	Entity that is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing
UserData1 to 5	Contains customized part data when that data could not be accommodated in the fields commonly used like Lot No.., Vendor, Part number and quantity.
Operator	Name of the operator who mounted the reel on the machine
Mount Time	Date and Time when the operator mounted the reel on the machine
Mount Operator	Name of the operator who mounted the reel on the machine
Mount Time	Date and Time when the operator mounted the reel on the machine.
Check in Operator	Name of the operator who checked in the component reel in an offline storage area
Check in Time	Date and Time when the component reel was checked in an offline storage area
Feeder barcode	Barcode that uniquely identifies the feeder containing the component reel
Status	It refers to the state of mounting i.e. whether the reel is mounted or unmounted.
Estimated Quantity	Quantity of parts remaining in the reel (identified by a unique material barcode/ material ID) after usage
Initial Quantity	Initial quantity of the parts existing in the reel (identified by a unique material barcode/material ID) before being used for production operations
Part Class	It refers to part shape that has a corresponding vision program used by the machine to recognize component shape. Some of the part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	It refers to identification alias created by the customer, barcoded on each reel, which maps to a part number.

● Printer Material Usage Count Report

Item	Description
Material Barcode	Barcode of the reel that could be mounted on the machine or existing in the offline storage area
Part No	Name assigning to a component. Each component is uniquely identified by this number.

Item	Description
Material	Type of screen printer parts.
Current Use Count	Number of use count so far.
Warning Count	Number of times that warning is displayed.
Max. Use Count	The number of times that is set as limit of use.
Comment	Comment registered on the material edit screen (for screen printer).

- Material Adjustment Report

Item	Description
Time	Date and Time when quantity adjustment was performed for a reel barcode.
Location of adjustment	Feeder location consisting of slot and subslot of particular table that part count is adjusted.
Material Barcode	Barcode of the reel for which part quantity was adjusted.
Part No	Part number of the component existing in the reel.
Vendor	Unique identifier for the part manufacturer.
Lot No	Lot number of the component existing in the reel for which part adjustment was carried out. Lot number is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing
UserData1	User data of the component existing the reel for which part adjustment was carried out. It refers to the customized part data when that data could not be accommodated in the fields commonly used like Lot No.., Vendor, Part number and quantity
Work Order Name	Work order name corresponding to which reel was used in production operations. A work order contains important information related to production schedule of a product like start date / end date of production, quantity of boards produced on that date etc.
Type	Type of component adjustment.
Old Quantity	Initial quantity of parts existing in the reel prior to adjustment
Adjustment	Quantity added to/subtracted from the initial part quantity.
Auto Decrement	Whether the number of materials is automatically subtracted by PanaCIM or not is displayed. Because adjustment of quantity is manually adjusted by using the mobile scanner, the value is normally ‘0’.
New Quantity	Part quantity after adjustment quantity was added to / subtracted from the initial part quantity
Operator	Name/barcode of the operator who adjusted the part quantity for a material barcode
Reason	Reason for adjusting the part quantity
Part Class	It refers to part shape that has a corresponding vision program used by the machine to recognize component shape. Some of the part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	It refers to identification alias created by the customer, barcoded on each reel, which maps to a part number.
UserData2 to 5	Contains customized part data when that data could not be accommodated in the fields commonly used like Lot No.., Vendor, Part number and quantity.

- Material Usage Report

Item	Description
Material Barcode	Unique barcode assigned to each reel for which material usage is being tracked.
Part No	Part number of the component existing in the reel.
Vendor	Unique identifier for the part manufacturer.
Lot No.	Lot number of the component existing in the reel.
Initial Quantity	Initial quantity of parts existing in the reel prior to being used for production operations.
Current Quantity	Current quantity of parts existing in the reel after reel was used in production operations.
Timestamp	Date and Time when the current quantity of parts was updated.
Machine	Name of the placement machine where the material barcode was last used.
Quantity	Quantity added to/subtracted from initial quantity.
Remaining Quantity	Remaining quantity after added to/subtracted from initial quantity.
Description	Operation that entailed reel usage.
Reason	Cause for material usage, usually production of PCB that requires placing parts on the PCB
Part Class	It refers to part shape that has a corresponding vision program used by the machine to recognize component shape. Some of the part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	It refers to identification alias created by the customer, barcoded on each reel, which maps to a part number.
UserData1 to 5	Contains customized part data when that data could not be accommodated in the fields commonly used like Lot No.., Vendor, Part number and quantity.

- Material Position History Report

Item	Description
Material Barcode	Unique barcode assigned to a part reel whose movement is being tracked.
Part No	Part number of the component existing in the reel whose movement is being tracked. Each component placed on the PCB is uniquely identified by the part number.
Vendor	Unique identifier for the part manufacturer.
Lot No	Lot number associated with part reel. Lot number is an entity used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing.
Location	If a reel is being used in production operations, it contains the name of the production zone where the reel is located. If a reel is located in offline storage area, it refers to name of material area.
Storage Unit	Name of the L1/L2 storage unit.
Storage Sub Unit	Sub storage unit number of the storage unit. Sub storage unit number is populated only for L1 storage unit.
Time In	Date and Time when a part reel was checked in to the production zone / offline storage area.
Time Out	Date and Time when a part reel was moved out of the production zone/offline storage unit. If a part reel is located in a particular location, text “Still on Location” is shown.

Item	Description
Duration	Duration for which part reel was existing in a particular location.
Operator	Name of the operator who moved the material to a particular location.
Main Reel	Barcode of the reel existing in the main memory
Splice Reel	Barcode of the reel existing in the splice memory
Splicing Reel 2 to 3	For stacking stick feeder.
Part Class	It refers to part shape that has a corresponding vision program used by the machine to recognize component shape. Some of the part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	It refers to identification alias created by the customer, barcoded on each reel, which maps to a part number.
UserData1 to 5	Contains customized part data when that data could not be accommodated in the fields commonly used like Lot No.., Vendor, Part number and quantity.

- Feeder Position History Report

Item	Description
Feeder barcode	Unique barcode label of a feeder
Location	It could either be name of the production zone or an offline feeder storage/maintenance area.
Storage Unit	Name of the L1/L2 storage unit
Storage Sub Unit	Sub storage unit number of the storage unit. Sub storage unit number is populated only for L1 storage unit
Time In	Date and Time when a feeder was checked in to the production zone or offline storage/maintenance area.
Time Out	Date and Time when a feeder was moved out of the production zone or offline storage/maintenance area.
Duration	Time period for which a feeder remained in a particular location.
Operator	Name of the operator who moved the material to a particular location.

- Material Movement Report

Item	Description
Material Travel Time	Time when the material was moved from one location to another
From Location	Source location from which material was moved
To Location	Destination location to which material was moved
Material Barcode	Unique barcode label assigned to the reel
Part No	Unique part number of a component
Vendor	Unique identifier for the part manufacturer.
Lot No	Lot number to which a part reel belongs. Lot number is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing
UserData1	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Part Quantity	Part quantity at the time material movement was carried out
Part Class	It refers to part shape that has a corresponding vision program used by the machine to recognize component shape. Some of the

	part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	It refers to identification alias created by the customer, barcoded on each reel, which maps to a part number.
UserData2 to 5	Contains customized part data when that data could not be accommodated in the fields commonly used like Lot No.., Vendor, Part number and quantity.

- Material Edit Report

Item	Description
Edit Date and Time	Date and Time when materials are edited.
Material Barcode	Unique barcode label assigned to the reel
Part No	Unique part number of a component
Estimated Quantity	This is the quantity in the reel entered during setup, or the expected current quantity from which the number of components consumed for production after setup has been subtracted.
Details	Details of materials to edit.

- Material Registration Report

Item	Description
Registration Date	Date when a material is newly registered.
Material Barcode	Unique barcode label assigned to the reel
Part No	Unique part number of a component
Vendor	Unique identifier for the part manufacturer.
Lot No	Lot number to which a part reel belongs. Lot number is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing
UserData1	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Initial Quantity	Initial quantity of the parts existing in the reel (identified by a unique material barcode/material ID) before being used for production operations
Estimated Quantity	This is the quantity in the reel entered during setup, or the expected current quantity from which the number of components consumed for production after setup has been subtracted.
Data Source	Source of the data import
Operation	Operation perform to register materials.
Operator	Name of the operator who operated.
Part Class	It refers to part shape that has a corresponding vision program used by the machine to recognize component shape. Some of the part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	It refers to identification alias created by the customer, barcoded on each reel, which maps to a part number.
UserData2 to 5	Contains customized part data when that data could not be accommodated in the fields commonly used like Lot No.., Vendor, Part number and quantity.

- Materials Summary Report

Item	Description
Part No	Unique part number of a component

Quantity	Total quantity counted by part number.

- Part Position Report

Item	Description
Part No	Unique part number of a component.
Material Barcode	Unique barcode label assigned to the reel.
Quantity	Predeicted current quantity.
Z/PU #	Specifies the feeder location at which reel is mounted. If reel is stored in an offline storage area, it will be blank. Z number is for Panasert machine while PU number is used for CM machines.
Zone/Area Name	Name of the production zone (if mounted on the machine) or the material area (if stored in the offline storage area).
Storage Unit	Name of the L1/L2 storage unit.
Storage Sub Unit	Sub storage unit number of the storage unit. Sub storage unit number is populated only for L1 storage unit.

- Material Management Report

Item	Description
Z/PU #	Specifies the feeder location at which reel is mounted. If reel is stored in an offline storage area, it will be blank. Z number is for Panasert machine while PU number is used for CM machines.
Slot No	Slot number at which a part is located。
SubSlot	In case of double feeder, thin-type single feeder or auto loading feeder (spliceless feeder), subslot corresponds to the side that components are supplied (L or R). L means left and R means right.
Part In Time	Time when a part is supplied
Part In Operator	Name of the operator who supplied the part.
Location	It could either be name of the production zone or an offline feeder storage/maintenance area.
Status	It refers to the state of mounting i.e. whether the reel is mounted or unmounted.
Storage Unit	Name of the L1/L2 storage unit
Storage Sub Unit	Sub storage unit number of the storage unit. Sub storage unit number is populated only for L1 storage unit
Zone Status	Zone status.
Item Status	Item status.
Part No	Part number of the component. Each component placed on the PCB is identified by a unique part number.
Lot No	Lot number linked to a part. Lot number is an entity that is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing.
Initial Quantity	Initial quantity of the parts existing in the reel (identified by a unique material barcode/material ID) before being used for production operations
Material Barcode	Unique barcode label assigned to the reel
Estimated Quantity	This is the quantity in the reel entered during setup, or the expected current quantity from which the number of components consumed for production after setup has been subtracted.
Update Part Quantity	Time when part quantity was updated.
Reserved Quantity	Quantity reserved by work order.

3.4 Report Item

Item	Description
Work Order Name	Name of work order.
Cart ID	Unique identifier of the presetup cart.
Splicing Reel	In case a reel was spliced to an existing reel, it contains the barcode of the spliced reel.
Spliced Lot No	Lot number of a spliced reel.
Splicing Predicted Quantity	Part quantity of a reel entered when the reel was spliced.
Splicing Total Quantity	Total of predicted quantity and splicing predicted quantity.
Feeder Registration Date	Feeder registration date
Parts Supplied	Number of parts supplied so far.
Total parts Supplied	Total number of parts supplied
Feeder Update	Feeder update time
Last Maintenance	Time when last feeder maintenance is performed.
Feeder barcode	Barcode label that uniquely identifies a feeder

■ Material Control - MSD

● MSD Transaction Report

Item	Description
Material Barcode	Unique barcode of the MSD reel
Part No	Part number of the MSD component
Lot No	Lot number of the MSD component. Lot number is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing.
MSL	Level of the MSD component. This columns contains one of the following values corresponding to each MSD level **1:** Indicates that maximum allowable floor life time is unlimited at the stated temperature and relative humidity conditions. **2:** Indicates that maximum allowable floor life time is 1 year at the stated temperature and relative humidity conditions. **2a:** Indicates that maximum allowable floor life time is 4 weeks at the stated temperature and relative humidity conditions. **3:** Indicates that maximum allowable floor life time is 168 hours at the stated temperature and relative humidity conditions. **4:** Indicates that maximum allowable floor life time is 72 hours at the stated temperature and relative humidity conditions. **5:** Indicates that maximum allowable floor life time is 48 hours at the stated temperature and relative humidity conditions. **5a:** Indicates that maximum allowable floor life time is 24 hours at the stated temperature and relative humidity conditions. **6:** Indicates that baking is mandatory before use. After bake, it must be reflowed within the time limit specified on the label.
Check-in Time	Date and Time when a reel consisting of MSD components was checked in to the location
Location	Location where the part reel consisting of MSD component was checked in. Usually it is the name of the production zone or the name of the material area
Operator	Name of the operator who checked in the MSD part reel into a location.
Floor life time remaining	Remaining time that a MSD component can be used for production at the time when it is brought in. Each time it is taking out from the baking unit, the dry box or a moisture barrier bag, the floor life counter activates, and when it is stored in the baking unit, the dry box or the moisture barrier bag, the counter stops. **Note:** This time is in the format **hours: minutes: seconds**.
Shelf life time remaining	Remaining time for which a MSD component could be reused (at the time it was checked into the location) after being exposed to moisture over the course of production runs. Shelf life is reduced because MSD components are exposed to moisture each time they are taking out from the baking unit or dry box and used for production. **Note:** This time is in the format **hours: minutes: seconds**.
Status	It refers to the state of MSD component which could be one of the following: **Sealed:** MSD component is sealed in the moisture barrier bag and not being used for production even once. **Packaged:** MSD component has been taken out of moisture barrier bag and exposed to moisture due to being used in production but currently checked in dry box to prevent exposure to moisture. **Unpackaged:** MSD component is checked out of dry box and is exposed to moisture due to being used in current production

Item	Description
	cycle. **Empty reel:** The reel contains no MSD component because the reel is exhausted.
UserData1 to 5	Contains customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.

- Latest MSD Report / Expired MSD Report

Item	Description
Material Barcode	Unique barcode label assigned to the reel
Part No	Unique part number of the MSD component
Vendor	Unique identifier for the part manufacturer.
UserData1	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Lot No	Lot number to which a MSD part reel belongs. Lot number is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing
MSL	Level of the MSD component. This columns contains one of the following values corresponding to each MSD level **1:** Indicates that maximum allowable floor life time is unlimited at the stated temperature and relative humidity conditions. **2:** Indicates that maximum allowable floor life time is 1 year at the stated temperature and relative humidity conditions. **2a:** Indicates that maximum allowable floor life time is 4 weeks at the stated temperature and relative humidity conditions. **3:** Indicates that maximum allowable floor life time is 168 hours at the stated temperature and relative humidity conditions. **4:** Indicates that maximum allowable floor life time is 72 hours at the stated temperature and relative humidity conditions. **5:** Indicates that maximum allowable floor life time is 48 hours at the stated temperature and relative humidity conditions. **5a:** Indicates that maximum allowable floor life time is 24 hours at the stated temperature and relative humidity conditions. **6:** Indicates that baking is mandatory before use. After bake, it must be reflowed within the time limit specified on the label.
Timestamp	Date and Time when MSD component was checked into current location.
Current Location	Current location of MSD part reel which could be name of the production zone (if used in current production cycle) or the name of material area (if stored in an offline storage unit).
Operator	Name of the operator who checked in the MSD reel at current location.
Baking Count	The number of time that a MSD component is checked into the baking unit
Floor Life Time Remaining	Remaining time that a MSD component can be used for production cycle after it is brought into the current location. Each time it is taking out from the baking unit, the dry box or a moisture barrier bag, the floor life counter activates, and when it is stored in the baking unit, the dry box or the moisture barrier bag, the counter stops. **Note:** This time is in the format **hours: minutes: seconds**.

Item	Description
Shelf Life Time Remaining	Remaining time for which a MSD component could be reused (at the time it was checked into the location) after being exposed to moisture over the course of production runs. Shelf life is reduced because MSD components are exposed to moisture each time they are taking out from the baking unit or dry box and used for production. **Note:** This time is in the format **hours: minutes: seconds**.
Initial Quantity	Number of parts presently existing in the MSD reel at the time it was moved to its current location
Main Reel	Material barcode of the reel existing in the main memory
Splicing Reel	Material barcode of the reel which is spliced to the main reel.
Splicing Reel 2 to 3	For stacking stick feeder.
Part Class	It refers to part shape that has a corresponding vision program used by the machine to recognize component shape. Some of the part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	It refers to identification alias created by the customer, barcoded on each reel, which maps to a part number.
UserData2 to 5	Contains customized part data when that data could not be accommodated in the fields commonly used like Lot No.., Vendor, Part number and quantity.

- MSA Part List

Item	Description
Part No	Part number of the MSD component
MSD Level	Level of MSD component. This column contains one of the following values corresponding to each MSD level. **1:** Indicates that maximum allowable floor life time is unlimited at the stated temperature and relative humidity conditions. **2:** Indicates that maximum allowable floor life time is 1 year at the stated temperature and relative humidity conditions. **2a:** Indicates that maximum allowable floor life time is 4 weeks at the stated temperature and relative humidity conditions. **3:** Indicates that maximum allowable floor life time is 168 hours at the stated temperature and relative humidity conditions. **4:** Indicates that maximum allowable floor life time is 72 hours at the stated temperature and relative humidity conditions. **5:** Indicates that maximum allowable floor life time is 48 hours at the stated temperature and relative humidity conditions. **5a:** Indicates that maximum allowable floor life time is 24 hours at the stated temperature and relative humidity conditions. **6:** Indicates that baking is mandatory before use. After bake, it must be reflowed within the time limit specified on the label.

- Baking Operation History

Item	Description
Material Barcode	Unique barcode label assigned to the reel
Part No	Unique part number of the MSD component
Vendor	Unique identifier for the part manufacturer.
MSD Level	Level of MSD component. This column contains one of the following values corresponding to each MSD level.

	1: Indicates that maximum allowable floor life time is unlimited at the stated temperature and relative humidity conditions. **2:** Indicates that maximum allowable floor life time is 1 year at the stated temperature and relative humidity conditions. **2a:** Indicates that maximum allowable floor life time is 4 weeks at the stated temperature and relative humidity conditions. **3:** Indicates that maximum allowable floor life time is 168 hours at the stated temperature and relative humidity conditions. **4:** Indicates that maximum allowable floor life time is 72 hours at the stated temperature and relative humidity conditions. **5:** Indicates that maximum allowable floor life time is 48 hours at the stated temperature and relative humidity conditions. **5a:** Indicates that maximum allowable floor life time is 24 hours at the stated temperature and relative humidity conditions. **6:** Indicates that baking is mandatory before use. After bake, it must be reflowed within the time limit specified on the label.
Check-in Time	Date and time when a MSD component is checked in to the backing unit
Check-out Time	Date and time when a MSD component is checked out from the backing unit
Check In Operator	Name of an operator who checked a MSD component into the backing unit.
Check Out Operator	Name of an operator who checked a MSD component out from the backing unit.
Storage Unit Name	Name of checked-in backing unit
Floor Life Time Remaining	Remaining time that a MSD component can be used for production cycle after it is brought into the current location. Each time it is taking out from the baking unit, the dry box or a moisture barrier bag, the floor life counter activates, and when it is stored in the baking unit, the dry box or the moisture barrier bag, the counter stops. **Note:** This time is in the format **hours: minutes: seconds**.
Shelf Life Time Remaining	Remaining time for which a MSD component could be reused (at the time it was checked into the location) after being exposed to moisture over the course of production runs. Shelf life is reduced because MSD components are exposed to moisture each time they are taking out from the baking unit or dry box and used for production. **Note:** This time is in the format **hours: minutes: seconds**.

■ Material Control – Material Allocation

● Material Allocation Reservation Report

Item	Description
Work Order Name	Name of the work order. A work order contains important information related to production schedule of a product like start date / end date of production, quantity of boards produced on that date etc.
BOM product Name	It is the base product name.
Board Name	Name of the product for which shortage exists.
PCB Lot Size	This is pattern quantity specified in the work order.
Line Name	Name of the route for which material shortages exist in an upcoming work order
Equipment	Name of the equipment on which production corresponding to a work order is scheduled to run.
Lane	Lane number of the placement machine. Placement machine can have one lane or two lanes. In each lane, single product will be manufactured.
Location	Feeder location at which shortage exists
Carriage	Name of a cart when a feeder is installed to the presetup cart. When a feeder is installed to the placement machine, it is blank.
Required Part Number	Part number for which shortage exists
Quantity Pattern	Refers to number of patterns existing in the panel.
Estimated Required Quantity	Total part quantity that is required to successfully execute the work order.
Buffer Quantity	Additional part quantity existing in the supporting areas over and above the required threshold to take care of production contingencies
Estimated Total Quantity	Sum total of Estimated required quantity and buffer quantity
Shortage Quantity	Part quantity that is neither existing in the production zones nor supporting areas and which needs to be backfilled by ordering this quantity from the warehouse.

● Material Allocation Reservation Report

Item	Description
Work Order Name	Name of upcoming work order for whose production material quantity has been reserved.
Board Name	Name of the product that will be manufactured in the current production run corresponding to a work order.
Pattern Quantity	Number of PCBs of a product that will be manufactured in the current production run corresponding to a work order.
Line Name	Name of the route on which material quantity has been reserved as part of production run for upcoming work order.
Equipment	Name of the placement machine on which production will run.
Z/PU #	Feeder location at which a reel (identified by a material ID) is present.
Slot No	Machine slot number to which reel (identified by material ID) supplies part.
SubSlot	In case of double feeder, thin-type single feeder or auto loading feeder (spliceless feeder), it corresponds to subslot that a reel supplies components. L (left subslot) or R (right subslot).
Required Part Name	Name of parts required to successfully execute the work order.

Item	Description
Reserved Substitute Part	Name of substitute parts to successfully execute the work order.
Estimated Required Quantity	Total part quantity that is required to successfully execute the work order. Successful work order execution ensures that all the panels are produced as there is sufficient quantity of parts existing at each feeder location to fulfill production requirements.
Buffer Quantity	Additional part quantity existing in the supporting areas over and above the required threshold to take care of production Contingencies..
Estimated Total Quantity	Sum total of Estimated required quantity and buffer quantity.
Material Barcode	Unique reel barcode of a part reel.
Reserved Quantity	Part quantity reserved for a work order. Reserved quantity exists on the production floor / supporting areas and hence is readily available for upcoming work order. This excludes the part quantity existing in the spliced reel.
Splicing Reel	Barcode of the reel that is spliced to an existing reel.
Splicing Reel 2 to 3	For stacking stick feeder.
Splice Reserved Quantity	Quantity of parts existing in the spliced reel.
Feeder barcode	Unique barcode assigned to the feeder that is supplying parts to a particular slot and subslot.
Location	Name of the production zone (if mounted on the machine) or the material area (if stored in the offline storage area)
Storage Unit	Name of the L1/L2 storage unit.
Sub Unit	Sub storage unit number associated with a storage unit.
Picking Operator	Name of the operator who performs picking.
Picking Time	Time when picking is performed.

- Material Allocation Setup Report

Item	Description
Previous Work Order	Work order whose production was previously running on this route before changeover was initiated
Previous Program Name	Name of the product that was being manufactured on this route before changeover was initiated
Previous Pattern Quantity	Number of PCBs of a product that were manufactured as part of previous production run
Current Work Order	Name of upcoming work order whose production is about to get underway
Current Product Name	Name of the product that will be manufactured in the current production run corresponding to a work order
Current Pattern Quantity	Number of PCBs of a product to be manufactured as part of current production run
Line Name	Name of the route on which changeover has been initiated to manufacture the current product
Equipment	Name of the placement machine on which production will run for the upcoming work order
Z/PU #	Feeder location at which a reel (identified by a material ID) is present
Slot No	Machine slot number to which reel (identified by material ID) supplies part
SubSlot	In case of double feeder, thin-type single feeder or auto loading feeder (spliceless feeder), it corresponds to subslot that a reel supplies components. L (left subslot) or R (right subslot).
Current Part No	Part number that is currently present at a feeder location

Item	Description
Required Part Number	Part number that is supposed to be present at a feeder location as per the generated program file
Action	Presetup operation. Normally part replacement.
Estimated Required Quantity	Total part quantity that is required to successfully execute the work order. Successful work order execution ensures that all the panels are produced as there is sufficient quantity of parts existing at each feeder location to fulfill production requirements.
Buffer Quantity	Additional part quantity existing in the supporting areas over and above the required threshold to take care of production contingencies
Estimated Total Quantity	Sum total of Estimated required quantity and buffer quantity
Material Barcode	Unique reel barcode of a part reel
Reserved Quantity	Part quantity reserved for a work order. Reserved quantity exists on the production floor / supporting areas and hence is readily available for upcoming work order. This excludes the part quantity existing in the spliced reel.
Splicing Reel	Barcode of the reel that is spliced to an existing reel
Splicing Reel 2 to 3	For stacking stick feeder.
Splice Reserved Quantity	Quantity of parts existing in the spliced reel.
Feeder barcode	Unique barcode assigned to the feeder that is supplying parts to a particular slot and subslot.
Location	Name of the production zone (if mounted on the machine) or the material area (if stored in the offline storage area)
Storage Unit	Name of the L1/L2 storage unit
Sub Unit	Sub storage unit number associated with a storage unit.

- Material Allocation Production Order History Report

Item	Description
Work Order Name	Name of the work order created
Line Name	Name of the route for which work order history is being viewed
Setup Mix	Name of the mix that contains all the manufacturing data to produce individual PCBs of a product on the route
Product Name	Name of the individual product in a mix. PCBs will be produced for an individual product by fetching different types of electronic components from feeders attached to placement machines and placing them at the designated locations on the board.
Pattern Quantity	Number of PCBs of a product that were manufactured as part of production run corresponding to a work order
Bard Cycle Time (second)	Time required in seconds to produce a single PCB of a product
Schedule Start Time	Date and Time when production started on a route corresponding to a work order
Schedule End Time	Date and Time when production ended on a route corresponding to a work order
#Reserved	Quantity of materials reserved corresponding to a work order. This is the material quantity that has been dedicated to a work order production run and no other work order can utilize this quantity until this work order is deleted.
#Storage	Material quantity that was neither existing in the production zones nor supporting areas and which needs to be backfilled by ordering this quantity from the warehouse
#Picked	Material quantity that has been picked, scanned and loaded to

Item	Description
	presetup cart or feeder for movement to production zones
Status	Status of the work order. Work order can have one of the following status within the context of this report: **Closed:** It signifies that production has completed for a work order. **Created:** It signifies that work order is created but no reservations have been done. **Reserved:** It signifies that Materials have been reserved for a work order but production has not begun. **Picked:** It signifies that reserved materials have been picked and moved to production floor. **Production:** It signifies that production has begun for a work order.

- Allocatable Component Report

Item	Description
Work Order Name	Name of the work order to produce.
Part No	Unique part number of a component
Estimated Required Quantity	Total part quantity that is required to successfully execute the work order. Successful work order execution ensures that all the panels are produced as there is sufficient quantity of parts existing at each feeder location to fulfill production requirements.
Reserved Quantity	Part quantity reserved for a work order. Reserved quantity exists on the production floor / supporting areas and hence is readily available for upcoming work order. This excludes the part quantity existing in the spliced reel.
Shortage Quantity	Part quantity that is neither existing in the production zones nor supporting areas and which needs to be backfilled by ordering this quantity from the warehouse
Material Barcode	Unique reel barcode of a part reel
Estimated Quantity	This is the quantity in the reel entered during setup, or the expected current quantity from which the number of components consumed for production after setup has been subtracted.
Creation Time	Time when the data is created.
Location	Name of the production zone (if mounted on the machine) or the material area (if stored in the offline storage area)

■ Traceability

● Panel to Component Traceability Report

Item	Description
Line Name	Name of the route on which panel was built
Zone Name	Name of the zone on which production occurred when the panel was built
Equipment Name	Name of the placement machine that placed parts on the panel
Lane	Lane number of the placement machine. Placement machine can have one lane or two lanes. In each lane, single product will be manufactured.
Stage No.	Stage number of a modular machine. Modular machines consisting of two tables will have a single stage while modular machines consisting of four tables have two stages. For example in CM602, Table 1 and 2 comprise stage 1 while Table 3 and 4 comprise stage 2. For placement machine consisting of two stages, some components will be placed on board in stage 1 while rest of the components will be placed on the board in stage 2.
Product Name	Name of the product whose component placements exist on the specified panel model number / serial number.
Setup Name	Name of the mix to which product belonged.
Machine File Name	Name of a program file associated with a particular placement machine. Placement machine uses this file for placing components on the PCB.
Model ID	Model number of the panel. The Model Code is used to validate the incoming panel is from the same Product as the Product in current production.
Panel Barcode	Unique barcode assigned to each panel.
Side	Indicates side (Top or Bottom) of the Panel where a part was placed
Serial No	Serial number of the panel. Serial Number is a unique identifier for each panel used for traceability. All traceability data is linked to the Panel Serial Number.
Pattern Serial No	Serial Number of the each pattern existing on the pattern. A pattern is defined as single printed circuit existing on the panel. There is one to one correspondence between product name and pattern.
Z/PU #	Feeder location at which a component reel was present.
Slot No	Slot number at which a part is located。
SubSlot	In case of double feeder, thin-type single feeder or auto loading feeder (spliceless feeder), subslot corresponds to the side that components are supplied (L or R). L means left and R means right.
Material Barcode	Unique barcode of the reel used for placing parts on the panel in order to produce an individual printed circuit board corresponding to a product.
Part No.	Part number of a component. Each component placed on the PCB is uniquely identified by the part number.
Expected Part	Part supposed to be present at a particular location based upon the program downloaded to the machine.
Lot No	Lot number linked to a part. Lot number is an entity that is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing.
Vendor	Unique identifier for the part manufacturer.

Item	Description
Feeder barcode	Identifier that uniquely identifies a feeder
Quantity	Number of parts existing in the reel at the time it was mounted to the machine for production
Time On	Date and Time when a part reel was mounted to the placement machine for production
Time Off	Date and Time when a part reel was unloaded from the placement machine
Mount Operator	Name of the operator who mounted the part reel on the placement machine
UserData1	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Work Order Name	Name of the work order corresponding to which a product, used a particular part number, was manufactured
Part Class	It refers to part shape that has a corresponding vision program used by the machine to recognize component shape. Some of the part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	It is an identification alias created by the user and barcoded on each reel, which maps to a part number.
UserData2 to 5	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Tape Guide Serial	Tape guide serial No for the auto load feeder.

- Barcode Trace Report

Item	Description
Line Name	Name of the route on which panel was built
Zone Name	Name of the zone on which production occurred when the panel was built
Equipment Name	Name of the placement machine that placed parts on the panel
Lane	Lane number of the placement machine. Placement machine can have one lane or two lanes. In each lane, single product will be manufactured.
Stage No.	Stage number of a modular machine. Modular machines consisting of two tables will have a single stage while modular machines consisting of four tables have two stages. For example in CM602, Table 1 and 2 comprise stage 1 while Table 3 and 4 comprise stage 2. For placement machine consisting of two stages, some components will be placed on board in stage 1 while rest of the components will be placed on the board in stage 2.
Product Program Name	Name of the product whose component placements exist on the specified panel model number / serial number.
Setup Name	Name of the mix to which product belonged.
Model ID	Model number of the panel. The Model Code is used to validate the incoming panel is from the same Product as the Product in current production.
Panel Barcode	Unique barcode assigned to each panel.
Side	Indicates side (Top or Bottom) of the Panel where a part was placed
Serial No	Serial number of the panel. Serial Number is a unique identifier for each panel used for traceability. All traceability data is linked to the Panel Serial Number.

Item	Description
Pattern Serial No	Serial Number of the each pattern existing on the pattern. A pattern is defined as single printed circuit existing on the panel. There is one to one correspondence between product name and pattern.
Time Off	Date and Time when a part reel was unloaded from the placement machine.
Z/PU #	Feeder location at which a component reel was present.
Slot No	Slot number at which a part is located。
SubSlot	In case of double feeder, thin-type single feeder or auto loading feeder (spliceless feeder), subslot corresponds to the side that components are supplied (L or R). L means left and R means right.
Part No.	Part number of a component. Each component placed on the PCB is uniquely identified by the part number.
Expected Part	Part supposed to be present at a particular location based upon the program downloaded to the machine.
Reference	Reference designator of the component for which trace information was generated
Lot No	Lot number linked to a part. Lot number is an entity that is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing.
Vendor	Unique identifier for the part manufacturer.
UserData1	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Part Class	It refers to part shape that has a corresponding vision program used by the machine to recognize component shape. Some of the part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	It is an identification alias created by the user and barcoded on each reel, which maps to a part number.
Material Barcode	Unique barcode of the reel used for placing parts on the panel in order to produce an individual printed circuit board corresponding to a product.
UserData2 to 5	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Pattern No	Number of each pattern.
Mount Operator	Name of the operator who mounted the part reel on the placement machine.
Lot Identification Letter	Letters to identify the lot number.
Identification Result	It indicates identification result of the lot number.
Pattern Barcode	Unique barcode assigning to each pattern.
Pattern Identifier	Identifier for each pattern existing on the pattern.

- Component to Panel Traceability Report

Item	Description
Line Name	Name of the route on which panel was built
Zone Name	Name of the zone on which production occurred when the panel was built
Equipment Name	Name of the placement machine that placed parts on the panel

Lane	Lane number of the placement machine. Placement machine can have one lane or two lanes. In each lane, single product will be manufactured.
Stage No.	Stage number of a modular machine. Modular machines consisting of two tables will have a single stage while modular machines consisting of four tables have two stages. For example in CM602, Table 1 and 2 comprise stage 1 while Table 3 and 4 comprise stage 2. For placement machine consisting of two stages, some components will be placed on board in stage 1 while rest of the components will be placed on the board in stage 2.
Product Name	Name of the product that was manufactured using a particular part number
Setup Name	Name of the mix to which product (that used a particular number) belonged
Machine File Name	Name of a program file associated with a particular placement machine. Placement machine uses this file for placing components on the PCB.
Model ID	Model number of the panel. The Model Code is used to validate the incoming panel is from the same Product as the Product in current production.
Panel Barcode	Unique barcode assigned to each panel
Side	Indicates side (Top or Bot
Serial No	Serial number of the panel. Serial Number is a unique identifier for each panel used for traceability. All traceability data is linked to the Panel Serial Number.
Pattern Serial No	Serial Number of the each pattern existing on the pattern. A pattern is defined as single printed circuit existing on the panel.
Z/PU #	Feeder location at which a part used to build a product was present.
Slot No	Slot number at which a part is located。
SubSlot	In case of double feeder, thin-type single feeder or auto loading feeder (spliceless feeder), subslot corresponds to the side that components are supplied (L or R). L means left and R means right.
Material Barcode	Unique barcode of the reel used for manufacturing a product
Part No	Part number of the component. Each component placed on the PCB is identified by a unique part number.
Expected Part	Part supposed to be present at a particular location based upon the program downloaded to the machine
Lot No	Lot number linked to a part. Lot number is an entity that is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing.
Vendor	Unique identifier for the part manufacturer.
Feeder barcode	Identifier that uniquely identifies a feeder
Quantity	Number of parts existing in the reel at the time it was mounted to the machine for production
Time On	Date and Time when a part reel was mounted to the placement machine for production
Time Off	Date and Time when a part reel was unloaded from the placement machine
Mount Operator	Name of the operator who mounted the part reel on the placement machine
UserData1	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor,

	Part number and quantity.
Work Order Name	Name of the work order corresponding to which a product, used a particular part number, was manufactured
Part Class	It refers to part shape that has a corresponding vision program used by the machine to recognize component shape. Some of the part class values are 1 (Leadless chip), 12 (BGA), 5 (PLCC), 4 (SOP/QFP).
MCID	An MCID is an identification alias, barcoded on each reel, which maps to a part number.
UserData2 to 5	It contains the customized part data when that data could not be accommodated in the fields commonly used like Lot No., Vendor, Part number and quantity.
Tape Guide Serial	Tape guide serial No for the auto load feeder.

- Panel Trace Report

Item	Description
Line Name	Name of the route on which panel was built
Zone Name	Name of the zone on which production occurred when the panel was built
Equipment Name	Name of the placement machine that placed parts on the panel
Lane	Lane number of the placement machine. Placement machine can have one lane or two lanes. In each lane, single product will be manufactured.
Stage No.	Stage number of a modular machine. Modular machines consisting of two tables will have a single stage while modular machines consisting of four tables have two stages. For example in CM602, Table 1 and 2 comprise stage 1 while Table 3 and 4 comprise stage 2. For placement machine consisting of two stages, some components will be placed on board in stage 1 while rest of the components will be placed on the board in stage 2.
Panel Barcode	Unique barcode assigned to each panel
Serial No	Serial number of the panel. Serial Number is a unique identifier for each panel used for traceability. All traceability data is linked to the Panel Serial Number.
Pattern Serial No	Serial Number of the each pattern existing on the pattern. A pattern is defined as single printed circuit existing on the panel.
Model ID	Model number of the panel. The Model Code is used to validate the incoming panel is from the same Product as the Product in current production.
Entry Time	Date and Time when the panel entered the machine.
Exit Time	Date and Time when the panel exited the machine.
Product Name	Name of the product whose component placements exist on the panel. These component placements define a single printed circuit.
Machine File Name	Name of a program file associated with a particular placement machine. Placement machine uses this file for placing components on the PCB.
Work Order Name	Name of the work order corresponding to which a product, which used a particular part number, was manufactured.
Humidity	(Normally not used)
Temperature	(Normally not used)
Audit result	Audit result of the process audit machine.

- Panel Hold Trace Report

Item	Description
Equipment Name	Name of the placement machine for which production of panel was put on hold
Start Time	Date and Time when panel production was put on hold
End Time	Date and Time when panel production again resumed
Duration (DD:HH:MM:SS)	Total duration in Days: Hours: Minutes: Seconds for which panel production remained on hold
Reason	Reason for putting panel production on hold
Start Operator	Name of the operator in the shift when production of panels was put on hold
End Operator	Name of the operator in the shift when production of panels was resumed
Comment	Additional information pertaining to panel hold

- Inspection /Repair Trace Report

Item	Description
Serial No	Serial number of the panel. Serial Number is a unique identifier for each panel used for traceability. All traceability data is linked to the Panel Serial Number.
Pattern Serial No	Serial Number of the each pattern existing on the pattern. A pattern is defined as single printed circuit existing on the panel.
Program	Program contains the details of the composition of the product to manufacture
Result	Indicates if Trace was successful. This column contains OK/NG.
Type	Traceability type. (Audit, Repair, etc.)
Time	Date and Time when panel was released into machine after verifying that model code matches that of product currently in production
Machine	Name of the machine that was repaired
Operator	Name / barcode of the operator who performed the presetup
Reference	Reference designator of the component for which trace information was generated
Reason Code	Reason the machine was repaired
Part No	Part number of the component supplied by the feeder
Lot No.	Lot No. which component scanned during presetup. Lot No. is used to keep track of parts that are placed at different points in time in order to know when a particular group of parts was used in the manufacturing.

- Placement Load Measurement Report

Item	Description
Line Name	Name of the route on which panel was built
Equipment Name	Name of the placement machine that placed parts on the panel
Lane	Lane number of the placement machine. Placement machine can have one lane or two lanes. In each lane, single product will be manufactured.
Stage No.	Stage number of a modular machine. Modular machines consisting of two tables will have a single stage while modular machines consisting of four tables have two stages. For example in CM602, Table 1 and 2 comprise stage 1 while Table 3 and 4

Item	Description
	comprise stage 2. For placement machine consisting of two stages, some components will be placed on board in stage 1 while rest of the components will be placed on the board in stage 2.
Panel Barcode	Unique barcode assigned to each panel
Model ID	Model number of the panel. The Model Code is used to validate the incoming panel is from the same Product as the Product in current production.
Serial No	Serial number of the panel. Serial Number is a unique identifier for each panel used for traceability. All traceability data is linked to the Panel Serial Number.
Table No	Table number counted from the head of the line
Head	Position of a head (1: Front head, 2: Rear head).
Head Type	Head type 1: 12-nozzle 2: 8-nozzle 3: 2-nozzle 4: Load 8-nozzle 9: 16-nozzle 10: 3-nozzle 11: 14-nozzle 12: Light weight 16-nozzle 0: None
Nozzle No	Nozzle format
Nozzle Holder Access	Address of a nozzle holder. Ones place: Row Tens place: Column Hundreds place: Head number
Nozzle Changer Address	Address of a nozzle change. Ones place: Row Tens place: Column Hundreds place: Head number
Placement Load%	The ratio of measured value to the reference load (%).
Judgment Threshold Value (Warning)	The threshold value, judged as warning, based on the change ratio of the measured value to the reference load.
Judgment Threshold Value (NG)	The threshold value, judged as NG, based on the change ratio of the measured value to the reference load.
Measurement Time	Date and Time of measurement

■ Feeder Maintenance

● Tagged Feeder Repair Report

Item	Description
Feeder barcode	Identifier that uniquely identifies a feeder
Last Maintenance	Time when the maintenance was last performed on the feeder
Tag Operator	Name of the operator who tagged the feeder for maintenance.
Tag Time	Time when the feeder was tagged for maintenance
Tag Reason	Reason for tagging the feeder for maintenance
Location	Work area where the feeder was being used at the time of tagging it for maintenance
Slot No	Slot number when the feeder was tagged for maintenance.
Z/PU #	Feeder position when it was tagged for maintenance.

● Feeder Repair Count Report

Item	Description
Feeder barcode	Identifier that uniquely identifies a feeder
SubSlot	In case of double feeder, thin-type single feeder or auto loading feeder (spliceless feeder), subslot (left or right) where the feeder is placed.
Components Fed	Total components supplied by the feeder in the current production run until it was taken off from production for repairs
Total components supplied	Total components supplied by the feeder between the time when the feeder was last repaired and current time when it is again sent for repairs
Placement Count	Total number of times that components were placed in the current production run until it was taken off from production for repairs
Total Placement Count	Total number of times that the machine was placed using the feeder between the time when the feeder was last repaired and current time when it is again sent for repairs
Pick Miss	Total number of times that pickup was missed in the feeder in the current production run until it was taken off from production for repairs
Total Pickup Miss Count	Total number of times that pickup was missed in the feeder between the time when the feeder was last repaired and current time when it is again sent for repairs
Pickup Error Count	Total number of pickup errors occurred in the feeder in the current production run until it was taken off from production for repairs
Total Pickup Error Count	Total number of pickup errors occurred in the feeder between the time when the feeder was last repaired and current time when it is again sent for repairs.
Shape Error Count	Total number of shape errors occurred in the feeder in the current production run until it was taken off from production for repairs
Total Shape Error Count	Total number of shape errors occurred in the feeder between the time when the feeder was last repaired and current time when it is again sent for repairs
Recognition Error Count	Total number of recognition errors occurred in the feeder in the current production run until it was taken off from production for repairs.
Total Recognition Error Count	Total number of recognition errors occurred in the feeder between the time when the feeder was last repaired and current time when it is again sent for repairs
Repair No	The number of times that a feeder has been repaired.

Last Maintenance	Date and Time that last maintenance of the feeder was done.

- Feeder Repair History Report

Item	Description
Feeder barcode	Identifier that uniquely identifies a feeder.
Check In Time	Time when a particular feeder was checked in for maintenance.
Maintenance Time	Time for which a particular feeder was under maintenance
Duration (DD:HH:MM:SS)	Total time in days, hours, minutes and seconds for which a feeder was not in production due to maintenance. It is calculated as the sum of Check In Time and Maintenance Time.
Zone Name	PanaCIM zone where the fee
Subslot Name	In case of double feeder, thin-type single feeder or auto loading feeder (spliceless feeder), subslot (left or right) where the feeder is placed.
Part Supply	Total number of parts supplied before maintenance was done.
Placement Count	The number of times parts are placed before maintenance was done.
Pickup Miss	Total number of times that pickup was missed before maintenance was done
Pickup Error	Total number of pickup errors occurred before maintenance was done.
Shape Error	Total number of shape errors occurred before maintenance was done.
Recognition Error	The number of recognition errors occurred before maintenance was done.
Operator Name	Name of the operator who checked in the feeder for repairs as part of overall maintenance of that feeder
Repair Type	Nature of repairs performed on a particular feeder. For example, there could be some technical fault which required urgent repairs or it could be just a routine maintenance after feeder has reached the threshold of supplying the required number of components
Repair Performed	Brief description of repairs performed on the feeder. For example, if there was a mechanical fault, it would be mentioned as to what was done to resolve this issue.
Tag Time	Time when the feeder was tagged indicating that it is not be used in production
Tag Reason	Exact cause for tagging the feeder. For example, feeder was tagged because there was an upcoming maintenance or a mechanical fault which needs urgent repairs.
Tag Operator	Operator who tagged the feeder for maintenance
User Comment	-

- Feeder Report

Item	Description
Location	Current location of a feeder.
Feeder barcode	Identifier that uniquely identifies a feeder
Serial No.	Another identifier for the feeder in the form of a number that is unique to a particular feeder. Unlike Feeder Barcode, serial number cannot be scanned.
Supply Method	Indicates how the components are fed to the machine. It could be in the form of reels, trays, bulk, stick etc.
Supply Type	Indicates the source from where the program file originated. For

Item	Description
	example, for product transfer from PT200, supply type is NAVI. If program file originated from PanaPRO, supply type contains the name of the feeder (for example, FA0010).
Machine Group	Type of machines where the feeder can be mounted. For example, it could be CM series of modular machines, DT series of modular machines, NPM placement machines or Radial part insertion machines (RL131, RL132).
Table Size	Dimensions of machine table to which feeder is attached
Cart Type	Type of carriage a particular feeder can be mounted on
Tape Width	Width of the tape consisting of electronic components
Main Pitch	Distance between two feeder slots
Sub pitch	Distance between a feeder slot and subslot
Cutter Location	Location of a feeder cutter.
Reel Diameter	Diameter of the tape reel
Handle Color	Color of a feeder handle
Num Sub Slots	Number of feeder subslots. A double feeder, thin-type single feeder or auto loading feeder (spliceless feeder) will have two subslots and hence accommodates two different tape reels while a feeder with one reel will have no subslots.
Part No	Part number of electronic component that is present in the tape reel
Vendor	A unique identifier for the manufacturer of electronic component that is supplied by the feeder
Model ID	Model number specific to a feeder. It provides yet another way to differentiate between different feeders based upon this attribute.
Tag Type	Tag type attached to a feeder. For example, feeder could be tagged due to a routine maintenance or it could be tagged because it has developed a snag and hence needs urgent repairs.
Maintenance Time	Time for which feeder is under maintenance after being tagged
Left Pickup Count After Maintenance	Total number of pickup count of parts on the left feeder after maintenance was done.
Right Pickup Count After Maintenance	Total number of pickup count of parts on the right feeder side after maintenance was done.
Start Date of Use	Date when a feeder was started to use.
Left Lifetime Pickup Count	Total number of pickup count of parts on the left feeder counted from start date.
Right Lifetime Pickup Count	Total number of pickup count of parts on the right feeder counted from start date.

■ Production Analysis

● Component Usage Report/Component Usage Report (incl. costs)

Item	Description
Part No.	Part number of a component. Each component placed on the PCB is uniquely identified by the part number.
Part Cost *1	Part cost
Pickup Count	Total number of parts that were picked up for placement
Pickup Cost *1	The value that the pickup count is converted into the cost.
Placement Count	Total number of parts that were placed successfully on boards
Placement Cost *1	The value that the placement count is converted into the cost.
Pickup Error Count	Total number of parts that were not picked up for placement
Pickup Miss (Count)	Total number of times pickup was cancelled/rejected by the line scan sensor for over thickness measurement failures. For example, part is standing on end or part is thicker than the thickness tolerance. Part is standing on its side and exceeds thickness tolerance.
Recognition Error Count	Total number of missed placements caused by a recognition error (the part was not recognized)
Shape Error Count	Total number of component shape errors other than the ones for Reference Number 50, 57 (Tantalum capacitor) and 58 (Tantalum capacitor) depending upon the Reference number one of the recognition algorithms (like Four-point block matching, One-point block matching and Outline recognition) are invoked by the machine . Reference Numbers 1-46 uses Four-point block matching algorithm, reference numbers 47-50 uses one-point block matching algorithm while reference numbers 51-250 uses Outline recognition reference number.
Height Miss (Count)	Total number of missed placements caused by virtue of invalid component height
Total Scraps	Total number of parts not placed on the PCB due to shape errors, pick being missed etc. and hence considered scrap during a production cycle.
Scrap Cost *1	The value that the number of scraps are converted into the cost
Scrap Percentage %	Percentage of parts considered scrap during a production cycle.
Placement Percentage %	Percentage of parts that that were placed successfully out of the total placement attempts that are made

*1 Display only Component Usage Report (incl. costs)

● PCB Production Report

Item	Description
BOM Product	It refers to the name of the base product.
Program Name	It refers to the product name existing within Product Setup module of PanaCIM.
Side	It refers to the side of the PCB (top or bottom).
Line Name	It refers to the name of the route on which PCB was manufactured.
Equipment Name	Name of the placement machine used for manufacturing the PCB
Lane	Lane number of the placement machine. Placement machine can have one lane or two lanes. In each lane, single product will be manufactured.
Stage No.	Stage number of a modular machine. Modular machines consisting of two tables will have a single stage while modular machines consisting of four tables have two stages. For example

Item	Description
	in CM602, Table 1 and 2 comprise stage 1 while Table 3 and 4 comprise stage 2. For placement machine consisting of two stages, some components will be placed on board in stage 1 while rest of the components will be placed on the board in stage 2.
Panel Quantity	Number of panels produced during production run
Pattern Quantity	Number of panels produced during production run
Pattern Quantity (Calculated Value)	Number of patterns calculated based upon the actual number of panels produced during production run. Defective panels are excluded for the purpose of calculating this value.

- Work Order Usage Report

Item	Description
Work Order Name	Name of a work order
Work Order Quantity	Number of panels produced as part of production cycle corresponding to a work order
Equipment	Name of the Placement machine used in the production.
Part No	Part number of a component. Each component placed on the PCB is uniquely identified by a part number.
Quantity (Per Panel)	Number of parts of a particular kind required to produce a single panel
Quantity Required	Number of parts of a particular kind required to produce all the panels
Quantity Picked	Number of parts of a particular kind that were picked for being placed on a PCB
Quantity Placed	Number of parts of a particular kind that were actually placed on a PCB. It is to be noted that not all the parts picked by a nozzle are actually placed on the board. For example, anomaly might be detected after a part is picked and recognized by a line camera due to which this part is discarded.
Difference (Pick–Place)	Difference of numbers of parts picked and number of parts placed corresponding to a part number

- Event Trace Report/Event Trace Summary Report

Item	Description
Equipment Name *2	Name of the placement machine on which the event occurred
Lane	Lane number of the placement machine. Placement machine can have one lane or two lanes. In each lane, single product will be manufactured.
Stage No *2	Stage number of a modular machine. Modular machines consisting of two tables will have a single stage while modular machines consisting of four tables have two stages. For example in CM602, Table 1 and 2 comprise stage 1 while Table 3 and 4 comprise stage 2. For placement machine consisting of two stages, some components will be placed on board in stage 1 while rest of the components will be placed on the board in stage 2.
Event Time	Date and Time at which event occurred.
Event *2	Name of the event that occurred. For example, Part exhaust, short supply, etc.
Z/PU #	Machine feeder location at which event occurred
Slot No	Slot number at which event occurred
SubSlot	In case of double feeder, thin-type single feeder or auto loading feeder (spliceless feeder), it refers to subslot (Left or Right) at which event occurred.

Item	Description
Error Code	Unique code associated with each error. An error is defined as event that causes production bottleneck for example, short supply, part exhaust etc.
Count *2	The number of events occurred.

*2 Display items of Event Trace Summary Report

- Down Time Report/Down Time Summary Report

Item	Description
Equipment Name *3	Name of the placement machine for which downtime occurred
Start Time	Date and Time at which downtime started
End Time	Date and Time at which downtime ended
Duration (DD:HH:MM:SS)	Duration of downtime in days, hours, minutes and seconds.
Reason *3	Specifies the exact cause of downtime.
Operator Name	Name of the operator who stopped the machine.
Duration *3	Total duration of downtime in days, hours, minutes and seconds.

*3 Display items of Down Time Summary Report

- State Trace Report/State Trace Summary Report

Item	Description
Equipment *4	Name of the placement machine for which state change is recorded
Status *4	State of a placement machine at a particular moment in time
Start Time	Date and Time at which machine entered a particular state
End Time	Date and Time at which machine ceased to exist in a particular state
Duration (DD:HH:MM:SS) *4	Total duration of state in days, hours, minutes and seconds

*4 Display items of State Trace Summary Report

- Part Supply Patrol Report

Item	Description
Line Name	Route Name.
Equipment Name	Placement machine name.
MC	Machine number.
Table number.	Table number.
Slot number	Slot number
SubSlot	In case of double feeder, thin-type single feeder or auto loading feeder (spliceless feeder), it represents subslot (left or right).
Z/PU #	Feeder location on the machine.
Part No	Name of a part
Part Status	Part status when a part is supplied (Emergency warning/Warning/Part Shortage)
Result Time (second)	Time spent for supply
Timestamp	Time when a part is supplied

■ Production Monitoring and Dispatch

● PDM Event History Report

Item	Description
Line Name	Name of the route for which event was generated
Machine	Name of the machine corresponding to which event was generated
Location	Feeder location corresponding to which machine event was generated. For example, if Part exhaust event occurred, feeder location at which part exhaust occurred would be populated in this column.
Event Description	Description of machine event if configured in the PMD
Message	Message that is displayed corresponding to a machine event. For example, if part exhaust occurred, “PART EXHAUST” would be populated.
Product Name	Name of the product during whose production run machine event was generated
Operator Name	Name of the operator who acknowledged the PMD event
Occurrence Time	Date and Time when the event first occurred
Acknowledgement Time	Date and Time when the machine event was acknowledged by the operator
Time to Acknowledgement	Time taken for acknowledge of the event
Start Time	Date and Time when the event happened. For example, if part exhaust occurs at a location, this column contains the time when part exhaust occurred.
Time to Start	Time taken for starting of the event.
Finish Time	Date and Time when the event ended after being responded to. For example, in case of part exhaust, it refers to the Date and Time when exhausted part was replenished.
Time to Finish	Time taken for event process. For example, when parts are exhausted, it indicates between the time when part shortage is detected and the time when the parts are refilled.
Total Duration	Total time when an event is effective. For example, when parts are exhausted, it indicates the total time when parts in one location are exhausted.
Manual Closing	Specifies whether the event was manually closed i.e required operator intervention
Force Close Reason	Reason as to why the event had to be closed manually

● PDM Event Summary Report

Item	Description
Event Description	Indicates the event that was generated. For example, if tape splice warning occurred, event description would contain “Tape Splice”
Count	Number of times a particular event occurred on the route
Total Time to Acknowledge	Total time taken to acknowledge all the events of a particular type. For example, if 10 events of particular type were generated, this would be total time taken to acknowledge all the events.
Average Time to Acknowledge	Average time taken to acknowledge a single event after multiple events of same type were generated
Total Time to Finish	Total time taken to respond to all the events of a particular type
Average Time to Finish	Average time taken to respond to an event when more than one same type events were generated.

- Work in Progress Report

Item	Description
Line Name	Name of the route for which production is currently underway
Work Order Name	Name of the work order for which production is currently underway
Product Name	Name of the product for which production is currently underway
Cycle Time	Actual time required to produce a single PCB based upon current production run
Quantity	Quantity of PCBs planned to be produced as part of production run
Scheduled Start Time	Scheduled start time of production
Actual Start Time	Actual start time of production
Expected Finish Time	Time when the production is estimated to finish
Comment	Comments added related to the work order.

- Upcoming Work Orders Report

Item	Description
Line Name	Name of the route for which production is scheduled to begin
Product Name	Name of the product for which production is scheduled to begin
Work Order Name	Name of the work order for which production is scheduled to begin
Cycle Time	Estimated time required to produce a single PCB
Quantity	Quantity of PCBs planned to be produced as part of production run
Scheduled Start Time	Date and Time when production was actually supposed to begin
Actual Start Time	Date and Time when the production is estimated to start after being delayed (for example, production of previous work order has not completed by scheduled end time)
Expected Finish Time	Time when the production is estimated to finish
Comment	Comments added related to the work order.

- Line Down Time Report

Item	Description
Line Name	Name of the route for which production is scheduled to begin in near future
Start Time	Time when the downtime started
End Time	Time when the downtime ended. User can change the end time to adjust for down times amount in each Work Order Time range
Duration (DD:HH:MM:SS)	Time period in Days: Hours: Minutes: Seconds for which downtime was in effect
Reason	Reason for initiating the downtime
Operator Name	Name of the operator who initiated the downtime

■ Enterprise Link

● Import History Report

Item	Description
Operation Type	Type of import operation.
Timestamp	Date and Time when import operation was done.
Data Source	Source of the data import
Status	Completion status of Enterprise Link Import
Status Message	Completion status message of Enterprise Link import
Duplicate Record	Number of duplicate records generated during Enterprise Link import operation
Processed Records	Number of records that were processed during Enterprise Link import operation
Error Records	Number of records that contained errors
Warning Records	Number of records that contained warnings

4. FUNCTION - [Production Analysis] - [Analytics]

This chapter explains function - Production Analysis - [Analytics].

4.1 Overview

To find [Production Analysis] – [Analytics], access PanaCIM EE Web application and select **Production Analysis** from the menu. Then select **Analytics** from the sub-menu.

The [Analytics] report of NewPA [Production Analysis] shows the data of production status down to the route level to help you analyze your production status. Further information can also be viewed by clicking the data concerned.

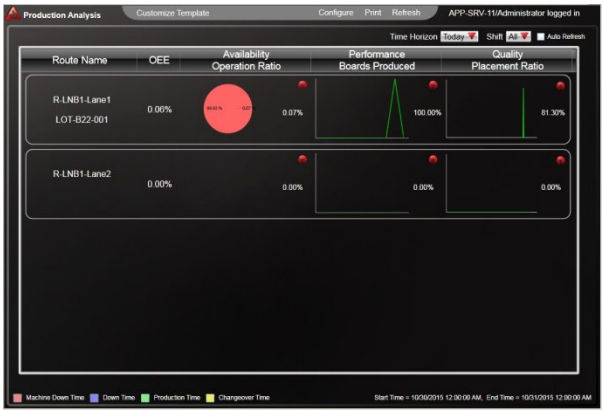




4.1.1 Production Analysis - [Analytics]

The table below summarizes the templates available for this function.

No	Template	Function
1	Administrator Panel	Shows the OEE (overall equipment effectiveness) and availability line by line.
2	Report Panel (for administrator)	Shows the placement status of each machine in the selected line.
3	Line Performance Report	Shows the performance of the selected line.
4	Production Summary	Shows the availability, number of panels produced and scrap rate line by line.
5	Production Summary - Level 1	Shows the number of panels produced and scrap rate of the selected line.
6	Scrap Data Report	Shows the tact time and placement rate of the selected line.
7	Factory Production	Shows the progress of the production, running rate and scrap rate line by line.*1



Administrator Panel



Report Panel (for administrator)



Line Performance Report



Production Summary



Scrap Data Report



Factory Production

4.2 Settings

4.2.1 Display Setting

From **Configuration** on the **Production Analysis** - **Analytics** screens, each template display settings can be made.

■ Global setting

Set the screen refresh interval, the threshold values of setting items and the display color of the threshold values.
Data less than the threshold value of each display item will not be shown.



Items include:

- Refresh Interval (min)
- Efficiency
- Availability
- Performance
- Quality
- Scrap Rate

<Display color>

- Running Rate (%)
- Scrap Rate

■ Display period setting

Target period can be set from the **Time Horizon** drop-down list on each **Production Analysis** - **Analytics** screens.
In addition, checking the **Auto Refresh** checkbox automatically refreshes the screen using the interval specified on the Configuration screen.



■ Administrator Panel

Set the line and items to be displayed on the Administrator Panel template.

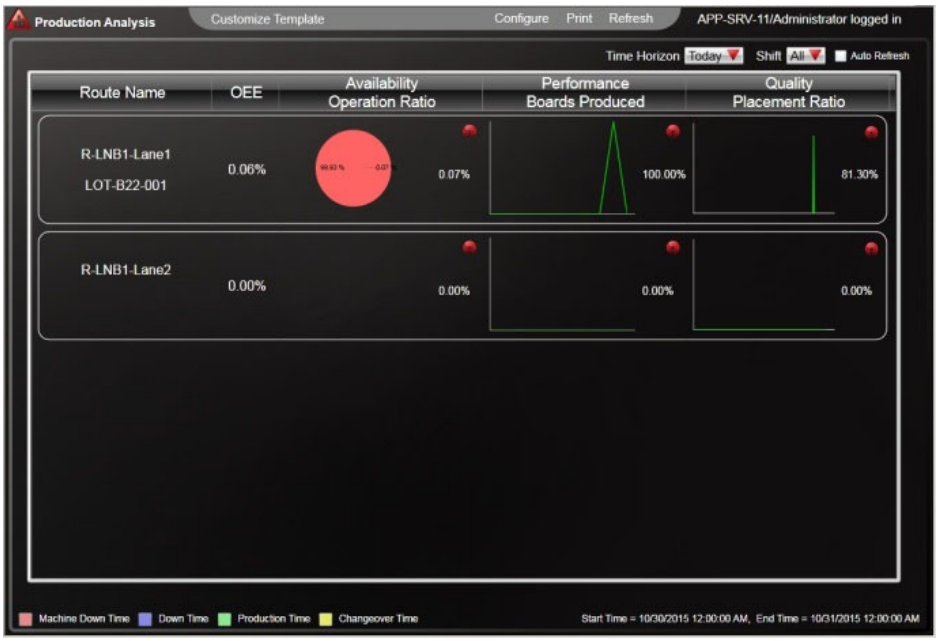


Items include:

- Route Name
- OEE
- Availability
  - Operation rate
- Performance
  - Boards Produced
  - Blocks Produced
  - Placement Count/CPH
  - Actual Boards vs Theoretical Boards
  - Actual Blocks vs Theoretical Blocks
- Quality
  - Placement Ratio

4.2.2 Display Items

■ Administrator Panel



Display item	Configuration element	
O.E.E. *1	O.E.E. Performance	Availability, Quality
Operation Ratio *1	Production Time [Ratio] Machine Down Time [Ratio]	Setup Time [Ratio]
Boards Produced *1*2	Boards Produced [Q'ty]	
Blocks Produced *1*2	Blocks Produced [Q'ty]	
Placement Count/CPH *1*2	Placement count/CPH	
Actual Boards vs Theoretical Boards 1*2	Boards Produced [Q'ty] Efficiency Boards	The number of theoretical boards
Actual Blocks vs Theoretical Blocks *1*2	Blocks Produced [Q'ty] Efficiency Blocks [Q'ty]	Theoretical Blocks
Placement Ratio *1	Placement Ratio	

*1 Select what to display on the administrator panel setting.
*2 Either one can be displayed.

The table below shows the list of defined elements, their descriptions and expressions.

Elements	Description	Expression
O.E.E.	Overall equipment effectiveness	Availability%*Performance% *Quality% (If there is no entry in Availability%, Performance% and Quality% or 0 is entered, then calculate using 100%.)
Availability[%]	The ratio of the total produced boards to the number of theoretical boards	SUM(Produced Boards)*Theoretical Cycle Time/Time Horizon*100

Elements	Description	Expression
Performance[%]	The ratio of the total produced boards to the total number of planned boards	SUM(Produced Boards)/SUM(Planned boards)*100
Quality[%]	=Placement ratio The ratio of the total placement count to the total pickup count	SUM(Place Count) /SUM(Pickup Count)*100
Production Time [Ratio]	The ratio of the total production times to the time Horizon (total planned down time, down time, setup time and production time)	SUM(Production Time)/ { SUM(Planned Down Time)+SUM(Down Time)+SUM(Setup Time)+SUM(Production Time) } *100
Setup Time [Ratio]	The ratio of the total setup time to the time horizon (total planned down time, down time, setup time, and total production time)	SUM(Setup Time)/ { SUM(Planned Down Time)+SUM(Down Time)+SUM(Setup Time)+SUM(Production Time) } *100
Machine Down Time [Ratio]	The ratio of the total machine down time to the time horizon (total planned down time, down time, setup time, and total production time)	SUM(Machine Down Time)/ { SUM(Planned Down Time)+SUM(Down Time)+SUM(Setup Time)+SUM(Production Time) } *100
Boards Produced [Q'ty]	Total number of boards produced	SUM(Boards Produced)
Blocks Produced [Q'ty]	Total number of blocks produced	SUM(Blocks Produced)
Placement Count/(CPH)	Total time that the machine placed per hour	SUM(Place count)/SUM(Real Run $Time_{[sec]}$)*3600
Theoretical Boards	The number of boards that can theoretically be produced during a time horizon	Time Horizon/Theoretical Cycle Time
Efficiency Boards	The number of boards considering efficiency to theoretical boards	Time Horizon/Theoretical Cycle Time*Efficiency/100
Theoretical Blocks	The number of blocks that can theoretically be produced during a time horizon	Time Horizon/Theoretical Cycle Time*AVE(Pattern Quantity/Board Quantity)
Efficiency Blocks [Q'ty]	The number of blocks considering efficiency to theoretical boards	Time Horizon/Theoretical Cycle Time*AVE(Pattern Quantity/Board Quantity)*Efficiency/100
Placement Ratio	The ratio of the total placement count to the total pickup count	SUM(Place count)/SUM(Pickup Count)*100

- Level I Supervisor View



Display item	Elements	
View	OR Average CPH	Average Cycle Time
List	Bottleneck Machine Worst Tact Time Best Tact Time Average Tact Time	Worst M/C Cycle Time Best M/C Cycle Time Average M/C Cycle Time
Boards/Blocks Produced *1	Boards Produced [Q'ty]	Blocks Produced [Q'ty]
Placement/ Pickup Ratio *1	Placement/ Pickup Ratio	
Components Placed	Components Placed	
Actual Boards vs Theoretical Boards *1	Boards Produced [Q'ty] Efficiency Boards	The number of theoretical boards
Actual Blocks vs Theoretical Blocks *1	Blocks Produced [Q'ty] Efficiency Blocks [Q'ty]	Theoretical Blocks
Setup Time	Changeover (Actual Time)	
Production Time *2	Real Run Time Next Process Wait [Time]	Prev Procs Wait [Time]
Total Stop Time *2	Previous Process Wait Time Single and Cycle Stop Time Board Transport Error Stop Time Restart Impossible Stop Time Chip Recognition Error Stop Time Transfer Unit Parts Drop Stop Time Bond Recognition Error Time	Next Proc Wait Time Single and Cycle Stop Error Time PCB Recog Error Time Chip Pickup Error Stop Time Chip Placement Error Stop Time Excess Judgement Stop Time (Inspection Head) Material Exhaust Error Stop Time
Power On Time *2	Running Time Maintenance Time Unit Adjust Time	Changeover Time Recipe Edit Time Other Time

Display item	Elements	
Scrap	By Production [Scrap Count] *3 By Chip [Scrap Count] *3	Scrap Count By Board *3 Scrap By Chip% *3

*1 Show any one
*2 Show any one
*3 Show any one

The table below shows the list of defined elements, their descriptions and expressions.

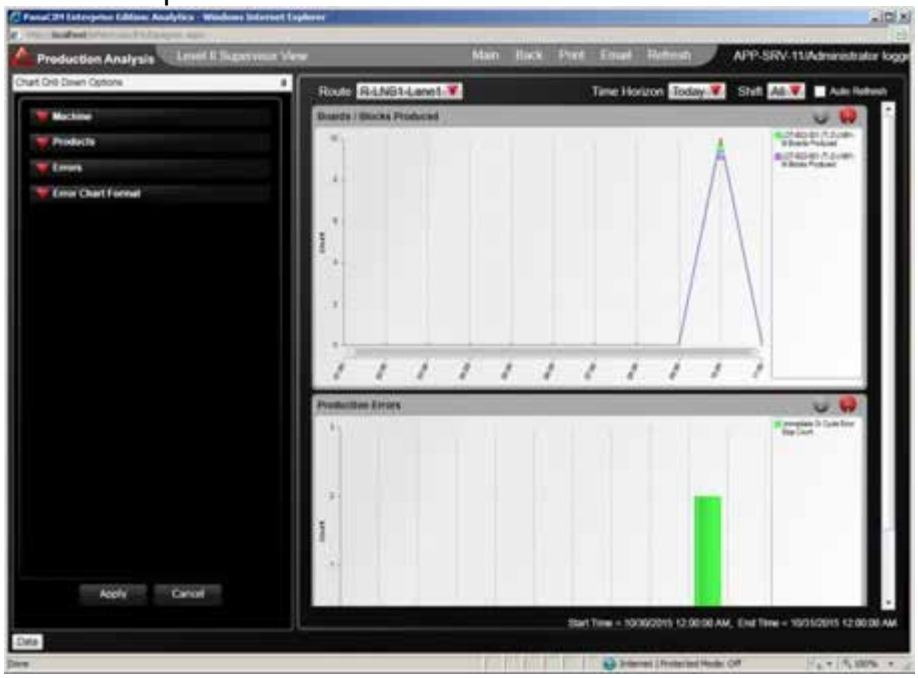
Elements	Description	Expression
OR	The ratio of the total production time to the time horizon (total planned down time, down time, setup time and production time)	SUM(Production Time)/ { SUM(Planned Down Time)+SUM(Down Time)+SUM(Setup Time)+SUM(Production Time) } *100
Average Cycle Time	The average time taken to place a single panel	SUM(Real Run Time)/SUM(Produced Count)
Average CPH	The average placement count per hour	SUM(Place count)/SUM(Real Run Time [sec])*3600
Bottleneck Machine	A factor of the machine in determining the production speed	
Worst M/C Cycle Time	The maximum time taken to place a single board	MAX (Cycle Time)
Worst Tact Time	The maximum time taken to produce a single board	MAX(Tact Time)
Best M/C Cycle Time	The minimum time taken to place a single board	MIN (Cycle Time)
Best Tact Time	The minimum time taken to produce a single board	MIN(Tact Time)
Average M/C Cycle Time	The average time taken to place a single board	AVE (Cycle Time)
Average Tact Time	The average time taken to produce a single board	AVE(Tact Time)
Boards Produced [Q'ty]	Total number of boards produced	SUM(Produced Count)
Blocks Produced [Q'ty]	Total number of blocks produced	SUM(Blocks Produced)
Placement/Pickup Ratio	=Placement Ratio The ratio of the total placement count to the total pickup count	SUM(Place Count)/SUM(Pickup Count)*100
Components Placed	=Placement Count Total number of times the machine placed components	SUM(Place Count)

Elements	Description	Expression
Theoretical Boards	The number of boards that can theoretically be produced during a time horizon	Time Horizon/Theoretical Cycle Time
Efficiency Boards	The number of boards considering efficiency to theoretical boards	Time Horizon/Theoretical Cycle Time*Efficiency/100
Theoretical Blocks	The number of blocks that can theoretically be produced during a time horizon	Time Horizon/Theoretical Cycle Time*AVE(Pattern Quantity/Board Quantity)
Efficiency Blocks [Q'ty]	The number of blocks considering efficiency to theoretical boards	Time Horizon/Theoretical Cycle Time*AVE(Pattern Quantity/Board Quantity)*Efficiency/100
Changeover (Actual Time)	Total time taken for setup	SUM(Changeover End Time-Changeover Start Time)
Real Run Time	Total board recognition time, placement/audit time and transport time (they are not included in case of alternative placement lane)	SUM(Real Run Time)
Previous Process Wait [Time]	Total time that the machine was waiting for the previous table or machine to finish processing a panel	SUM(Prev Proc Wait Time)
Next Process Wait [Time]	Total time that the machine was waiting for the next table or machine to finish processing a panel	SUM(Next Proc Wait Time)
Single and Cycle Stop Time	Total time that the machine made single or cycle stop caused by operation button press action	SUM(Single Cycle Stop Time)
Single and Cycle Stop Error Time	Total time that the machine made single or cycle stop when the machine automatically detect an error (However, if it is included in other error, it is not included in this item)	SUM(Single Cycle Stop Error Time)
Board Transport Error Stop Time	Total single stop time caused by board transport error (any errors caused on the conveyor)	SUM(Transport Error Time)
Board Recognition Error Stop Time	Total time that the machine made single stop caused by board recognition error (the type of board fiducial is not	SUM(PCB Recog Error Time)

Elements	Description	Expression
	recognized)	
Restart Impossible Stop Time	Total time that the machine stopped while waiting for error correction	SUM(Start Imp Stop Time
Chip Pickup Error Stop Time	Total time that the machine made single stop caused by chip pickup error (a chip is not picked up for placement)	SUM(Chip Pickup Error Time)
Chip Recognition Error Stop Time	Total time that the machine stopped caused by chip recognition error (When an error occurs three times in a row in the same feeder, it results in recognition error)	SUM(Chip Recog Error Time)
Chip Placement Error Stop Time	Total time that the machine made single stop caused by placement error	SUM(Chip Mount Error Time)
Transfer Unit Parts Drop Stop Time	Total time that the machine made single stop caused by transfer unit parts drop error	SUM(Transfer Unit Parts Drop Stop Time)
Excess Judgement Stop Time(Inspection Head)	Total time that the machine stopped caused by excess judgment	SUM(Excess Judgement Stop Time)
Bond Recognition Error Stop Time	Total time that the machine stopped caused by bond recognition error.	SUM(Bond Recog Error Time)
Material Exhaust Error Stop Time	Total time that the machine made cycle stop when materials (bond, etc.) are exhausted	SUM(Material Exhaust Error Stop Time)
Running Time	Total time between the start and end of automatic operation	SUM(Run Time)
Changeover Time	・Setup Time Total time that setup related menus are displayed	SUM(Setup Time)
Maintenance Time	Total time that maintenance related menus are displayed	SUM(Maintenance Time)
Recipe Edit Time	Total time that production data related menus are displayed	SUM(Production Data Edit Time)
Unit Adjust Time	Total time that machine related menu are displayed	SUM(Unit Adjust Time)
Other Time	Total time that the home screen is displayed	SUM(Power on Time)-SUM(Setup Menu Time)-SUM(Prod Management Menu Time) -SUM(Maintenance Time) -SUM(Recipe Edit Time)

Elements	Description	Expression
		-SUM(Unit Adjust Time) -SUM(Run Time)
By Production [Scrap Count]	The total number of chips that are judged as scraps without placing on the board for each production lot	SUM(Pickup Count)-SUM(Place Count)
Scrap Count By Board	The total number of chips that are judged as scraps without placing on the board for each board.	SUM(Pickup Count) -SUM(Place Count)
By Chip [Scrap Count]	The total number of chips that are judged as scraps without placing on the board for each chip	SUM(Pickup Count) -SUM(Place Count)
By Chip [Scrap Count] %	The ratio of the total number of chips that are judged as scraps without placing on the board for each chip, to the total pickup count	{ SUM(Pickup Count)-SUM(Place Count) }/SUM(Pickup Count)*100

- Level II Supervisor View



Display item	Elements	
Boards/Blocks Produced *4	Boards Produced [Q'ty]	Blocks Produced [Q'ty]
Placement/Pickup Ratio *4	Placement/Pickup Ratio	
CPH *4	Components Placed	
Current setup *4	Changeover Time	
Production Time *4	Real Run Time Next Process Wait [Time]	Previous Process Wait [Time]
Total Stop Time *4	Previous Process Wait Time Single and Cycle Stop Time Board Transport Error Stop Time Restart Impossible Stop Time Chip Recognition Error Stop Time Transfer Unit Parts Drop Stop Time Bond Recognition Error Stop Time	Next Process Wait Time Single and Cycle Stop Error Time PCB Recognition Error Time Chip Pickup Error Stop Time Chip Placement Error Stop Time Excess Judgement Stop Time(Inspection Head) Material Exhaust Error Stop Time
Scrap Count by Board *4	By Board [Scrap Count]	
Scrap Count by Chip *4	By Chip [Scrap Count]	
Production Error	Restart Impossible Stop Count Chip Ejection Error [Count] Chip Recognition Error [Count] Chip Pickup Error Count Placement Point Recognition Error Count Bond Elongation Error Count Mask Recognition Error Count Bond Recognition Error Count	Single or Cycle Error Stop Count Chip Placement Error [Count] Chip Erection Error [Count] Board Transport Error Count Board Recognition Error Count Paste Countup Error [Count] Feedback Error Count Other Error Count

*4 Items are displayed based on the selection of items in the Level I Supervisor View

The table below shows the list of defined elements, their descriptions and expressions.

Elements	Description	Expression
Boards Produced [Q'ty]	Total number of boards produced	SUM(Boards Produced)
Blocks Produced [Q'ty]	Total number of blocks produced	SUM(Pattern Quantity)
Placement/Pickup Ratio	=Placement Ratio The ratio of the total placement count to the total pickup count	SUM(Place Count)/SUM(Pickup Count)*100
Components Placed	=Placement Count Total number of times the machine placed chips	SUM(Place Count)
Changeover (Actual Time)	Total time taken for setup	SUM(Changeover End Time-Changeover Start Time)
Real Run Time	Total board recognition time, placement/audit time and transport time (they are not included in case of alternative placement lane)	SUM(Real Run Time)
Previous Process Wait [Time]	Total time that the machine was waiting for the previous table or machine to finish processing a panel	SUM(Prev Proc Wait Time)
Next Process Wait [Time]	Total time that the machine was waiting for the next table or machine to finish processing a panel	SUM(Next Proc Wait Time)
Single and Cycle Stop Time	Total time that the machine made single or cycle stop caused by operation button press action	SUM(Single Cycle Stop Time)
Single and Cycle Stop Error Time	Total time that the machine made single or cycle stop when the machine automatically detect an error (However, if it is included in other error, it is not included in this item)	SUM(Single Cycle Stop Error Time)
Board Transport Error Stop Time	Total single stop time caused by board transport error (any errors caused on the conveyor)	SUM(Transport Error Time)
Board Recognition Error Stop Time	Total time that the machine made single stop caused by board recognition error (the type of board fiducial is not recognized)	SUM(PCB Recog Error Time)

Elements	Description	Expression
Restart Impossible Stop Time	Total time that the machine stopped while waiting for error correction	SUM(Start Imp Stop Time
Chip Pickup Error Stop Time	Total time that the machine made single stop caused by chip pickup error (a chip is not picked up for placement)	SUM(Chip Pickup Error Time)
Chip Recognition Error Stop Time	Total time that the machine stopped caused by chip recognition errors (When an error occurs three times in a row in the same feeder, it results in recognition error)	SUM(Chip Recog Error Time)
Chip Placement Error Stop Time	Total time that the machine made single stop caused by placement error	SUM(Chip Mount Error Time)
Transfer Unit Parts Drop Stop Time	Total time that the machine made single stop caused by transfer unit parts drop error	SUM(Transfer Unit Parts Drop Stop Time)
Excess Judgement Stop Time(Inspection Head)	Total time that the machine stopped caused by excess judgment	SUM(Excess Judgement Stop Time)
Bond Recognition Error Stop Time	Total time that the machine stopped caused by bond recognition error.	SUM(Bond Recog Error Time)
Material Exhaust Error Stop Time	Total time that the machine made cycle stop when materials (bond, etc.) are exhausted	SUM(Material Exhaust Error Stop Time)
Scrap Count By Board	The total number of chips that are judged as scraps without placing on the board for each board.	SUM(Pickup Count) -SUM(Place Count)
By Chip [Scrap Count]	The total number of chips that are judged as scraps without placing on the board for each chip	SUM(Pickup Count) -SUM(Place Count)
Restart Impossible Stop Count	The total number of times that the machine stopped while waiting for error correction	SUM(Start Imp Stop Count)
Single or Cycle Error Stop Count	The total number of times that the machine made single or cycle stop when the machine automatically detect an error (However, if it is included in other error, it is not included in this item)	SUM(Single Cycle Stop Count)
Chip Ejection Error [Count]	Total number of times the machine stopped caused by errors when the machine ejects defect chips	SUM(Chip Eject Error Count)
Chip Placement Error [Count]	Total number of times the machine made single stop	SUM(Chip Place Error Count)

Elements	Description	Expression
	caused by placement error	
Chip Recognition Error [Count]	Total number of times the machine stopped caused by chip recognition error (It results in recognition error when an error occurs three times in a row in the same feeder)	SUM(Chip Recog Error Count)
Chip Erection Error [Count]	Total number of times the machine stopped caused by chip erection error	SUM(Chip Erection Error Count)
Chip Pickup Error Count	Total number of times the machine stopped caused by chip pickup error (the chip was not picked up for placement)	SUM(Chip Pickup Error Count)
Board Transport Error Count	Total number of times the machine made single stop caused by board transport error (any error caused on the conveyor)	SUM(Transport Error Count)
Placement Point Recognition Error Count	Total number of times that the machine stopped caused by placement point recognition error (the type of local placement point/pattern fiducial is not recognized)	SUM(Place Pt Recog Error Count
Board Recognition Error Count	Total number of times the machine made single stop caused by board recognition error (the type of board fiducials were not recognized)	SUM(Board Recog Error Count)
Board Elongation Error Count	Total number of times the machine stopped caused by board elongation error (the type of board fiducials were not recognized)	SUM(Board Elong Error)
Paste Countup Error [Count]	Total number of times the machine stopped caused by paste countup error	SUM(Paste Countup Error Count)
Mask Recognition Error Count	Total number of times the machine made single stop caused by mask/solder recognition error (stencil recognition error)	SUM(Mask Recog Error Count)
Feedback Error Count	Total number of times the machine made cycle stop caused by feedback stop information	SUM(Feedback Error Count)
Bond Recognition Error	Total number of times the	SUM(Bond Recog Error

Elements	Description	Expression
	machine stopped caused by bond recognition error	Count)
Other Error Count	Total number of times the machine stopped caused by other errors	SUM(Other Error Count)

- Level III Supervisor View

Display item	Elements	
Placement/Pickup Ratio (Feeder) *5	Placement/Pickup Ratio	
Placement/Pickup Ratio (Nozzle) *5	Placement/Pickup Ratio	
CPH (By Feeder) *5	Components Placed	
CPH (By Nozzle) *5	Components Placed	
Production Time Analysis *6	Previous Process Wait Count Single and Cycle Stop Count	Next Process Wait Count
Total Stop Time Breakdown *6	Previous Process Wait Count Single and Cycle Stop Count	Next Process Wait Count
Scrap Count By Board/Feeder *5	By Board [Scrap Count]	
Scrap Count By Board/Nozzle *5	By Board [Scrap Count]	
Scrap Count By Chip/Feeder *5	By Chip[Scrap Count]	
Scrap Count By Chip/Nozzle *5	By Chip[Scrap COunt]	
Production Error	Restart Impossible Stop Count Chip Ejection Error [Count] Chip Recognition Error [Count] Chip Pickup Error Count Placement Point Recognition Error Count Bond Elongation Error Count Mask Recognition Error Count Bond Recognition Error Count	Single or Cycle Error Stop Count Chip Placement Error [Count] Chip Erection Error [Count] Board Transport Error Count Board Recognition Error Count Paste Countup Error [Count] Feedback Error Count Other Error Count

*5 Items are displayed by selecting either “Feeder” or “Nozzle” of “Select from the following drill down” in Chart Drill Down Options on the Level II Supervisor View and selecting display items.
*6 Items are displayed based on the selection of items in the Level II Supervisor View

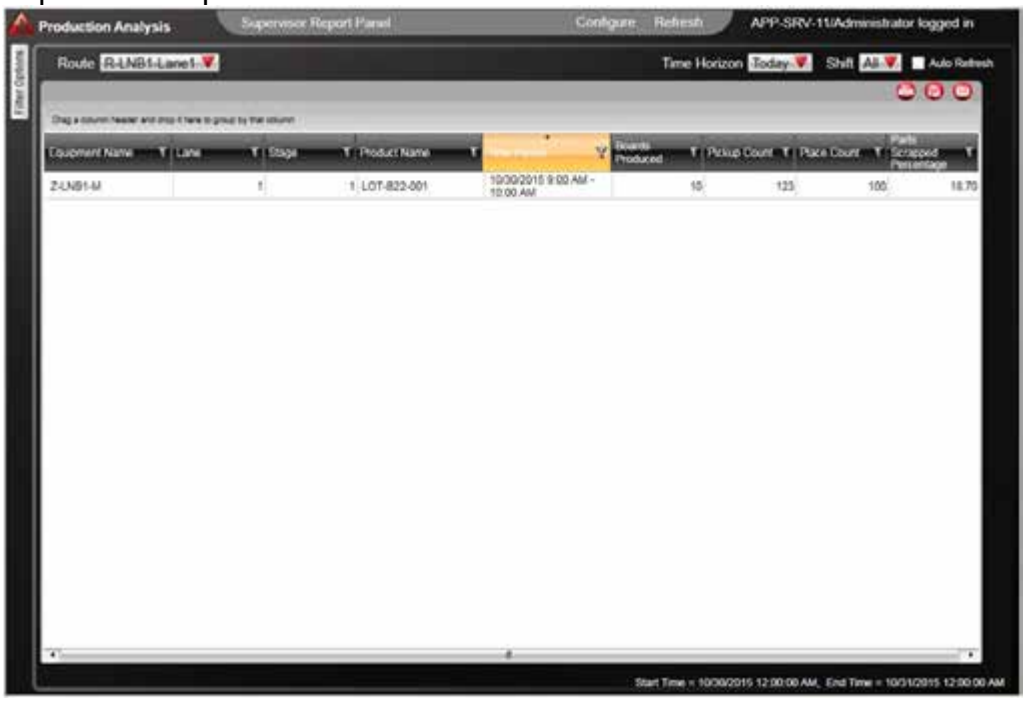
The table below shows the list of defined elements, their descriptions and expressions.

Elements	Description	Expression
Placement/Pickup Ratio	=Placement ratio The ratio of the total placement count to the total pickup count	SUM(Place Count) /SUM(Pickup Count)*100
Components Placed	=Placement Count Total number of times the machine placed chips	SUM(Place Count)

Elements	Description	Expression
Previous Process Wait Count	Total number of times that the machine was waiting for the previous table or machine to finish a process	SUM (Prev Proc Wait Count)
Next Process Wait Count	Total number of times that the machine was waiting for the next table or machine to finish a process	SUM(Next Proc Wait Count)
Single and Cycle Stop Count	Total number of times that the machine made single or cycle stop caused by operation button press action	SUM(Single Cycle Stop Count
Scrap Count By Board	The total number of chips that are judged as scraps without placing on the board for each board.	SUM(Pickup Count) -SUM(Place Count)
By Chip [Scrap Count]	The total number of chips that are judged as scraps without placing on the board for each chip	SUM(Pickup Count) -SUM(Place Count)
Restart Impossible Stop Time	Total time that the machine stopped while waiting for error correction	SUM(Start Imp Stop Time)
Single or Cycle Error Stop Count	Total time that the machine made single or cycle stop when the machine automatically detect an error (However, if it is included in other error, it is not included in this item)	SUM(Single Cycle Stop Count)
Chip Ejection Error [Count]	Total number of times the machine stopped caused by errors when the machine ejects defect chips	SUM(Chip Eject Error Count)
Chip Placement Error [Count]	Total number of times the machine made single stop caused by placement error	SUM(Chip Mount Error Count)
Chip Recognition Error [Count]	Total number of times the machine stopped caused by chip recognition error (It results in recognition error when an error occurs three times in a row in the same feeder)	SUM(Chip Recog Error Count)
Chip Erection Error [Count]	Total number of times the machine stopped caused by chip erection error	SUM(Chip Erection Error Count)
Chip Pickup Error Count	Total number of times the	SUM(Chip Pickup Error

Elements	Description	Expression
	machine stopped caused by chip pickup error (the chip was not picked up for placement)	Count)
Board Transport Error Count	Total number of times the machine made single stop caused by board transport error (any error caused on the conveyor)	SUM(Transport Error Count)
Board Recognition Error Count	Total number of times that the machine stopped caused by placement point recognition error (the type of local placement point/pattern fiducial is not recognized)	SUM(Place Pt Recog Error Count
Board Recognition Error Count	Total number of times the machine made single stop caused by board recognition error (the type of board fiducials were not recognized)	SUM(Board Recog Error Count)
Board Elongation Error Count	Total number of times the machine stopped caused by board elongation error (the type of board fiducials were not recognized)	SUM(Board Elong Error)
Paste Countup Error [Count]	Total number of times the machine stopped caused by paste countup error	SUM(Paste Countup Error Count)
Mask Recognition Error Count	Total number of times the machine made single stop caused by mask/solder recognition error (stencil recognition error)	SUM(Mask Recog Error Count)
Feedback Error Count	Total number of times the machine made cycle stop caused by feedback stop information	SUM(Feedback Error Count)
Bond Recognition Error	Total number of times the machine stopped caused by bond recognition error	SUM(Bond Recog Error Count)
Other Error Count	Total number of times the machine stopped caused by other errors	SUM(Other Error Count)

■ Supervisor Report Panel



Display item	Elements	
List	Machine name. Stage No. Period Pickup Count Parts scrapped Percentage	Lane Product Name Boards Produced Place Count

The table below shows the list of defined elements, their descriptions and expressions.

Elements	Description	Expression
Equipment Name.	Machine name used for production	-
Lane [Number]	Lane number used for production (Front lane:1,Rear lane:2)	-
Stage	Stage number	-
Product Name	Board name to be produced	-
Time Period	Period of time that boards are actually produced in the time horizon	-
Boards Produced [Q'ty]	Total number of boards produced	SUM(Produced Count)
Pickup Count	Total number of times the machine picked up chips	SUM(Pickup Count)
Placement Count	= Placement Count Total number of times the machine placed	SUM(Place Count)
Parts Scrapped Percentage	The ratio of the total number of parts that are not placed on	{ SUM(Pickup

Elements	Description	Expression
	the board to the total pickup count	Count)-SUM(Place Count) } /SUM(Pickup Count)*100

■ Line Performance Report



Display item	Elements	
Production	#Products Blocks Produced [Q'ty] Placement Count Worst M/C Cycle Time Best M/C Cycle Time Average Tact Time	Boards Produced [Q'ty] Pickup Count Bottleneck Machine Worst Tact Time Average M/C Cycle Time Operation Ratio
Performance	CPH Pickup Ratio	Lot Run Time Placement Ratio
Stop Time	Changeover Time Next Process Wait Time Other Time	Previous Process Wait Time Exhaust Time
Operation Ratio	Production Time [Ratio] Other Time [Ratio]	Down Time [Ratio]
Actual Boards vs Theoretical Boards	Produced Bpard [Q'ty] Efficiency Boards	The number of theoretical boards
Actual Blocks vs Theoretical Blocks	Produced Blocks [Q'ty] Efficiency Blocks [Q'ty]	Theoretical Blocks [Q'ty]
Actual Changeover	Changeover (Actual Time)	

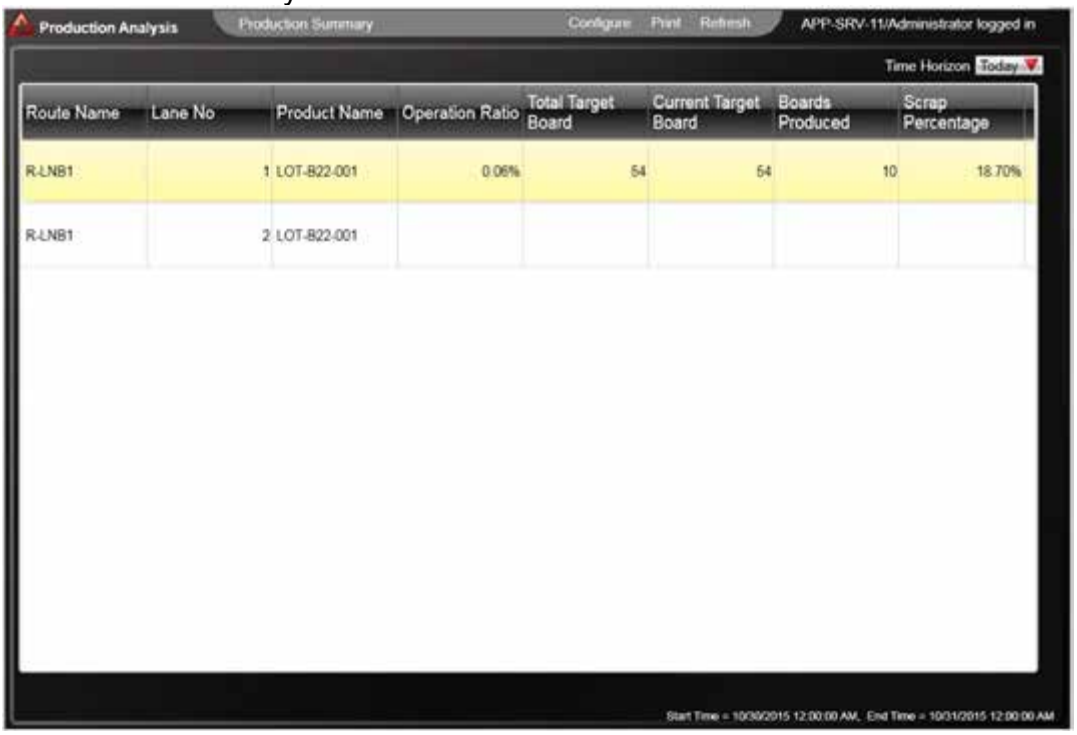
The table below shows the list of defined elements, their descriptions and expressions.

Elements	Description	Expression
#Production	The number of production lots	COUNT(Production Lot)
Boards Produced [Q'ty]	Total number of boards produced	SUM(Produced Count)
Blocks Produced [Q'ty]	Total number of blocks produced	SUM(Blocks Produced)
Pickup Count	Total number of times the machine picked up chips	SUM(Pickup Count)
Placement Count	Total number of times the machine placed	SUM(Placement Count)

Elements	Description	Expression
Bottleneck Machine	A factor of the machine in determining the production speed	
Worst M/C Cycle Time	The maximum time taken to mount a single board	MAX (Cycle Time)
Worst Tact Time	The maximum time taken to produce a single board	MAX(Tact Time)
Best M/C Cycle Time	The minimum time taken to mount a single board	MIN (Cycle Time)
Average M/C Cycle Time	The average time taken to mount a single board	AVE (Cycle Time)
Average Tact Time	The average time taken to produce a single board	AVE(Tact Time)
Operation Ratio	The ratio of the total real running time to the total running time	SUM(Real Run Time)/ SUM(RunTime)*100
CPH	Placement Count per hour	SUM(Place Count)/SUM(Real Run Time $_{[sec]}$)*3600
Lot Run Time	Lot produced period	Lot End Time-Lot Start Time
Pickup Ratio	The ratio of the total number of times that chips are picked up successfully to the total pickup count	{ SUM(Pickup Count)-SUM(Pickup Miss Count) }/SUM(Pickup Count)*100
Placement Ratio	The ratio of the total placement count to the total pickup count	SUM(Place Count)/SUM(Pickup Count)*100
Changeover Time	Total time taken for changeover	SUM(Changeover End Time-Changeover Start Time)
Previous Process Wait Time	Total time that the machine was waiting for the previous table or machine to finish processing a panel	SUM(PCB Wait Load Time)+SUM(Prev Proc Wait Time)
Next Process Wait Time	Total time that the machine was waiting for the next table or machine to finish processing a panel	SUM(Next Proc Wait Time)
Exhaust Time	Total time that the machine stopped while waiting for an exhausted chip to be supplied.	SUM (Chip Supply Wait Time)
Other Time	Total other time	SUM(Other Time)
Production Time [Ratio]	The ratio of the total real run time to the time horizon	SUM(Real Run Time)/Time Horizon*100
Down Time [Ratio]	The ratio of the total waiting	SUM(Wait Time)/Time

Elements	Description	Expression
	time to the time horizon	Horizon*100
Other Time [Ratio]	The ratio of the total real run time and time other than waiting time to the time horizon	{ Time Horizon-SUM(Real Run Time)-SUM(Wait Time) }/Time Horizon*100
Theoretical Boards	The number of boards that can theoretically be produced during a time horizon	Time Horizon/Theoretical Cycle Time
Efficiency Boards	The number of boards considering efficiency to theoretical boards	Time Horizon/Theoretical Cycle Time*Efficiency/100
Theoretical Blocks	The number of blocks that can theoretically be produced during a time horizon	Time Horizon/Theoretical Cycle Time*AVE(Pattern Quantity/Board Quantity)
Efficiency Blocks [Q'ty]	The number of blocks considering efficiency to theoretical boards	Time Horizon/Theoretical Cycle Time*AVE(Pattern Quantity/Board Quantity)*Efficiency/100
Changeover (Actual Time)	Total time taken for setup	SUM(Changeover End Time-Changeover Start Time)

■ Production Summary



Display item	Elements	
List	Route Name Product Name Total Target Board Board Produced	Lane No. Operation Ratio Current Target Board Scrap Percentage

The table below shows the list of defined elements, their descriptions and expressions.

Elements	Description	Expression
Route Name	Route Name	
Lane No	Lane number used for production (Front lane:1,Rear lane:2)	
[Current] Product Name	Currently producing board name	
Operation Ratio	The ratio of the total production time to the time horizon	SUM(Production Time)/Time Horizon*100
Total Target Board	The number of theoretical boards during the time horizon	Time Horizon/Average Cycle Time
Current Target Board	The number of theoretical boards between the start of time horizon and current time	(Current Time-Start of Time Horizon)/ Average Cycle Time
Board Produced	Total number of boards produced	SUM(Produced Count)
Scrap Percentage	The ratio of the total number of parts that are not placed on the board to the total pickup	{ SUM(Pickup Count)-SUM(Place Count) } /SUM(Pickup Count)*100

Elements	Description	Expression
	count	

■ Production Summary Level 1



Display item	Elements	
Actual Boards Vs Theoretical Lane 1	Boards Produced [Q'ty] Efficiency Boards	The number of theoretical boards
Lane 1 Top 10 Feeder Scrap	Scrap %	
Actual Boards Vs Theoretical Lane 2	Boards Produced [Q'ty] Efficiency Boards	The number of theoretical boards
Lane 2 Top 10 Feeder Scrap	Scrap %	

The table below shows the list of defined elements, their descriptions and expressions.

Elements	Description	Expression
Boards Produced [Q'ty]	Total number of boards produced	SUM(Produced Count)
Theoretical Boards	The number of boards that can theoretically be produced during a time horizon	Time Horizon/Theoretical Cycle Time
Efficiency Boards	The number of boards considering efficiency to theoretical boards	Time Horizon/Theoretical Cycle Time*Efficiency/100
Scrap Count %	・Scrap Percentage The ratio of the total number of parts that are not placed on the board to the total pickup count	{ SUM(Pickup Count)-SUM(Place Count) }/SUM(Pickup Count)*100

■ Scrap Data Report



Display item	Elements	
List	Feeder barcode Placement Ratio	Pickup Ratio

The table below shows the list of defined elements, their descriptions and expressions.

Elements	Description	Expression
Feeder Barcode	Feeder Serial	
Pickup Ratio	The ratio of the total number of times that chips are picked up correctly to the total pickup count	{ SUM(Pickup Count)-SUM(Pickup Miss Count) }/SUM(Pickup Count)*100
Placement Ratio	The ratio of the total placement count to the total pickup count	SUM(Place Count)/SUM(Pickup Count)*100

- Scrap Data Details Report



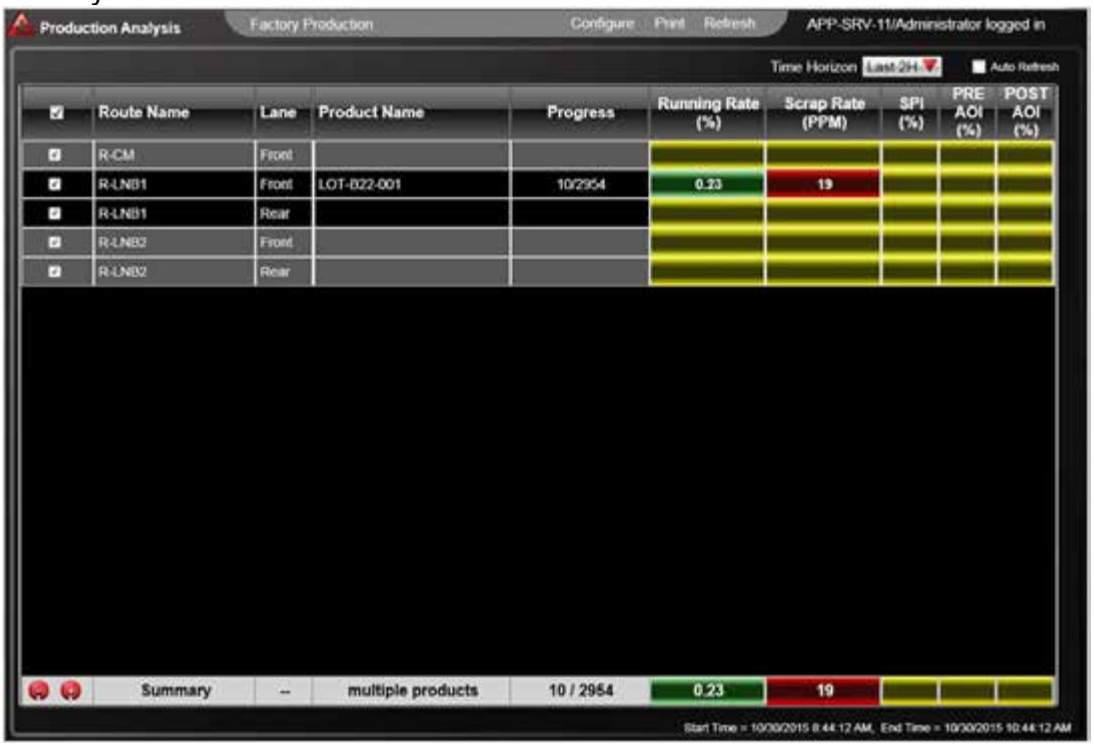
Display item	Elements	
List	Location Pickup Miss[Count] Height Miss [Count] Placement Count Placement Mistake[Count] Pickup Error [Count]	Number Reocg Error [Count] Pickup Count Drop Mistake [Count] Shape Error [Count]

The table below shows the list of defined elements, their descriptions and expressions.

Elements	Description	Expression
Location	Feeder Serial	
Number	Job ID	
Pickup Miss Count	Total number of pickup miss times (error caused by failure of chip thickness measurement so that the scan sense in line rejects chips)	SUM(Pickup Miss Count)
Reocgition Error [Count]	Total number of missed placements caused by a recognition error (the part was not recognized)	SUM(Recog Error Count)
Height Miss [Count]	・Number of height measurement errors Total number of height measurement errors (miss placements caused by virtue of invalid component height)	SUM(Height Miss Count)
Pickup Count	Total number of times the machine picked up chips	SUM(Pickup Count)
Placement Count	= Placement Count Total number of times the machine placed	SUM(Place Count)

Elements	Description	Expression
Drop Mistake [Count]	Total number of drop error counts	SUM(Drop Mistake Count)
Placement Mistake [Count]	=Placement Mistake Total number of placement error count	SUM(Mount Mistake Count)
Shape Error [Count]	Total number of shape error counts	SUM(Shape Error Count)
Pickup Error [Count]	Total number of pickup error (not picked up) counts	SUM(Pickup Error Count)

■ Factory Production



Display item	Elements	
List	Product Name Running Rate	Progress Scrap Rate

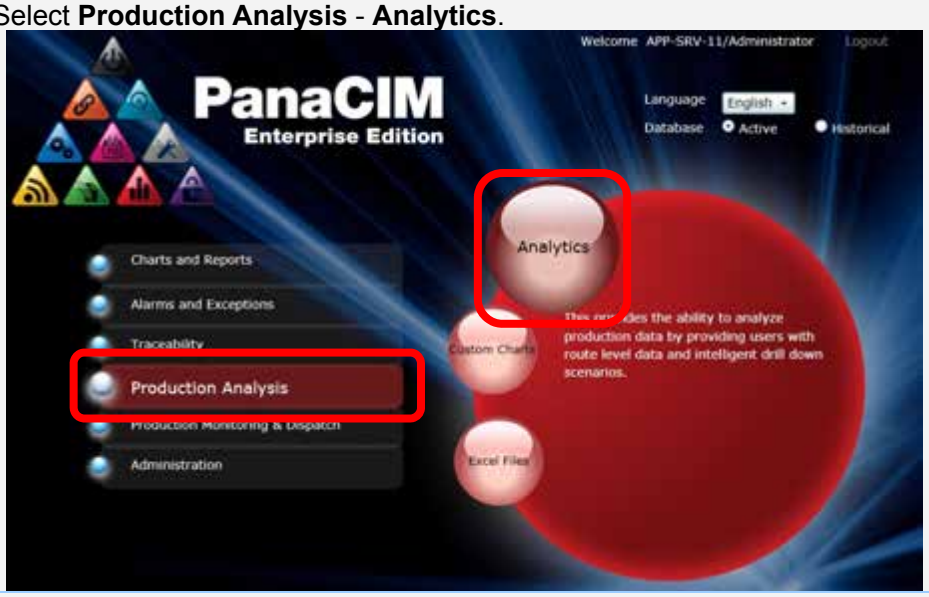
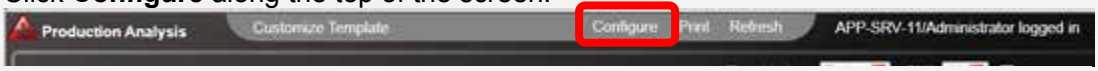
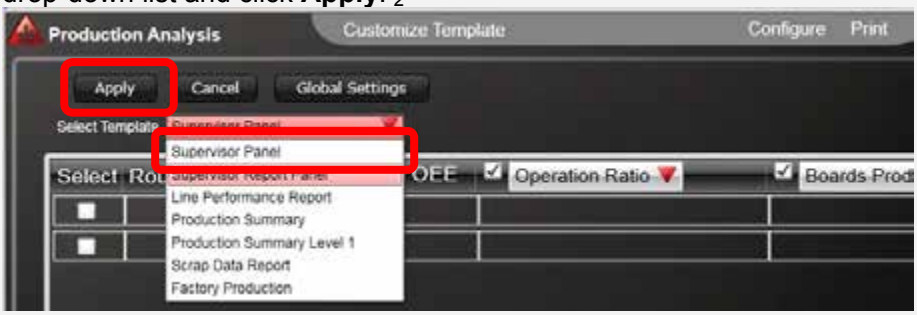
The table below shows the list of defined elements, their descriptions and expressions.

Elements	Description	Expression
Product Name	Name of a produced board	
Progress	Denominator: The number of boards produced Numerator: The number of theoretical boards	SUM(Boards Produced)/(Time Horizon/Cycle Time)
Running Rate	The ratio of the total real run time to the total running time	SUM(Real Run Time)/SUM(Run Time)*100
Scrap Rate (%)	The ratio of the total number of parts that are not placed on the board to the total pickup count	{ SUM(Pickup Count)-SUM(Place Count) }/SUM(Pickup Count)*100

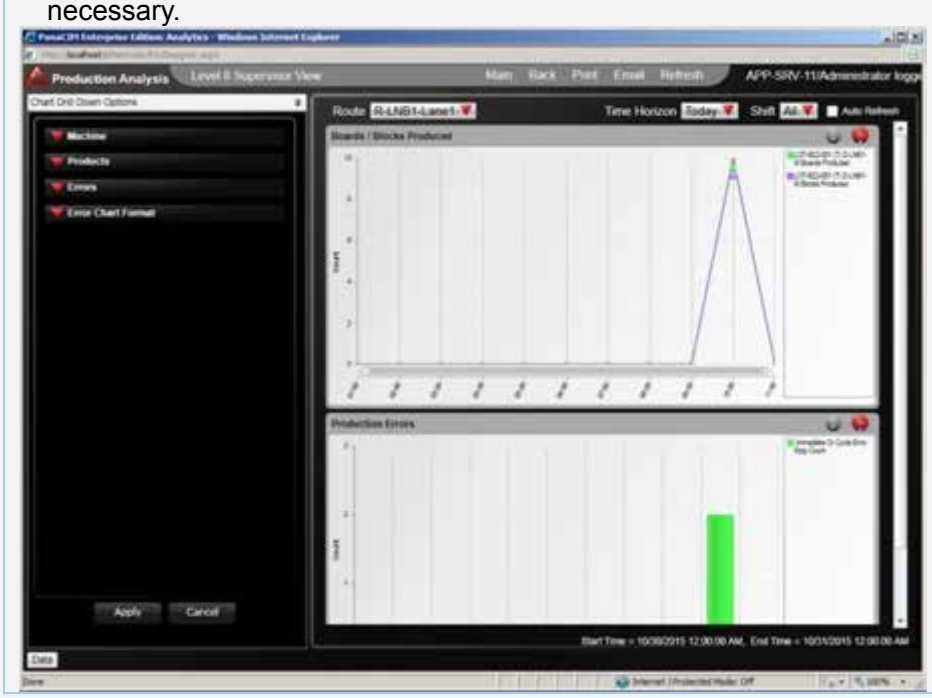
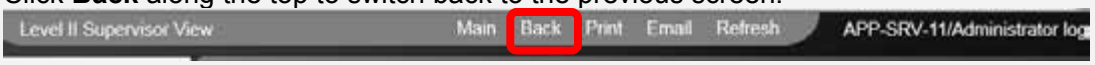
4.3 Operation

Select **Production Analysis** - **Analytics** from PanaCIM Web application to display the analysis templates.

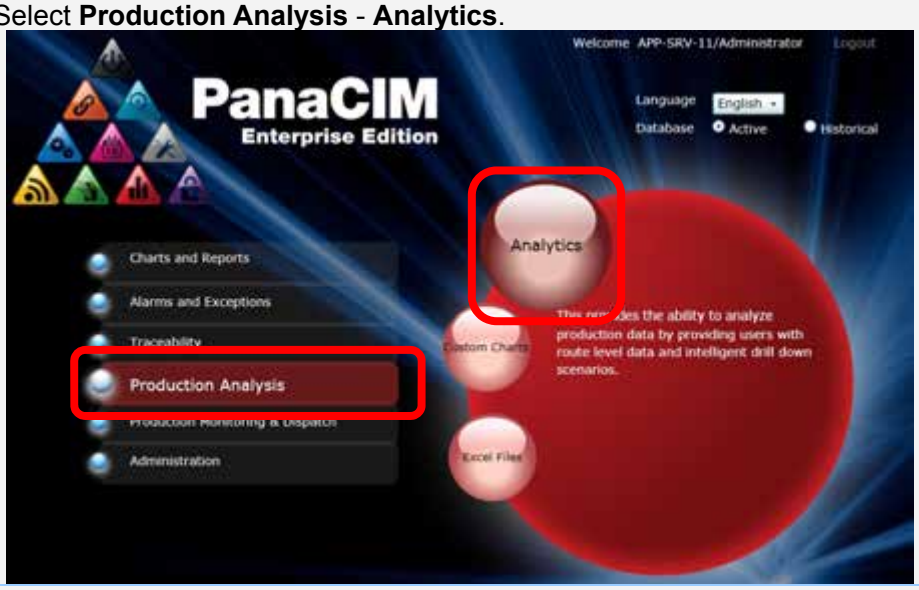
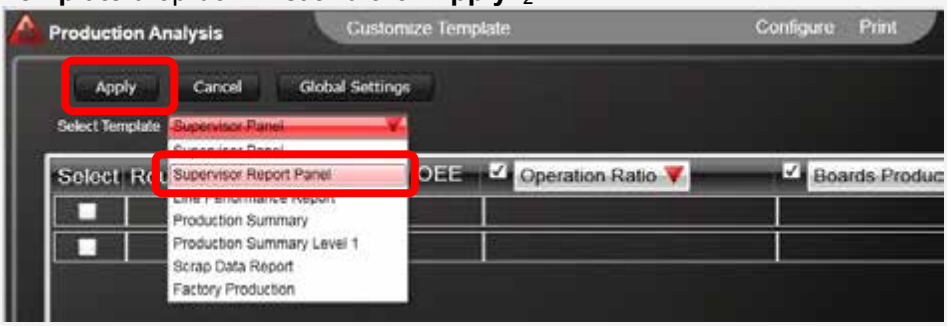
4.3.1 Administrator Panel

Step	Procedure
1	Launch PanaCIM Web application.
2	Select **Production Analysis** - **Analytics**. 
3	Click **Configure** along the top of the screen. 
4	On the screen shown below*1, select **Administrator Panel** from the Select Template drop-down list and click **Apply**.*2  *1: Refer to 4.2 Settings for details of the Configuration screen. *2: The template last used will be shown automatically when the [Analytics] is selected next time.

5	The Administrator Panel template populated with the items selected on the Configuration screen appears. 
6	Select the desired line from those shown on the Administrator Panel template.
7	The Level 1 Administrator View appears.  Items include: • Availability • Average cycle time • Average CPH • Cycle/Tact time list • Production data graph • Setup time graph • Waiting time graph • Scrap graph
8	Select the desired graph on the Level I Supervisor View. For selectable items, refer to the display items in the Level II Supervisor View.

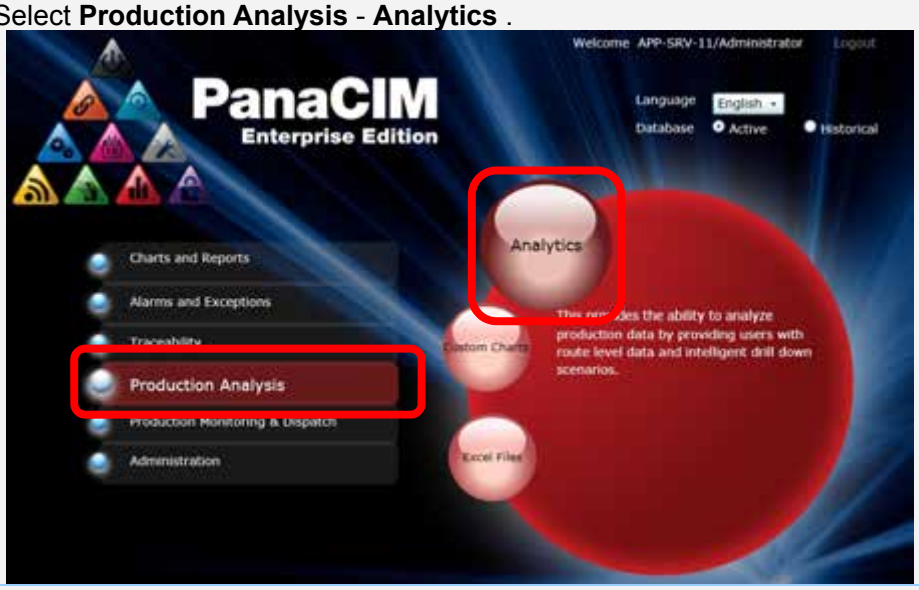
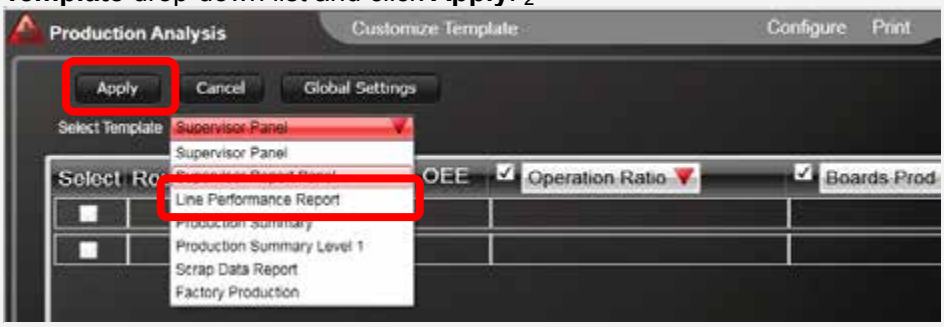
9	The Level II Supervisor View appears. Select the display items (Machine / Products / Production Time / Total Stop Time / Error / Error Chart Format / Parts / Part Graph Type) in the Chart Drill Down Options as necessary. 
10	Select your desired graph on the Level II Supervisor View. For selectable items, refer to selectable items, refer to the display items in the Level III Supervisor View. (Select either “Feeder” or “Nozzle” of “Select from the following drill down” in Chart Drill Down Options on the Level II Supervisor View before a graph is selected.)
11	The Level III Supervisor View appears. Select the display items (Machine / Feeder / Nozzle / Product / Configuration of Production Time / Configuration of Total Stop Time / Error / Error Chart Format / Parts Part Graph Type) in the Chart Drill Down Options as necessary.
12	Click **Back** along the top to switch back to the previous screen. 

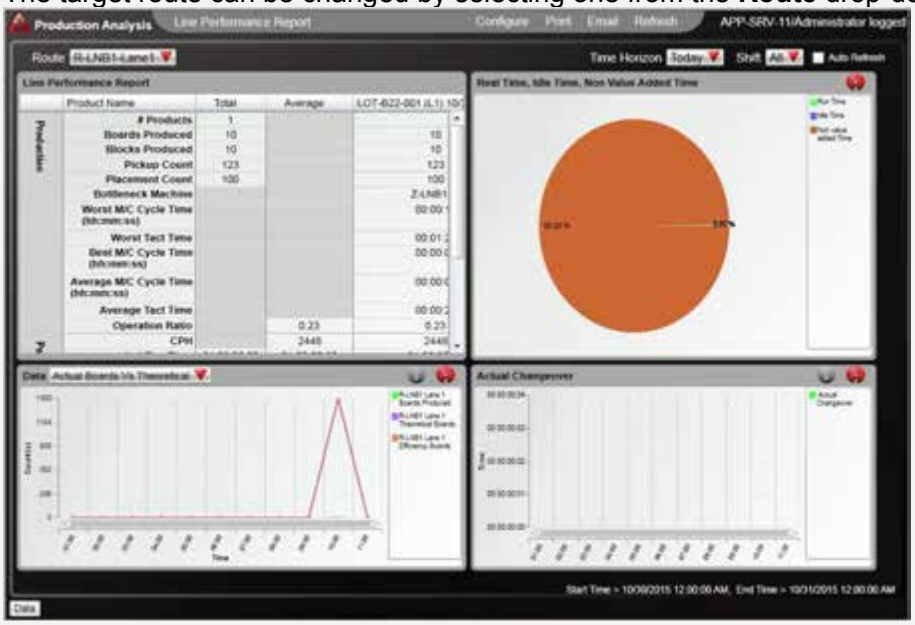
4.3.2 Report Panel (for Administrator)

Step	Procedure
1	Launch PanaCIM Web application.
2	Select **Production Analysis** - **Analytics**. 
3	Click **Configure** along the top of the screen. 
4	On the screen shown below*1, select **Report Panel (for Administrator)** from the **Select Template** drop-down list and click **Apply**.*2  *1: Refer to 8.2 Settings for details of the Configuration screen. *2: The template last used will be shown automatically when the [Analytics] is selected next time.

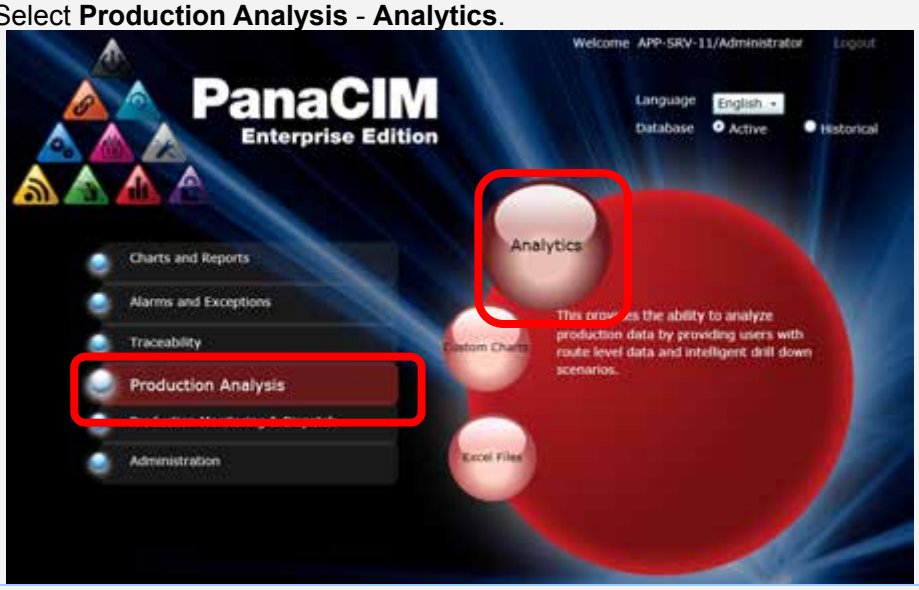
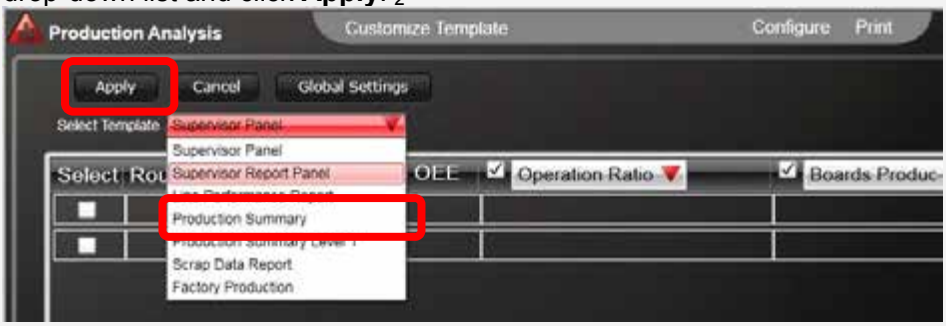
Step	Procedure
5	The Report Panel (for Administrator) template appears. The target route can be changed by selecting one from the **Route** drop-down list. 

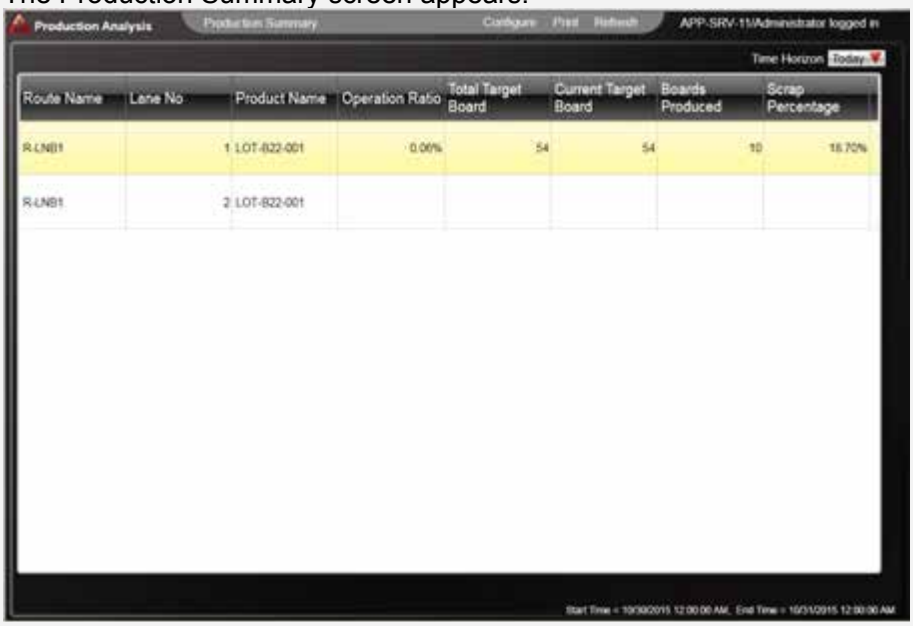
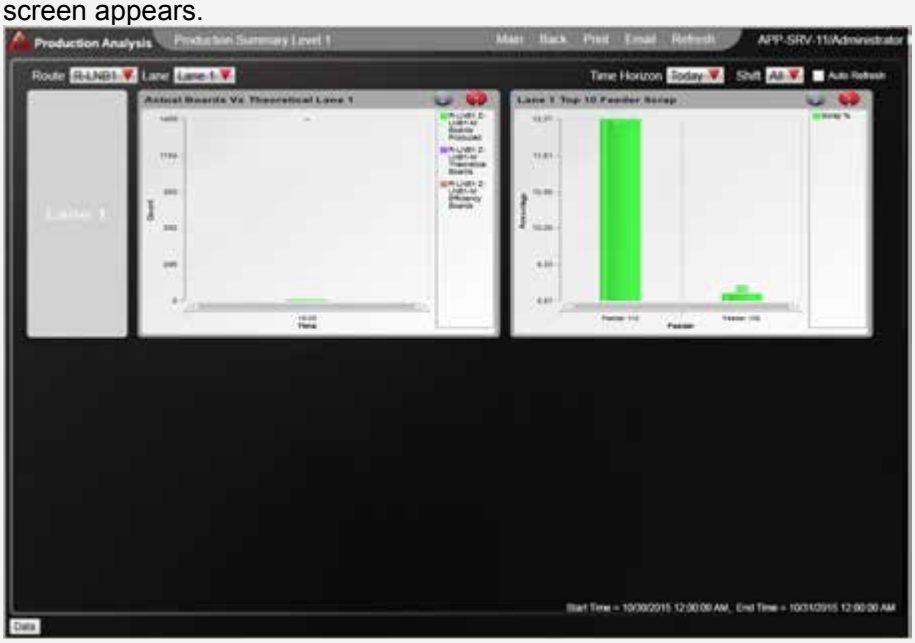
4.3.3 Line Performance Report

Step	Procedure
1	Launch PanaCIM Web application.
2	Select **Production Analysis** - **Analytics** . 
3	Click **Configure** along the top of the screen. 
4	On the screen shown below*1, select **Line Performance Report** from the **Select Template** drop-down list and click **Apply**.*2  *1: Refer to 4.2 Settings for details of the Configuration screen. *2: The template last used will be shown automatically when the [Analytics] is selected next time.

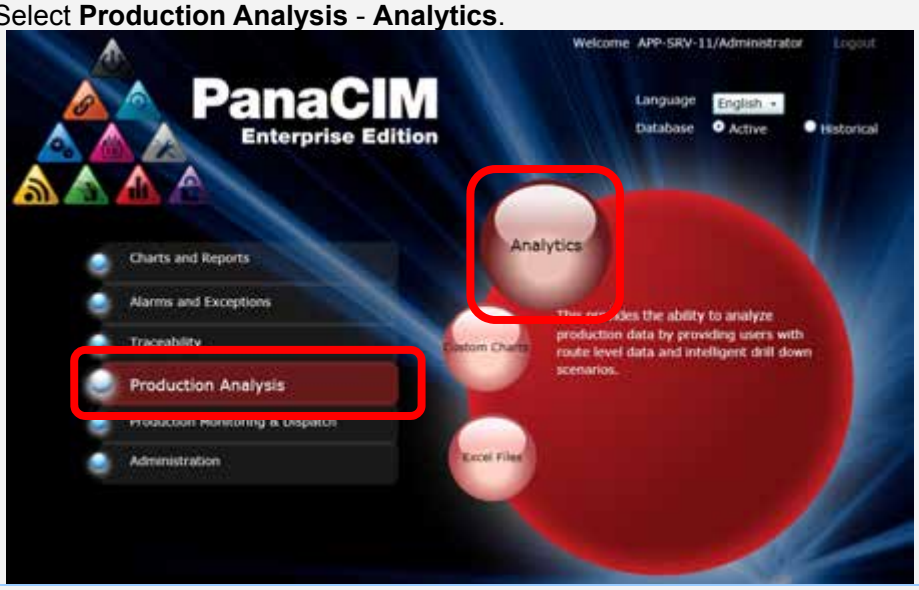
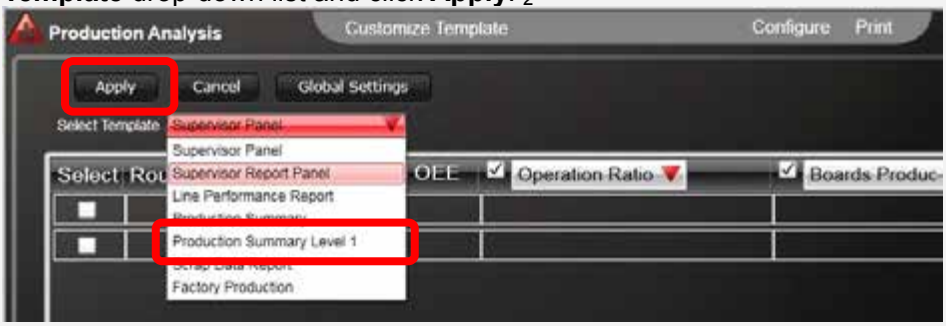
Step	Procedure
5	The Line Performance Report template appears. The target route can be changed by selecting one from the **Route** drop-down list. 

4.3.4 Production Summary

Step	Procedure
1	Launch PanaCIM Web application.
2	Select **Production Analysis** - **Analytics**. 
3	Click **Configure** along the top of the screen. 
4	On the screen shown below*1, select **Production Summary** from the **Select Template** drop-down list and click **Apply**.*2  *1: Refer to 4.2 Settings for details of the Configuration screen. *2: The template last used will be shown automatically when the [Analytics] is selected next time.

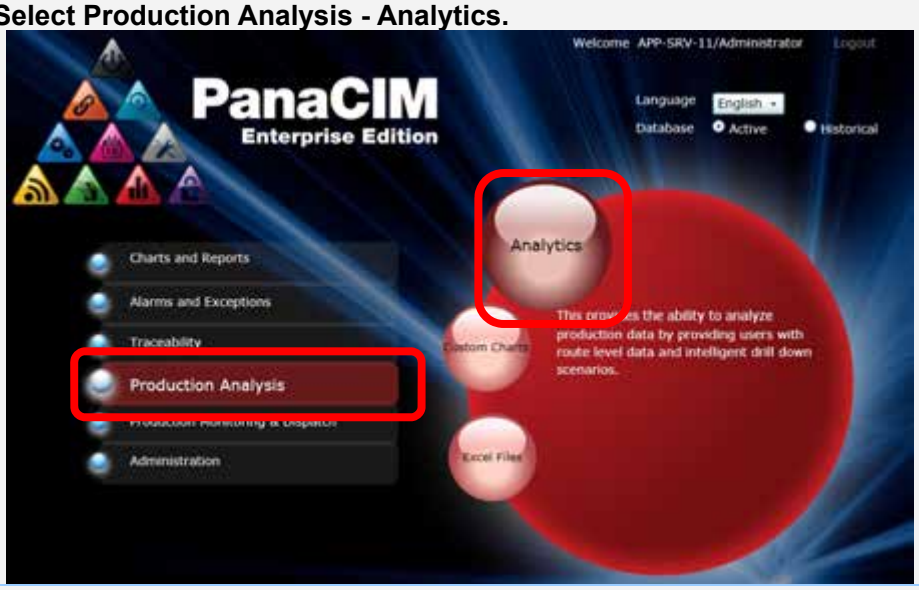
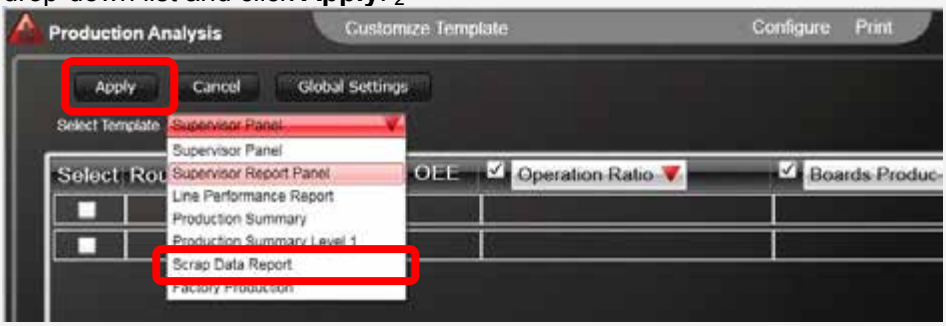
Step	Procedure
5	The Production Summary screen appears. 
6	Select a lane shown on the Production Summary screen.
7	The Level 1 screen filled with the lane information selected on the Production Summary screen appears. 
8	Click **Back** along the top to switch back to the previous screen.

4.3.5 Production Summary - Level 1

Step	Procedure
1	Launch PanaCIM Web application.
2	Select **Production Analysis** - **Analytics**. 
3	Click **Configure** along the top of the screen. 
4	On the screen shown below*1, select **Production Summary Level 1** from the **Select Template** drop-down list and click **Apply**.*2  *1: Refer to 8.2 Settings for details of the Configuration screen. *2: The template last used will be shown automatically when the NewPA - [Analytics] is selected next time.

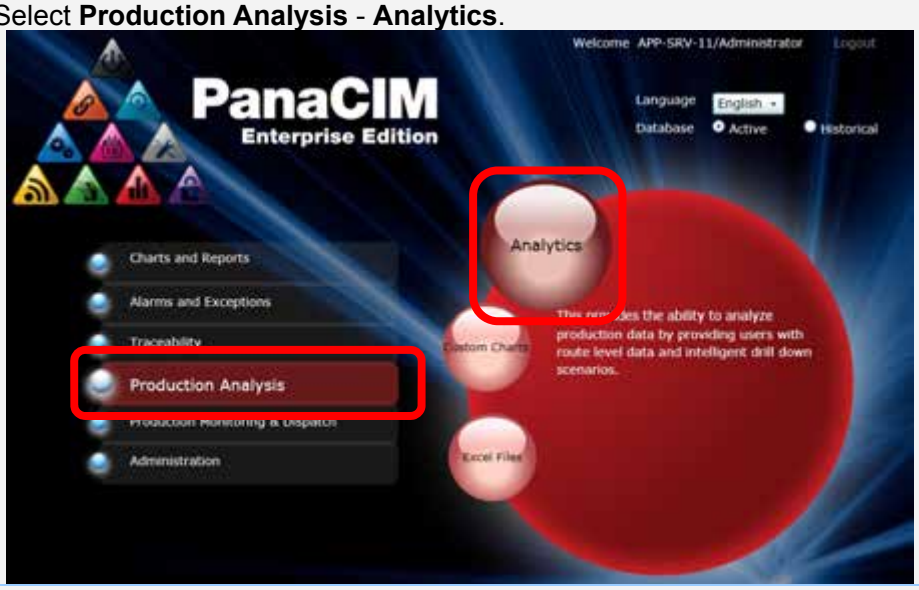
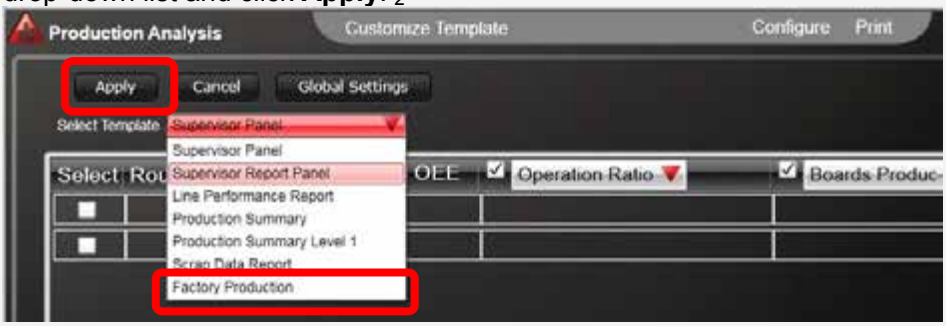
Step	Procedure
5	The Production Summary level 1 screen appears. The target lane can be changed by selecting ones from the **Route** and **Lane** drop-down lists. 

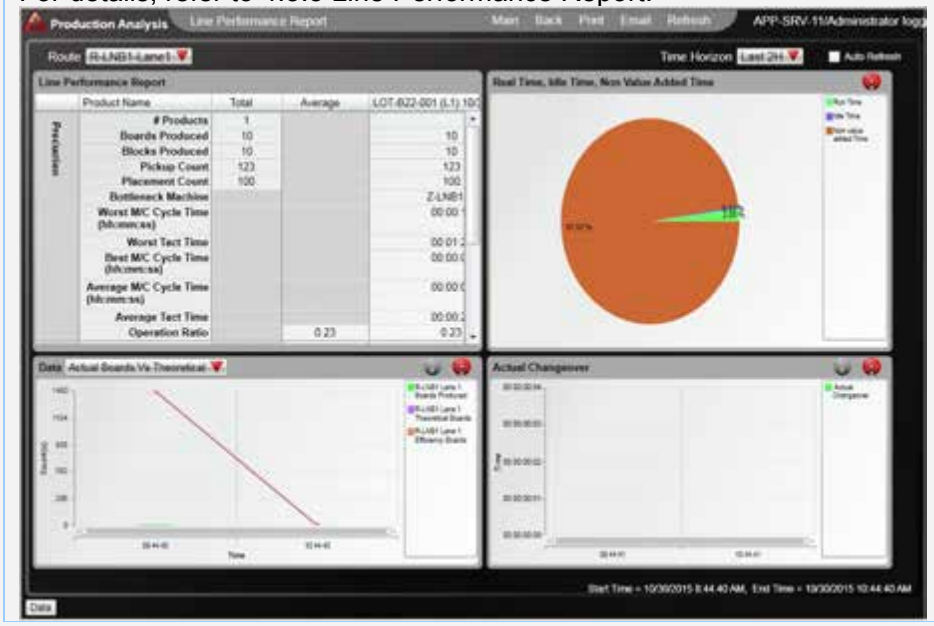
4.3.6 Scrap Data Report

Step	Procedure
1	Launch PanaCIM Web application.
2	**Select Production Analysis - Analytics.** 
3	**Click Configure along the top of the screen.** 
4	On the screen shown below*1, select **Scrap Data Report** from the **Select Template** drop-down list and click **Apply**.*2  *1: Refer to 4.2 Settings for details of the Configuration screen. *2: The template last used will be shown automatically when the [Analytics] is selected next time.

Step	Procedure
5	The Scrap Data Report screen appears. The target route can be changed by selecting one from the **Route** drop-down list. 
6	Select the desired list shown on the Scrap Data Report screen.
7	The Scrap Data Details Report screen appears. The target route can be changed by selecting one from the **Route** drop-down list. 
8	Click **Back** along the top to switch back to the previous screen.

4.3.7 Factory Production

Step	Procedure
1	Launch PanaCIM Web application.
2	Select **Production Analysis** - **Analytics**. 
3	Click **Configure** along the top of the screen. 
4	On the screen shown below*1, select **Factory Production** from the **Select Template** drop-down list and click **Apply**.*2  *1: Refer to 4.2 Settings for details of the Configuration screen. *2: The template last used will be shown automatically when the [Analytics] is selected next time.

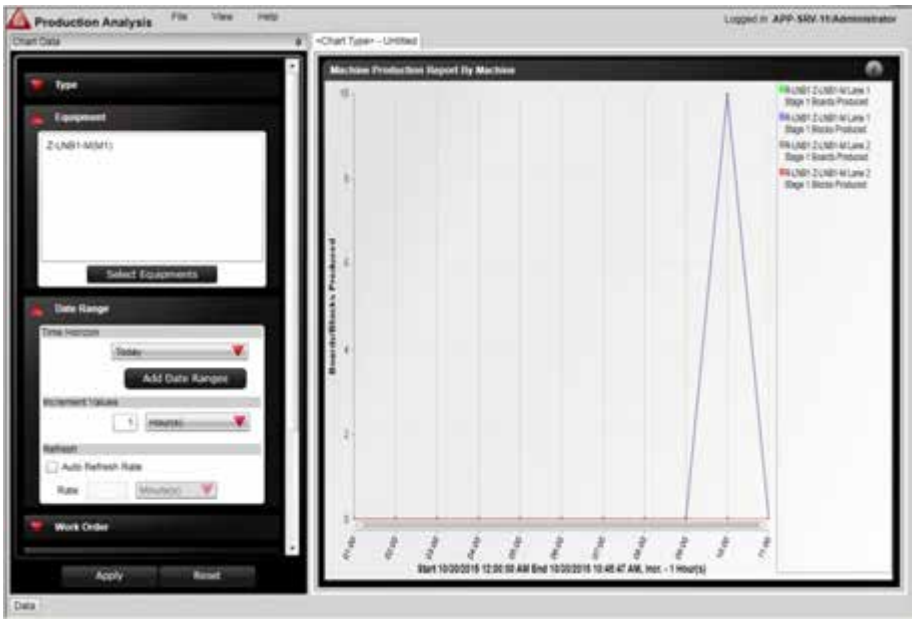
Step	Procedure
5	The Factory Production screen appears. 
6	Select the Production rate of any line shown on the Factory Production screen.
7	The Line Performance Report screen filled with the route information on the Factory Production screen appears. For details, refer to 4.3.3 Line Performance Report. 
8	Click **Back** along the top to switch back to the previous screen.

5. FUNCTION - [Production Analysis] - [Custom Chart]

This chapter explains function-Production Analysis - [Custom Chart].

5.1 Overview

To find [Production Analysis] – [Custom Charts], access PanaCIM EE Web application and select **Production Analysis** from the menu. Then select **Custom** from the balloon-menu.



5.1.1 Production Analysis - [Custom Charts]

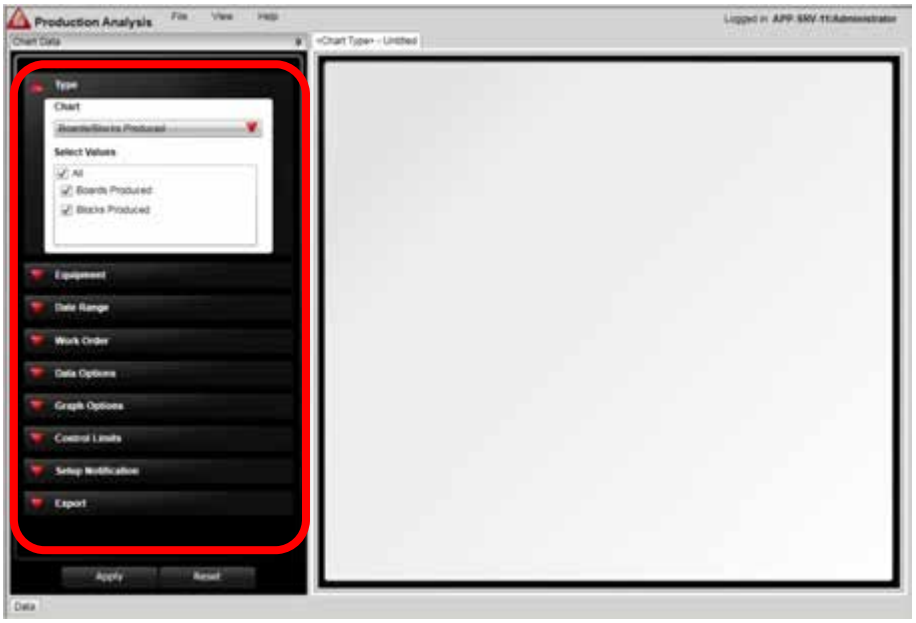
The table below shows the list of charts in Production Analysis - [Custom Charts].

No	Chart name
1	Boards/Blocks Produced
2	Total Run Time Report
3	Wait Time in Percentage
4	Error Stop Time in Percentage
5	Ratio Report
6	Machine State Analysis
7	Wait Count
8	Comprehensive Error Report
9	Error Report Count
10	Tact Time Report
11	Total Time Ratio Report
12	Pickup/Placement Count
13	Nozzle Report
14	Nozzle Ratio Report
15	Feeder/Part Report
16	Feeder Ratio Report by Disposal
17	Feeder Ratio Report by Pickup
18	Scrap Report
19	Work Cell Performance
20	PMD Operation Ratio
21	Machine Pickup/Placement Ratio
22	Feeder Pickup/Placement Ratio
23	Actual Boards vs Theoretical Boards
24	Actual Blocks vs Theoretical Blocks
25	Produced Boards vs Scrap
26	Error Stop Time vs Error Stop Count
27	Board Counter Monitor
28	Equipment Product Performance Report
29	Custom Chart

5.2 Settings

5.2.1 Filter Criteria

The table below is the chart display criteria set in Production Analysis - [Custom Charts].

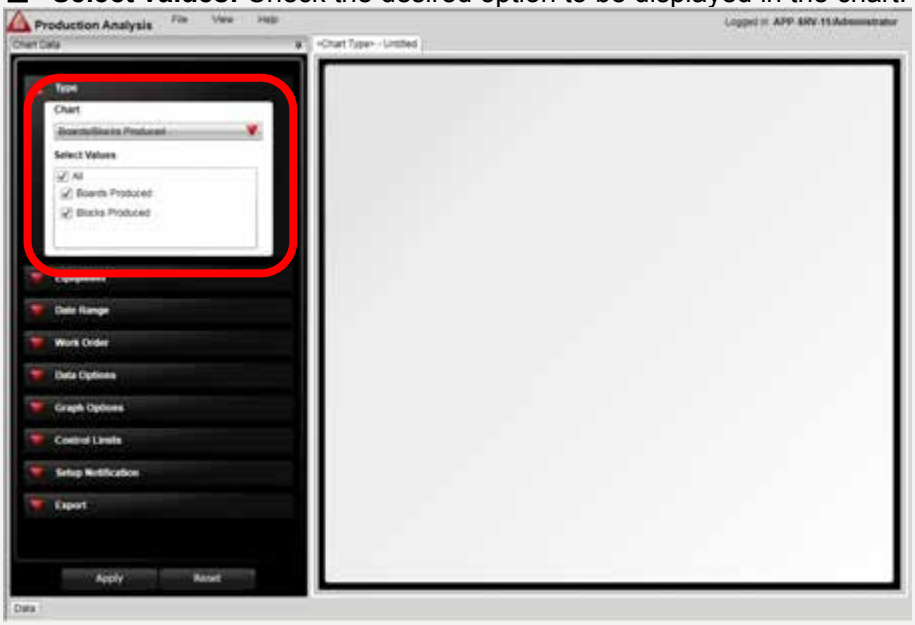


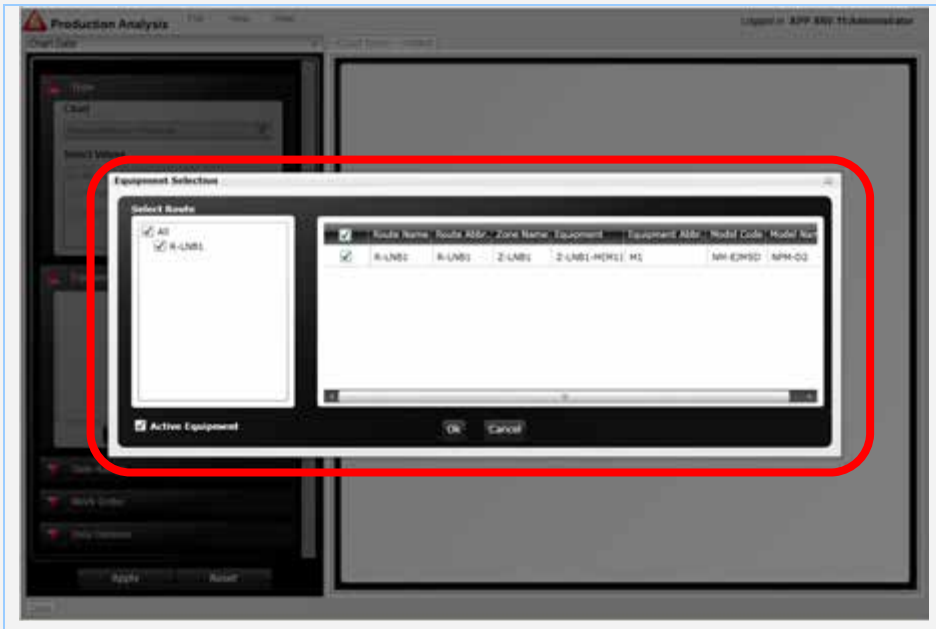
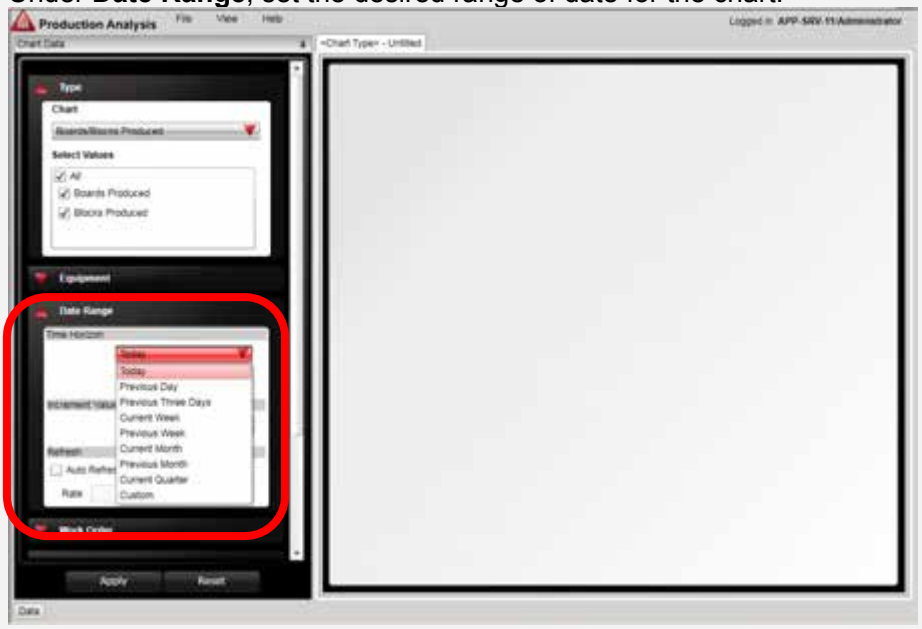
No.	Condition	Description
1	Type	Specify the type of chart. Refer to 9.1.1 NewPA [Custom Charts] for details.
2	Equipment	Specify the applicable equipment.
3	Nozzle	Specify the applicable nozzle on the equipment.
4	Feeder	Specify the applicable feeder on the equipment.
5	Date range	Specify the applicable period.
6	Work order	Specify the applicable work order.
7	Data option	Specify the group (Equipment, product data, work order, lane, stage) and product to be included in the chart.
8	Graph option	Set the title, property and type of the graph.
9	Control limits	Set either UCL or LCL.
10	Setup notification	Set whether or not to inform the message and condition.
11	Export	Set whether or not to export the chart, as well as the timing and options

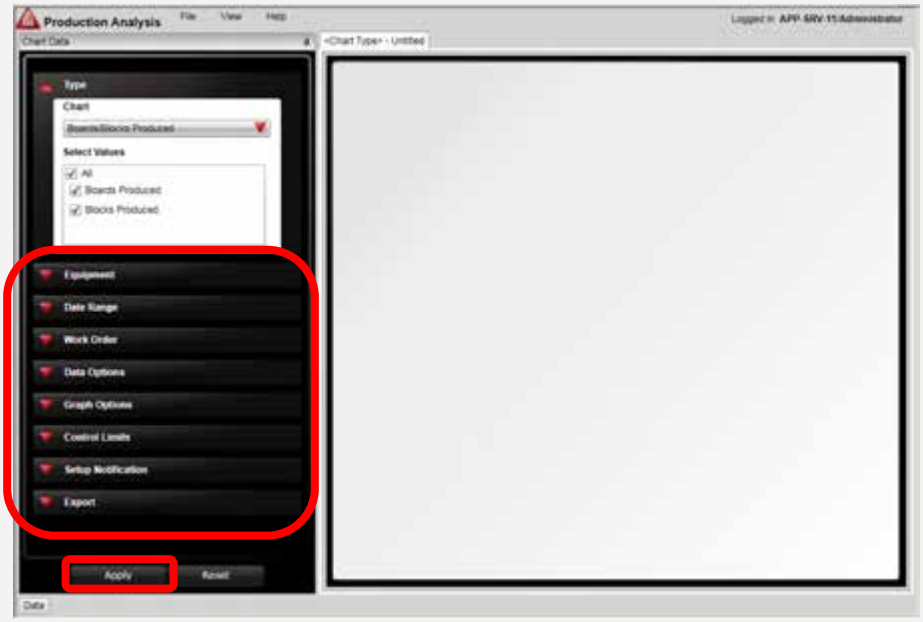
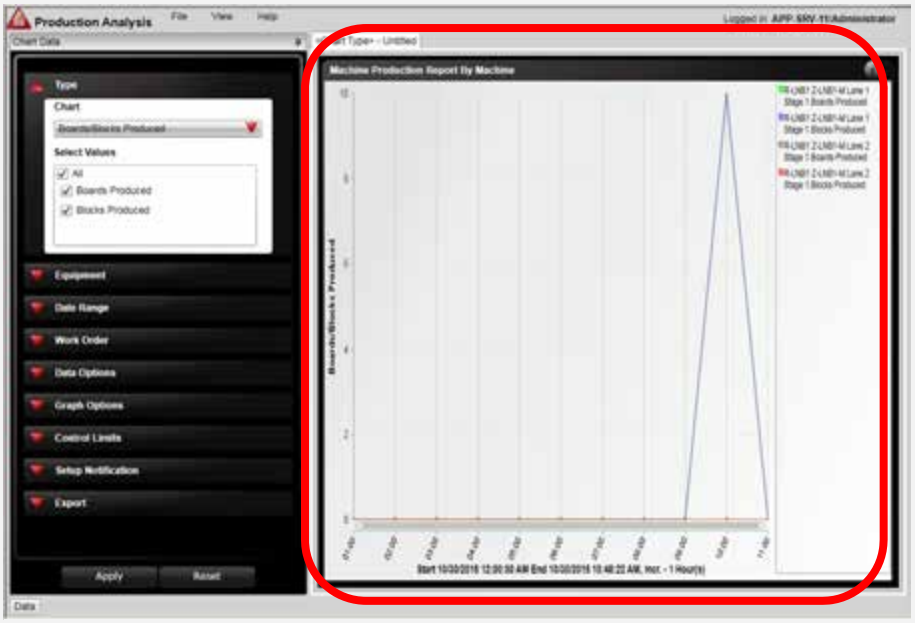
5.3 Operation

5.3.1 Operation Procedure

Select **Custom** from PanaCIM Web application to create a chart of the production analysis.

Step	Procedure
1	Launch PanaCIM Web application.
2	Select **Production Analysis** - **Custom**. 
3	Under **Type**, select and set the following items. ■ **Module:** Select the desired chart from the drop-down list. ■ **Select values:** Check the desired option to be displayed in the chart. 
4	Under **Equipment**, select the equipment for which the chart is to be created.

Step	Procedure
	
5	Under **Date Range**, set the desired range of date for the chart. 

Step	Procedure
6	Set other conditions *1, if any. After setting the conditions, click **Apply**.  *1: Refer to 9.2 Settings for details on the conditions of the chart.
7	A chart will be created and shown in the right pane of the screen.  • The settings made so far can be saved as a configuration file (template) by selecting **Save** in the **File** menu.

5.4 Chart Items

5.4.1 Chart Items

The table below shows the list of defined elements, their descriptions and expressions.

■ Boards/Blocks Produced

Item	Explanation	Expression
Boards Produced	Total number of boards produced	SUM(Boards Produced)
Blocks Produced	Total number of blocks produced	SUM(Blocks Produced)

■ Total Run Time Report

Item	Explanation	Expression
Real Run Time	Total of board recognition time, placement/audit time and transport time (they are not included in case of alternative placement lane)	SUM(Real Run Time)
Total Down Time	Total time that the machine stopped	SUM(Total Down Time)

■ Wait Time in Percentage

Item	Explanation	Expression
Previous Process	The ratio of total previous process time to the total running time	SUM(Prev Proc Wait Time)/SUM(Run Time)*100
Next Process	The ratio of total next process time to the total running time	SUM(Next Proc Wait Time)/SUM(Run Time)*100
Single and Cycle Stop	The ratio of total single and cycle top times to the total running time	SUM(Single Cycle Stop Time)/SUM(Run Time)*100
Chip Supply Wait	The ratio of total chip supply wait time to the total running time	SUM(Chip Supply Wait Time)/SUM(Run Time)*100
Solder Supply Wait (Counter)	The ratio of total solder supply wait time (Counter) to the total running time	SUM(Sldr Ctr Wait)/SUM(Run Time)*100
Solder Supply Wait (Sensor)	The ratio of total solder supply wait time (Sensor) to the total running time	SUM(Sldr Snsr Wait)/ SUM(Run Time)*100
Cleaning Paper Supply Wait	The ratio of total cleaning paper supply wait to the total running time	SUM(Cln Paper Wait Time)/ SUM(Run Time)*100

■ Error Stop Time in Percentage

Item	Explanation	Expression
Restart Impossible Stop Time	The ratio of the total restart impossible stop time to the total running time	SUM(Start Imp Stop Time)/ SUM(Run Time)*100
Single and Cycle Stop Error Time	The ratio of the total single and cycle stop error time to the total running time	SUM(Single Cycle Stop Error Time)/SUM(Run Time)*100

Item	Explanation	Expression
Chip Ejection Error Stop Time	The ratio of the total chip ejection error stop time to the total running time	SUM(Chip Eject Error Time)/SUM(Run Time)*100
Chip Recognition Error Stop Time	The ratio of the total chip recognition error stop time to the total running time	SUM(Chip Recog Error Time)/SUM(Run Time)*100
Chip placement Error Stop Time	The ratio of the total chip placement error stop time to the total running time	SUM(Chip Mount Error Time)/SUM(Run Time)*100
Chip Erection Error Stop Time	The ratio of the total chip erection error stop time to the total running time	SUM(Chip Errorect Error Time)/SUM(Run Time)*100
Chip Pickup Error Stop Time	The ratio of the total chip pickup error stop time to the total running time	SUM(Chip Pickup Error Time)/SUM(Run Time)*100
Board Transport Error Stop Time	The ratio of the total board transport error stop time to the total running time	SUM(Transport Error Time)/SUM(Run Time)*100
Placement Point Recognition Error Stop Time	The ratio of the total mounting point recognition error stop time to the total running time	SUM(Place Pt Recog Error Time)/SUM(Run Time)*100
Board Recognition Error Stop Time	The ratio of the total board recognition error stop time to the total running time	SUM(PCB Recog Error Time)/SUM(Run Time)*100
Board Elongation Error Stop Time	The ratio of the total board elongation error stop time to the total running time	SUM(PCB Elong Error Time)/SUM(Run Time)*100
Paste Coun up Error Stop Time	The ratio of the total paste count up error stop time to the total running time	SUM(Paste Countup Error Time)/SUM(Run Time)*100
Mask/Solder Recognition Error Stop Time	The ratio of the total mask/solder recognition error stop time to the total running time	SUM(Mask Recog Error Time)/SUM(Run Time)*100
Feedback Error Stop Time	The ratio of the total feedback error stop time to the total running time	SUM(Feedback Error Time)/SUM(Run Time)*100
Bond Recognition Error Stop Time	The ratio of the total bond recognition error stop time to the total running time	SUM(Bond Recog Error Time)/SUM(Run Time)*100
Other Error Stop Time	The ratio of the total other error stop time to the total running time	SUM(Oher Error Stop Time)/SUM(Run Time)*100

■ Ratio report

Item	Explanation	Expression
Machine Operation Ratio	The ratio of the total real run time to the total running time	SUM(Real Run Time)/ SUM(Run Time)*100
Machine Run Ratio	The ratio of the total real run time and previous process wait time to the total running time	{ SUM(Real Run Time)+SUM(Prev Proc Wait Time) }/SUM(Run Time)*100
Down Time Ratio	The ratio of the total stop time to the total running time	SUM(Stop Time)/SUM(Run Time)*100
Placement Ratio	The ratio of the total number of placement count to the total	SUM(Place Count)/SUM(Pickup

Item	Explanation	Expression
	pickup count	Count)*100

■ Machine State Analysis

Item	Explanation	Expression
Down Time	The ratio of the total stop time to the total running time	SUM(Stop Time)/SUM(Run Time)*100
Previous Process Wait	The ratio of total previous process time to the total running time	SUM(Prev Proc Wait Time)/SUM(Run Time)*100
Next Process Wait	The ratio of total next process time to the total running time	SUM(Next Proc Wait Time)/SUM(Run Time)*100
Single and Cycle Stop	The ratio of total single and cycle top times to the total running time	SUM(Single Cycle Stop Time)/SUM(Run Time)*100
Chip Supply Wait	The ratio of total chip supply wait time to the total running time	SUM(Chip Supply Wait Time)/SUM(Run Time)*100
Solder Supply Wait Count (Counter)	The ratio of total solder supply wait time (Counter) to the total running time	SUM(Sldr Ctr Wait)/SUM(Run Time)*100
Solder Supply Wait Count (Sensor)	The ratio of total solder supply wait time (Sensor) to the total running time	SUM(Sldr Snsr Wait)/ SUM(Run Time)*100
Cleaning Paper Supply Wait	The ratio of total cleaning paper supply wait to the total running time	SUM(Cln Paper Wait Time)/ SUM(Run Time)*100
Board Recognition	The ratio of total board recognition time to the total running time	SUM(Board Recog Time)/SUM(Run Time)*100
Chip Pickup Error Stop	The ratio of total chip pickup error stop time to the total running time	SUM(Chip Pickup Error Time)/SUM(Run Time)*100
Production Data Menu	The ratio of total production data edit related menu display time to the total running time	SUM(Production Data Edit Time)/SUM(Run Time)*100
Maintenance Menu	The ratio of total maintenance related menu display time to the total running time	SUM(Maintenance Time)/SUM(Run Time)*100
Error Stop Count Due To Other Stage Or Lane	The ratio of total error stop time caused by other stage and lane to the total running time	SUM(Error Count Due To Other Stage Lane)/SUM(Run Time)*100
Production Management Menu	The ratio of total production management related menu display time to the total running time	SUM(Production Manage Menu)/SUM(Run Time)*100
Unit Menu	The ratio of total unit related menu display time to the total running time	SUM(Unit Adjust Time)/SUM(Run Time)*100
Real Run Time	The ratio of total real run time to the running time	SUM(Real Run Time)/SUM(Run Time)*100

■ Wait Count

Item	Explanation	Expression
Down Time Count	Total number of times that the	SUM(Stop Count)

Item	Explanation	Expression
	machine stopped	
Previous Process Wait Count	Total number of times that the machine was waiting for the previous table or machine to finish a process	SUM (Prev Proc Wait Count)
Next Process Wait Count	Total number of times that the machine was waiting for the next table or machine to finish a process	SUM(Next Proc Wait Count)
Single and Cycle Stop Count	Total number of times that the machine made single or cycle stop caused by operation button press action	SUM(Single Cycle Stop Count
Chip Supply Wait Count	Total number of times that the machine stopped while waiting for an exhausted chip to be supplied	SUM(Chip Supply Wait Count)
Solder Supply Wait Count (Counter)	Total number of times the machine stopped while waiting for a re-supply of solder paste (replenishment triggered by a counter)	SUM(Sldr Ctr Wait Count)
Solder Supply Wait Count (Sensor)	Total number of times the machine stopped while waiting for a re-supply of solder paste (replenishment triggered by a sensor)	SUM(Sldr Snsr Wait Count)
Cleaning Paper Supply Wait Count	Total number of times the machine stopped while waiting for a cleaning paper exhaust to be replenished	SUM(Cln Paper Wait Count)

■ Comprehensive Error Report

Item	Explanation	Expression
Restart Impossible Stop	Total number of times that the machine stopped while waiting for error correction.	SUM(Start Imp Stop Count)
Single and Cycle Error Stop	Total number of times that the machine made single or cycle stop when the machine automatically detect an error (However, if it is included in other error, it is not included in this item)	SUM(Single Cycle Stop Count)
Chip Ejection Error	Total number of times the machine stopped caused by errors when the machine ejects defect parts	SUM(Chip Eject Error Count)
Chip Placement Error	Total number of times the machine made single stop caused by placement error	SUM(Chip Mount Error Count)
Chip Recognition Error	Total number of times the machine stopped caused by chip recognition error (It results in recognition error when an error occurs three times in a	SUM(Chip Recog Error Count)

Item	Explanation	Expression
	row in the same feeder)	
Chip Erection Error	Total number of times the machine stopped caused by chip erection error	SUM(Chip Erection Error Count)
Chip Pickup Error	Total number of times the machine stopped caused by chip pickup error (the chip was not picked up for placement)	SUM(Chip Pickup Error Count)
Board Transport Error	Total number of times the machine made single stop caused by board transport error (any error caused on the conveyor)	SUM(Transport Error Count)
Placement Point Recognition Error	Total number of times that the machine stopped caused by placement point recognition error (the type of local placement point/pattern fiducial is not recognized)	SUM(Place Pt Recog Error Count
Board Recognition Error	Total number of times the machine made single stop caused by board recognition error (the type of board fiducials were not recognized)	SUM(Board Recog Error Count)
Board Elongation Error	Total number of times the machine stopped caused by board elongation error (the type of board fiducials were not recognized)	SUM(Board Elong Error)
Paste Count up Error	Total number of times the machine stopped caused by paste count up error	SUM(Paste Countup Error Count)
Mask/Solder Recognition Error	Total number of times the machine made single stop caused by mask/solder recognition error (stencil recognition error)	SUM(Mask Recog Error Count)
Feedback Error	Total number of times the machine made cycle stop caused by feedback stop information	SUM(Feedback Error Count)
Bond Recognition Error	Total number of times the machine stopped caused by bond recognition error	SUM(Bond Recog Error Count)
Other Error Count	Total number of times the machine stopped caused by other errors	SUM(Other Error Count)
Error Caused by Other Stage /Error	Total number of times that the machine stopped caused by other stage or lane error (During production in dual alternative lane, if the lane 1 stopped caused by error such as pickup error, the lane 2 will also stopped caused by other lane error)	SUM(Waiting Count Caused By Other Lane)

■ Error Report Count

Item	Explanation	Expression
Restart Impossible Stop	Total number of times that the machine stopped while waiting for error correction.	SUM(Start Imp Stop Count)
Single and Cycle Error Stop	Total time that the machine made single or cycle stop when the machine automatically detect an error (However, if it is included in other error, it is not included in this item)	SUM(Single Cycle Stop Count)
Chip Ejection Error	Total number of times the machine stopped caused by errors when the machine ejects defect parts	SUM(Chip Eject Error Count)
Chip Placement Error	Total number of times the machine made single stop caused by placement error	SUM(Chip Mount Error Count)
Chip Recognition Error	Total number of times the machine stopped caused by chip recognition error (It results in recognition error when an error occurs three times in a row in the same feeder)	SUM(Chip Recog Error Count)
Chip Erection Error	Total number of times the machine stopped caused by chip erection error	SUM(Chip Erection Error Count)
Chip Pickup Error	Total number of times the machine stopped caused by chip pickup error (the chip was not picked up for placement)	SUM(Chip Pickup Error Count)
Board Transport Error Count	Total number of times the machine made single stop caused by board transport error (any error caused on the conveyor)	SUM(Transport Error Count)
Placement Point Recognition Error Count	Total number of times that the machine stopped caused by placement point recognition error (the type of local placement point/pattern fiducial is not recognized)	SUM(Place Pt Recog Error Count
Board Recognition Error Count	Total number of times the machine made single stop caused by board recognition error (the type of board fiducials were not recognized)	SUM(Board Recog Error Count)
Board Elongation Error Count	Total number of times the machine stopped caused by board elongation error (the type of board fiducials were not recognized)	SUM(Board Elong Error)
Paste Count up Error[Count]	Total number of times the machine stopped caused by paste count up error	SUM(Paste Countup Error Count)

Item	Explanation	Expression
Mask/Solder Recognition Error Count	Total number of times the machine made single stop caused by mask/solder recognition error (stencil recognition error)	SUM(Mask Recog Error Count)
Feedback Error	Total number of times the machine made cycle stop caused by feedback stop information	SUM(Feedback Error Count)
Bond Recognition Error	Total number of times the machine stopped caused by bond recognition error	SUM(Bond Recog Error Count)
Pickup Miss	Total number of pickup miss times (error caused by failure of chip thickness measurement so that the scan sense in line rejects chips)	SUM(Pickup Miss Count)
Reocgition Error Count	Total number of missed placements caused by a recognition error (the part was not recognized)	SUM(Recog Error Count)
Height Miss [Count]	・Number of height measurement errors Total number of height measurement errors (miss placements caused by virtue of invalid component height)	SUM(Height Miss Count)
Other Error Count	Total number of times the machine stopped caused by other errors	SUM(Other Error Count)
Solder Recognition Error	Total number of times the machine stopped caused by solder recognition error	SUM(Solder Recog Error Count)
Recognition Error 1 (MSF)	Total number of times the machine stopped caused by recognition error (MSF)	SUM(Recog Error Count1(MSF))
Recognition Error2 (Shape)	Total number of times the machine stopped caused by recognition error (shape)	SUM(Recog Error Count2(Shape)
Drop Mistake	Total number of times that the machine had drop mistake.	SUM(Drop Mistake Count)
Placement Error	Total number of times that the machine had placement error	SUM(Mount Error Count)

■ Tact Time Report

Item	Explanation	Expression
Tact Time	The average time taken to produce a single board	SUM(Real Run Time)/ SUM(Boards Produced)

■ Total Time Ratio Report

Item	Explanation	Expression
Running Rate (T)	The ratio of the total run time to the time horizon	SUM(Run Time)/Time Horizon*100
Machine Operation Ratio (T)	The ratio of the total	SUM(Real Run Time)/Time

Item	Explanation	Expression
	real run time to the time horizon	Horizon*100
Machine Run Ratio (T)	The ratio of the total real run time and previous process wait time to the total running time	{ SUM(Real Run Time)+SUM(Prev Proc Wait Time) }/SUM(Run Time)*100

■ Pickup/Placement Count

Item	Explanation	Expression
Placement Count	Total number of times the machine placed chips	SUM(Place Count)
Pickup Count	Total number of times the machine picked up chips	SUM(Pickup Count)

■ Nozzle Report / ■Feeder Part Report

Item	Explanation	Expression
Pickup Miss Count	Total number of pickup miss times (error caused by failure of chip thickness measurement so that the scan sense in line rejects chips)	SUM(Pickup Miss Count)
Reocgition Error Count	Total number of missed placements caused by a recognition error (the part was not recognized)	SUM(Recog Error Count)
Height Miss Count	・Number of height measurement errors Total number of height measurement errors (miss placements caused by virtue of invalid component height)	SUM(Height Miss Count)
Pickup Count	Total number of times the machine picked up chips	SUM(Pickup Count)
Placement Count	Total number of times the machine placed chips	SUM(Place Count)
Drop Mistake Count	Total number of drop error counts	SUM(Drop Mistake Count)
Placement Mistake Count	=Placement Mistake Total number of placement error counts	SUM(Mount Mistake Count)
Shape Error Count *1	Total number of shape error counts	SUM(Shape Error Count)
Pickup Error Count *1	Total number of pickup error (not picked up) counts	SUM(Pickup Error Count)

*1 Display only Feeder/Part Report

■ Nozzle Ratio Report / ■ Nozzle Ratio Report by Disposal
■ Feeder Ratio Report by Pickup

Item	Explanation	Expression
Scrap PPM (P chart)	The ratio of the total number of parts that are not placed on the board to the total pickup count for each aggregation unit (PPM)	{ SUM(Pickup Count)－SUM (Place Count) } /SUM(Pickup Count)*1000000
Scrap Ratio (P chart) *2	The ratio of the total number of parts that are not placed on the board to the total pickup count for each aggregation unit	{ SUM(Pickup Count)－ SUM(Place Count) } /SUM(Pickup Count)
Pickup Ratio (P chart) *3	The ratio of the total number of times that chips are picked up successfully to the total pickup count for each aggregation unit	SUM(Place Count)/ SUM(Pickup Count)
Scrap Percentage (P chart) *2	The ratio of the total number of parts that are not placed on the board to the total pickup count for each aggregation unit (%)	{ SUM(Pickup Count)－ SUM(Place Count) } /SUM(Pickup Count)*100
Pickup Percentage(P chart) *3	The ratio of the total number of times that chips are picked up successfully to the total pickup count for each aggregation unit (%)	SUM(Place Count)/ SUM(Pickup Count)*100
Scrap Count (NP-Chart)	The total number of chips that are judged as scraps without placing on the board for each aggregation unit	SUM(Pickup Count)－ SUM(Place Count)

*2 Display only Nozzle Ratio Report and Nozzle Ratio Report by Disposal
*3 Display only Feeder Ratio Report by Pickup

■ Scrap Report

Item	Explanation	Expression
Scrap Count	The total number of chips that are judged as scraps without placing on the board	SUM(Pickup Count)－ SUM(Place Count)

■ Work Cell Performance

Item	Explanation	Expression
Boards Produced (Boards)	Total number of boards produced	SUM(Boards Produced)
Blocks Produced (Blocks)	Total number of blocks produced	SUM(Blocks Produced)
Real Run Time	Total board recognition time, placement/audit time and transport time (they are not included in case of alternative placement lane)	SUM(Real Run Time)
Previous Process Wait Time *4	Total time that the machine was waiting for the previous table or machine to finish processing a panel	SUM(PCB Wait Load Time)+SUM(Prev Proc Wait Time)
Next Process Wait Time *4	Total time that the machine was waiting for the next table or machine to finish processing a	SUM(Next Proc Wait Time)

Item	Explanation	Expression
	panel	
Single and Cycle Stop Time *4	Total time that the machine made single or cycle stop caused by operation button press action	SUM(Single Cycle Stop Time)
Chip Supply Wait Time *4	Total time that the machine stopped while waiting for an exhausted chip to be supplied	SUM(Chip Supply Wait Time)
Solder Supply Wait Time (Counter) *4	Total time that the machine stopped while waiting for a re-supply of exhausted solder paste (Replenishment triggered by a counter)	SUM(Sldr Ctr Wait Time)
Solder Supply Wait Time (Sensor) *4	Total time that the machine stopped while waiting for a re-supply of exhausted solder paste (Replenishment triggered by a sensor)	SUM(Sldr Snsr Wait Time)
Cleaning Paper Supply Wait Time *4	Total time that the machine stopped while waiting for a re-supply of exhausted cleaning paper	SUM(Cln Paper Wait Time)
Restart Impossible Stop Time *4	Total time that the machine stopped while waiting for error correction	SUM(Start Imp Stop Time)
Single and Cycle Stop Error Time *4	Total time that the machine made single or cycle stop when the machine automatically detect an error (However, if it is included in other error, it is not included in this item)	SUM(Single Cycle Stop Error Time)
Chip Ejection Error Stop Time *4	Total time that the machine stopped caused by error when the machine ejects defect chips	SUM(Chip Eject Error Time)
Chip Placement Error Stop Time *4	Total time that the machine made single stop caused by placement error	SUM(Chip Mount Error Time)
Chip Recognition Error Stop Time *4	Total time that the machine stopped caused by chip recognition errors (When an error occurs three times in a row in the same feeder, it results in recognition error)	SUM(Chip Recog Error Time)
Chip Erection Error Stop Time *4	Total time that the machine stopped caused by chip erection errors	SUM(Chip Eject Error Time)
Chip Pickup Error Stop Time *4	Total time that the machine made single stop caused by chip pickup error (the chip was not picked up for placement)	SUM(Chip Pickup Error Time)
Board Transport Error Stop Time *4	Total time that the machine made single stop caused by board transport errors (any errors caused on the conveyor)	SUM(Transport Error Time)
Placement Point Recognition Error Stop Time *4	Total time that the machine stopped caused by placement	SUM(Mnt Pt Recog Error Time)

Item	Explanation	Expression
	point recognition error (the type of local placement point/pattern fiducial is not recognized)	
Board Recognition Error Stop Time *4	Total time that the machine made single stop caused by board recognition error (the type of board fiducial is not recognized)	SUM(PCB Recog Error Time)
Board Elongation Error Stop Time *4	=Board Elongation Error Stop Time Total time that the machine stopped caused by board elongation error (the type of board fiducials were not recognized)	SUM(PCB Elong Error Time)
Paste Count up Error Stop Time *4	=Paste Count up Error Stop Time Total time that the machine stopped caused by paste count up error	SUM(Paste Countup Error Time)
Mask/Solder Recognition Error Stop Time *4	Total time that the machine made single stop caused by mask/solder recognition error (error recognizing the stencil)	SUM(Mask Recog Error Time)
Feedback Error Stop Time *4	Total time that the machine made cycle stop caused by feedback stop information.	SUM(Feedback Error Time)
Bond Recognition Error Stop Time *4	Total time that the machine made cycle stop caused by bond recognition error.	SUM(Bond Recog Error Time)
(Other) Error Stop Time *4	Total time that the machine stopped caused by other errors	SUM(Other Error Time)
Previous Process Wait Count	Total number of times that the machine was waiting for the previous table or machine to finish a process	SUM (Prev Proc Wait Count)
Next Process Wait Count	Total number of times that the machine was waiting for the next table or machine to finish a process	SUM(Next Proc Wait Count)
Single and Cycle Stop Count	Total number of times that the machine made single or cycle stop caused by operation button press action	SUM(Single Cycle Stop Count
Chip Supply Wait Count	Total number of times that the machine stopped while waiting for an exhausted chip to be supplied	SUM(Chip Supply Wait Count)
Solder Supply Wait Count (Counter)	Total number of times the machine stopped while waiting for a re-supply of older paste (replenishment triggered by a counter)	SUM(Sldr Ctr Wait Count)
Solder Supply Wait Count (Sensor)	Total number of times the machine stopped while waiting for a re-supply of solder paste (replenishment triggered by a	SUM(Sldr Snsr Wait Count)

Item	Explanation	Expression
	sensor)	
Cleaning Paper Supply Wait Count	Total number of times the machine stopped while waiting for a cleaning paper exhaust to be replenished	SUM(Cln Paper Wait Count)
Restart Impossible Stop	Total number of times that the machine stopped while waiting for error correction.	SUM(Start Imp Stop Count)
Board Transport Error Count	Total number of times the machine made single stop caused by board transport error (any error caused on the conveyor)	SUM(Transport Error Count)
Pickup Count	Total number of times the machine picked up chips	SUM(Pickup Count)
Placement Count	Total number of times the machine placed	SUM(Place Count)
Placement Ratio	The ratio of the total placement count to the total pickup count	SUM(Place count)/SUM(Pickup Count)*100
Chip Recognition Error	Total number of times the machine stopped caused by chip recognition error (It results in recognition error when an error occurs three times in a row in the same feeder)	SUM(Chip Recog Error Count)
Placement Point Recognition Error Count	Total number of times that the machine stopped caused by placement point recognition error (the type of local placement point/pattern fiducial is not recognized)	SUM(Place Pt Recog Error Count
Board Recognition Error Count	Total number of times the machine made single stop caused by board recognition error (the type of board fiducials were not recognized)	SUM(Board Recog Error Count)
Mask/Solder Recognition Error Count	Total number of times the machine made single stop caused by mask/solder recognition error (stencil recognition error)	SUM(Mask Recog Error Count)
Bond Recognition Error	Total number of times the machine stopped caused by bond recognition error	SUM(Bond Recog Error Count)
Board Elongation Error Count	Total number of times the machine stopped caused by board elongation error (the type of board fiducials were not recognized)	SUM(Board Elong Error)
Tact Time	The average time taken to produce a single board	SUM(Real Run Time)/ SUM(Boards Produced)
Machine Operation Ratio	The ratio of the total real run time to the total running time	SUM(Real Run Time)/ SUM(Run Time)*100
Machine Run Ratio	The ratio of the total real run time and previous process wait time to the total running time	{ SUM(Real Run Time)+SUM(Prev Proc Wait Time) }/SUM(Run Time)*100

Item	Explanation	Expression
Part Replenishment Ratio	=Scrap Percentage The ratio of the total number of parts that are not placed on the board to the total pickup count	{ SUM(Pickup Count)－ SUM(Place Count) } /SUM(Pickup Count)*100

*4 If percentage is selected in data option, ratio of the total to the total running time is displayed

■ PMD Operation Ratio

Item	Explanation	Expression
Running Rate	The ratio of the total real run time to the total running time	SUM(Real Run Time)/SUM(Run Time)*100

■ Machine Pickup/Placement Ratio / ■Feeder Pickup/Placement Ratio

Item	Explanation	Expression
Placement Percentage	The rat of the total number of placement count to the total pickup count for each aggregation unit	SUM(Place Count)/SUM(Pickup Count)*100
Pickup Percentage	The ratio of the total number of times that chips are picked up successfully to the total pickup count for each aggregation unit	{ SUM(Pickup Count)-SUM(Pickup Miss) } /SUM(Pickup Count)*100

■ Actual Boards vs Theoretical Boards

Item	Explanation	Expression
Boards Produced	Total number of boards produced	SUM(Boards Produced)
Theoretical Boards	The number of boards that can theoretically be produced during a time horizon	Time Horizon/Theoretical Cycle Time

■ Actual Blocks vs Theoretical Blocks

Item	Explanation	Expression
Blocks Produced	Total number of blocks produced	SUM((Blocks Produced)
Theoretical Blocks	The number of blocks that can theoretically be produced during a time horizon	Time Horizon/Theoretical Cycle Time*Pattern Quantity/Board Quantity

■ Produced Boards vs Scrap

Item	Explanation	Expression
Boards Produced	Total number of boards produced	SUM(Boards Produced)
Scrap Count	The total number of chips that are judged as scraps without placing on the board	SUM(Pickup Count)-SUM (Place Count)

■ Error Stop Time vs Error Stop Count

Item	Explanation	Expression
Restart Impossible Stop Time	Total time that the machine stopped caused by error correction	SUM(Start Imp Stop Time)
Single and Cycle Stop Time	Total time that the machine	SUM(Single Cycle Stop Error

Item	Explanation	Expression
	made single or cycle stop when the machine automatically detect an error	Time)
Chip Ejection Error Stop Time	Total time that the machine stopped caused by error when the machine ejects defect chips	SUM(Chip Eject Error Time)
Chip Recognition Error Stop Time	Total time that the machine stopped caused by chip recognition errors (When an error occurs three times in a row in the same feeder, it results in recognition error)	SUM(Chip Recog Error Time)
Chip Placement Error Stop Time	Total time that the machine made single stop caused by placement error	SUM(Chip Mount Error Time)
Chip Erection Error Stop Time	Total time that the machine stopped caused by chip erection errors	SUM(Chip Eject Error Time)
Chip Pickup Error Stop Time	Total time that the machine made single stop caused by chip pickup error (the chip was not picked up for placement)	SUM(Chip Pickup Error Time)
Board Transport Error Stop Time	Total time that the machine made single stop caused by board transport errors (any errors caused on the conveyor)	SUM(Transport Error Time)
Placement Point Recognition Error Stop Time	Total time that the machine stopped caused by placement point recognition error (the type of local placement point/pattern fiducial is not recognized)	SUM(Mnt Pt Recog Error Time)
Board Recognition Error Stop Time	Total time that the machine made single stop caused by board recognition error (the type of board fiducial is not recognized)	SUM(PCB Recog Error Time)
Board Elongation Error Stop Time	Total time that the machine stopped caused by board elongation error (the type of board fiducials were not recognized)	SUM(PCB Elong Error Time)
Paste Count up Error Stop Time	Total time that the machine stopped caused by paste count up error	SUM(Paste Countup Error Time)
Mask/Solder Recognition Error Stop Time	Total time that the machine made single stop caused by mask/solder recognition error (error recognizing the stencil)	SUM(Mask Recog Error Time)
Feedback Error Stop Time	Total time that the machine made cycle stop caused by feedback stop information.	SUM(Feedback Error Time)
Bond Recognition Error Stop Time	Total time that the machine made cycle stop caused by bond recognition error.	SUM(Bond Recog Error Time)
(Other) Error Stop Time	Total time that the machine stopped caused by other errors	SUM(Other Error Time)

Item	Explanation	Expression
Error Count	Total number of times error occurred	SUM (Error Count)

■ Board Counter Monitor

Item	Explanation	Expression
Boards Produced	Total number of boards produced	SUM(Boards Produced)
Theoretical Boards	The number of boards that can theoretically be produced during a time horizon	Time Horizon/Theoretical Cycle Time
Boards Produced Summary		

■ Consolidated Production Report

● Line information

Item	Explanation	Expression
Spoilage Rate PPM	=Scrap PPM The ratio of the total number of parts that are not placed on the board to the total pickup count (PPM)	{ SUM(Pickup Count)-SUM(Place Count) }/SUM(Pickup Count)*1000000
Total Pickup Count	Total number of pickup count	SUM(Pickup Count)
Total Error Count	Total of each error	SUM (Error Count)
Pickup Error	Total number of pickup error (not picked up) counts	SUM(Pickup Error Count)
Recognition Error	Total number of recognition error count (parts are not recognized)	SUM(Recog Error Count)
Thickness Error	Total number of thickness error count	SUM(Thick Error Count)
Placement Error	Total number of placement error count	SUM(Mount Error Count)
Parts Drop Error	Total number of parts drop error count	SUM(Parts Drop Error Count)
Transfer Unit Parts Drop Error	Total number of times that the machine stopped caused by transfer unit part drop error	SUM(Transfer Unit Parts Drop Error Count)

● Board information

Item	Explanation	Expression
Board Quantity	Total number of boards produced	SUM(Boards Produced)
Blocks Quantity	Total number of blocks produced	SUM(Blocks Produced)
Running Rate	The ratio of total real run time to the running time	SUM(Real Run Time)/SUM(Run Time)*100
Run Time	Total time between the start and end of automatic operation	SUM(Run Time)
Real Run Time	Total board recognition time, placement/audit time and transport time (they are not included in case of alternative placement lane)	SUM(Real Run Time)

Item	Explanation	Expression
Total Stop Time	Total stop time caused by waiting or error	SUM(Stop Time)
Total Pickup Count	Total number of pickup count	SUM(Pickup Count)
Total Placement Count	=Placement Count Total number of times the machine placed	SUM(Place Count)
Total Error Count	Total l number of each error	SUM (Error Count)
Recognition Error	Total l number of recognition error count (parts were not recognized)	SUM(Recognition Error Count)
Placement Error	Total l number of placement error count	SUM(Mount Error Count)
Parts Drop Error	Total l number of parts drop error count	SUM(Parts Drop Error Count)
Transfer Unit Parts Drop Error	Total l number of transfer unit parts drop error count	SUM(Transfer Unit Parts Drop Error Count)
Pickup Error	Total l number of pickup error (parts were not picked up) count	SUM(Pickup Error Count)
Thickness Error	Total l number of thickness error count	SUM(Thick Error Count)

- Stop information

Item	Explanation	Expression
Power ON Time	Total time that the machine was activated	SUM(Power ON Time)
Run Time	Total time between the start and end of automatic operation	SUM(Run Time)
Real Run Time	Total of board recognition time, placement/audit time and transport time (they are not included in case of alternative placement lane)	SUM(Real Run Time)
Transport Time	Total time between loading a board and transferring it to the placement position	SUM(Transport Time)
Board Recognition Time	Total board recognition time	SUM(Board Recog Time)
Placement Time	Total placement time	SUM(Place Time)
Total Stop Time	Total time of waiting and stop time	SUM(Stop Time)
Previous Process Wait Time	Total time that the machine was waiting for the previous table or machine to finish processing a panel	SUM(Prev Proc Wait Time)
Next Process Wait Count	Total number of times that the machine was waiting for the next table or machine to finish a process	SUM(Next Proc Wait Count)
Next Process Wait Time	Total time that the machine was waiting for the next table or machine to finish processing a panel	SUM(Next Proc Wait Time)
Next Process Wait Count	Total number of times that the machine was waiting for the	SUM(Next Proc Wait Count)

Item	Explanation	Expression
	next table or machine to finish a process	
Chip Supply Wait Time	Total time that the machine stopped while waiting for an exhausted chip to be supplied	SUM(Chip Supply Wait Time)
Chip Supply Wait Count	Total number of times that the machine stopped while waiting for an exhausted chip to be supplied	SUM(Chip Supply Wait Count)
Cycle Stop Time	Total time that the machine made single or cycle stop caused by operation button press action	SUM(Single Cycle Stop Time)
Cycle Stop Count	Total number of times that the machine made single or cycle stop caused by operation button press action	SUM(Single Cycle Stop Count
Cycle Error Stop Time	Total time that the machine made single or cycle stop when the machine automatically detect an error (However, if it is included in other error, it is not included in this item)	SUM(Single Cycle Stop Error Time)
Cycle Error Stop Count	Total number of times that the machine made single or cycle stop when the machine automatically detect an error (However, if it is included in other error, it is not included in this item)	SUM(Single Cycle Error Count)
Transport Error Stop Time	Total time that the machine made single stop caused by board transport errors (any errors caused on the conveyor)	SUM(Transport Error Time)
Transport Error Stop Count	Total number of times the machine made single stop caused by board transport error (any error caused on the conveyor)	SUM(Transport Error Count)
Board Recognition Error Stop Time	Total time that the machine made single stop caused by board recognition error (the type of board fiducial is not recognized)	SUM(PCB Recog Error Time)
Board Recognition Error Stop Count	Total number of times the machine made single stop caused by board recognition error (the type of board fiducials were not recognized)	SUM(Board Recog Error Count)
No Restart Trouble Stop Time	Total time that the machine stopped caused by error correction	SUM(Start Imp Stop Time)
No Restart Trouble Stop Count	Total number of times the machine wad down while waiting for error correction.	SUM(Start Imp Stop Count)
Discharge Recognition Error Stop Time	Total time that the machine made single stop caused by	SUM(Discharge Recognition Error Stop Time)

Item	Explanation	Expression
	discharge recognition error	
Discharge Recognition Error Stop Count	Total number of times that the machine made single stop caused by discharge recognition error	SUM(Discharge Recognition Error Stop Count)
Chip Pickup Error Stop Time	Total time that the machine made single stop caused by chip pickup error (the chip was not picked up for placement)	SUM(Chip Pickup Error Time)
Chip Pickup Error Stop Count	Total number of times the machine stopped caused by chip pickup errors (the chip was not picked up for placement)	SUM(Chip Pickup Error Count)
Chip Recognition Error Stop Time	Total number of times the machine stopped caused by chip recognition errors (It results in recognition error when an error occurs three times in a row in the same feeder)	SUM(Chip Recog Error Time)
Chip Recognition Error Stop Count	Total number of times the machine stopped caused by chip recognition errors (It results in recognition error when an error occurs three times in a row in the same feeder)	SUM(Chip Recog Error Count)
Parts Drop Error Stop Time	Total time that the machine stopped caused by part drop error	SUM(Parts Drop Error Stop Time)
Parts Drop Error Stop Count	The total number of times that the machine stopped caused by part drop error	SUM(Parts Drop Error Stop Count)
Chip Placement Error Stop Time	Total time that the machine made single stop caused by placement error	SUM(Chip Mount Error Time)
Chip Placement Error Stop Count	Total number of times the machine made single stop caused by placement error	SUM(Chip Mount Error Count)
Chip Erection Error Stop Time	Total time that the machine stopped caused by chip erection error	SUM(Chip Errorect Error Time)
Chip Erection Error Stop Count	Total number of times the machine stopped cause by chip erection error	SUM(Chip Erection Error Count)
Transfer Unit Parts Drop Error Stop Time	Total time that the machine made single stop caused by transfer unit parts drop error	SUM(Transfer Unit Parts Drop Stop Time)
Transfer Unit Parts Drop Error Stop Count	Total number of times the machine made single stop caused by transfer unit parts drop error	SUM(Transfer Unit Parts Drop Error Stop Count)
Excess Judgement Stop Time (Inspection Head)	Total time that the machine stopped caused by excess judgment	SUM(Excess Judgement Stop Time (Inspection Head))
Excess Judgement Stop Count	Total number of times the machine stopped caused by excess judgement	SUM(Excess Judgement Stop Count)

Item	Explanation	Expression
Waiting Time Caused By Other Lane	Total time that the machine stopped caused by other stage or lane error (During production in dual alternative lane, if the lane 1 stopped caused by error such as pickup error, the lane 2 will also stopped caused by other lane error)	SUM(Waiting Time Caused By Other Lane)
Waiting Count Caused By Other Lane	Total number of times that the machine stopped caused by other stage or lane error (During production in dual alternative lane, if the lane 1 stopped caused by error such as pickup error, the lane 2 will also stopped caused by other lane error)	SUM(Waiting Count Caused By Other Lane)
Material Exhaust Error Stop Time	Total time that the machine made cycle stop when materials (bond, etc.) are exhausted	SUM(Material Exhaust Error Stop Time)
Material Exhaust Error Stop Count	The total number of times that the machine made cycle stop when materials (bond, etc) are exhausted	SUM(Material Exhaust Error Stop Count)
Changeover Time	Total time that setup related menus are displayed	SUM(Setup Time)
Maintenance Time	Total time that maintenance related menus are displayed	SUM(Maintenance Time)
Recipe Edit Time	Total time that production data related menus are displayed	SUM(Production Data Edit Time)
Unit Adjust Time	Total time that machine related menu are displayed	SUM(Unit Adjust Time)
Other Time	Total time that home screen is displayed	SUM(Power on Time)-SUM(Changeover Menu Time)-SUM(Prod Management Menu Time) -SUM(Maintenance Time) -SUM(Recipe Edit Time) -SUM(Unit Adjust Time)-SUM(Run Time)
Machine Previous Process Wait Time	Total time that the machine stopped while all lanes in production are waiting for previous process.	SUM(Machine Prev Proc Wait Time)
Machine Previous Process Wait Count	Total number of times that the machine stopped while all lanes in production are waiting for previous process.	SUM(Machine Prev Proc Wait Count)
Machine Next Process Wait Time	Total time that the machine stopped while all lanes in production are waiting for next process.	SUM(Machine Next Proc Wait Time)
Machine Next Process Wait Count	Total number of times that the machine stopped while all lanes in production are waiting for next process.	SUM(Machine Next Proc Wait Count)

- Pickups/(Machine/Table/Feeder/Nozzle Changer/Nozzle Holder)

Item	Explanation	Expression
Machine Name	Machine number and Lane name	
T *5 *6	Table number	
FA *6	Feeder address	
SS *6	Feeder side (L/R)	
Part Number *6	Part number installed to feeder	
H *7 *8	Head position (Front:1, Rear: 2)	
CA *7	Slot number of a nozzle changer that picked up nozzles are arranged	
HA *8	Nozzle holder address	
Pickup Count	The total number of pickup count for each count	SUM(Pickup Count)
Error Count	The total number of error count for each count	SUM (Error Count)
Spoilage Rate PPM	・Scrap PPM The ratio of the total number of each error count to the total of pickup count for each aggregation unit (PPM)	{ SUM(Pickup Count)-SUM(Place Count) }/SUM(Pickup Count)*1000000
Error Rate	・Scrap % The ratio of the total number of each error count to the total of pickup count for each aggregation unit (%)	{ SUM(Pickup Count)-SUM(Place Count) }/SUM(Pickup Count)*100
Pickup Error	Total number of pickup error count (the chip was not picked up for placement) for each count	SUM(Chip Pickup Error Count)
Recognition Error	Total number of recognition error count ((the part was not recognized) for each count	SUM(Recognition Error Count)
Thickness Error	Total number of thickness error count for each aggregation unit	SUM(Thick Error Count)
Placement Error	Total number of placement error count for each aggregation unit	SUM(Place Error Count)
Parts Drop Error	Total number of parts drop error count for each aggregation unit	SUM(Parts Drop Error Count)
Transfer Unit Parts Drop Error	Total number of transfer unit parts drop error for each aggregation unit	SUM(Transfer Unit Parts Drop Error Count)

*5：Display only for Pickups/Table
*6：Display only for Pickups/Feeder
*7：Display only for Pickups/Nozzle changer
*8：Display only for Pickups/Nozzle holder

= MEMO =

6. FUNCTION - [PRODUCTION ANALYSIS] - [EXCEL FILE]

This chapter explains function- [Production Analysis] - [Excel File].

6.1 Overview

Export production analysis charts to an Excel file so that you can even refer to it while offline.
To display production analysis charts, access PanaCIM EE “Web application” and select [Production Analysis] from the menu. Then select Excel file from the balloon menu.
The Excel file containing these charts can also be sent by E-mail to the address specified.

・Microsoft Office Excel 2010 or later version is required for PanaCIM application server.
・To export the charts to Excel file, you need to manually configure the setting in advance.

6.1.1 Production Analysis - [Excel File]

The Excel file includes the following worksheet.

- Intro
- Production Summary
- Production Summary-Data
- OEE
- Placement Summary
- Placement Summary-Data
- Boards Produced
- Machine Status
- Placement Info
- Sheet for each product (*)
- Glossary

* For weekly report or monthly report, the worksheets are created as board information, feeder information and nozzle information .

6.2 Settings

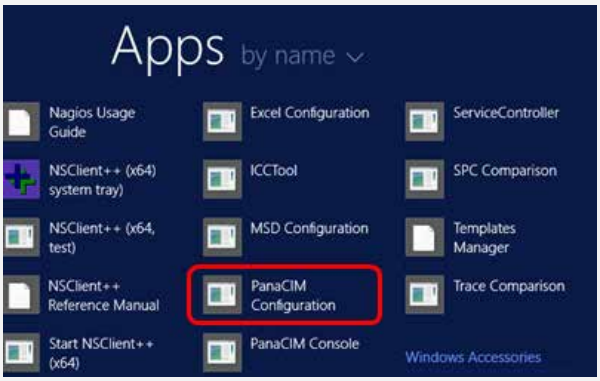
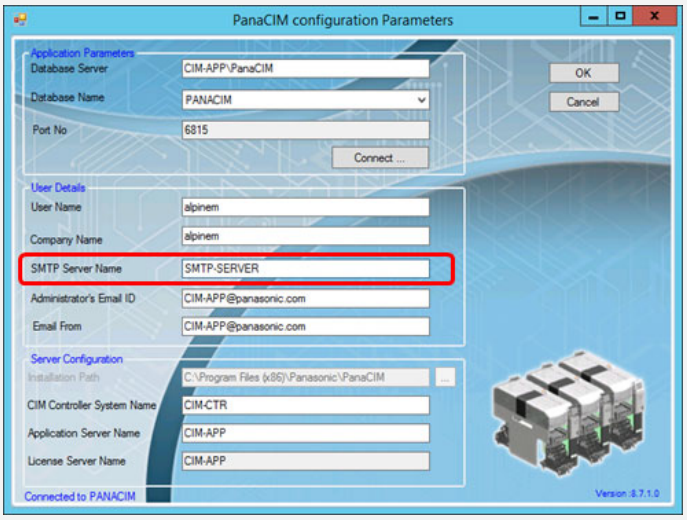
This section describes how to configure the setting when the data is exported into the Excel.
The data types exported into Excel are as follows.

- Daily report
- Shift
- Weekly report
- Monthly report
- Floor production status

<u>If you want to send the exported Excel file as an attachment, you need to prepare the SMTP server. The SMTP server must be registered in the PanaCIM application server in advance.</u>

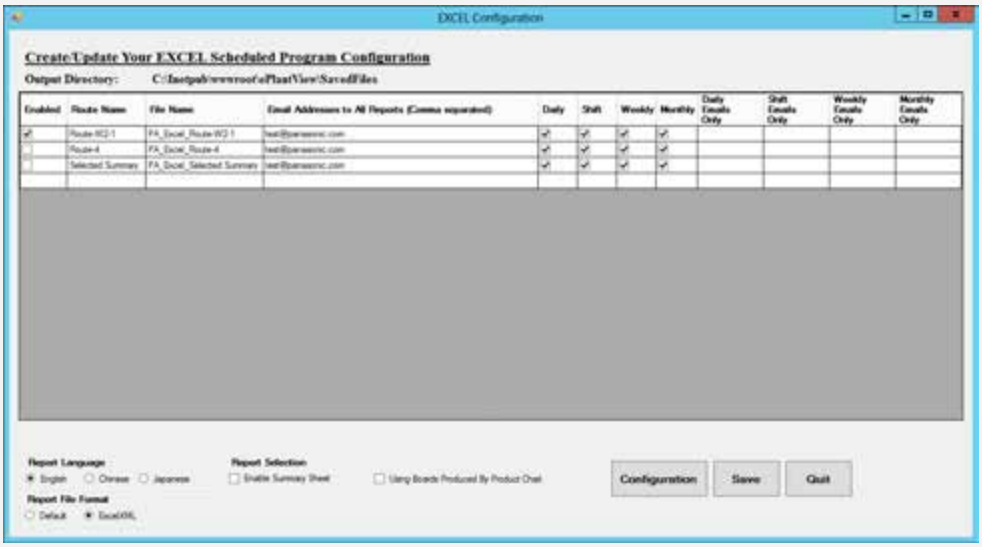
6.2.1 Register SMTP Server

The procedure for registering the SMTP server in PanaCIM application server is as follows.

Step	Procedure
1	Launch PanaCIM Configuration ・Select PanaCIM Configuration from the Windows start menu in PanaCIM Application Server 
2	Enter the name in the **SMTP Server Name** field in **User Details** and click **OK**. 

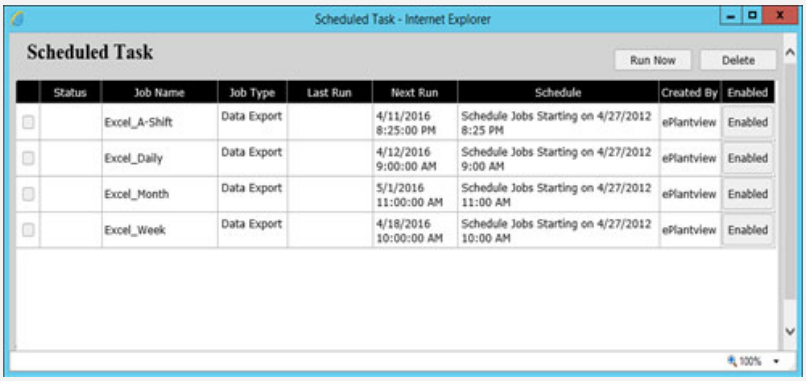
6.2.2 Registering / Updating Excel Export Settings

The procedure for setting a route, type, a mail transmission destination and language is as follows.

Step	Procedure
1	Launch Excel Configuration. ・PanaCIM Application Server Execute C:¥ePlantViewDLLs¥EXCELConfigure.exe  *Only routes whose production analysis licenses are registered are displayed.
2	Select the route that you want to export into Excel from the list and mark the checkbox of **Enabled**.
3	Enter the **email address** (required) where you want to send the exported Excel file. *Even if you do not send the file, you need to enter any arbitrary email address.
4	Select the types (Daily, Shift, Weekly, Monthly) of the Excel file and mark each checkbox. *Select (Selected Summary) and target course for the floor production status.
5	If you want to send the Excel file to the address other than specified in step3 according to the type, enter the individual email address (Daily Emails Only, Shift Emails Only, Weekly Emails Only, Monthly Emails Only).
6	Select language of the Excel file.
7	Select **Save** to register.

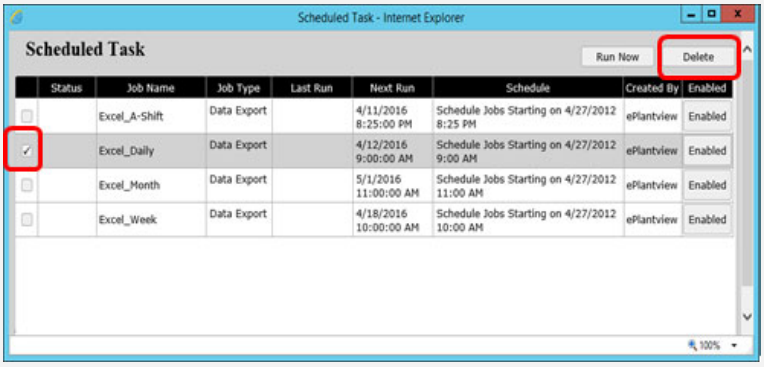
6.2.3 Confirming Registration Status of Excel Export Settings

You can confirm the status such as the next timing of exporting data into each Excel file and the previous export status.
The procedure for confirming the registration status of exporting data into the Excel file is as follows.

Step	Procedure
1	Launch ePlantView and select **Administration - Task Administration**. 
2	You can confirm the registration status of the Excel export settings from the Schedule Task. 

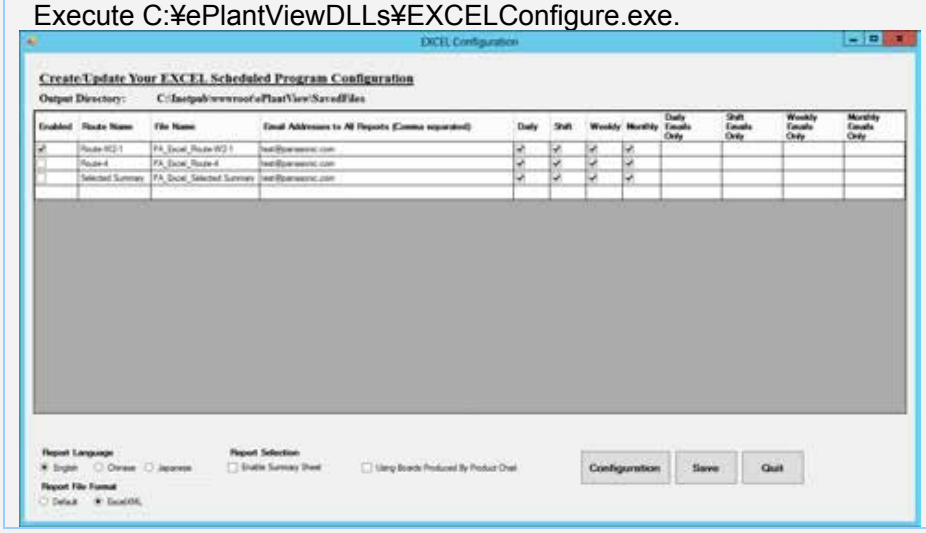
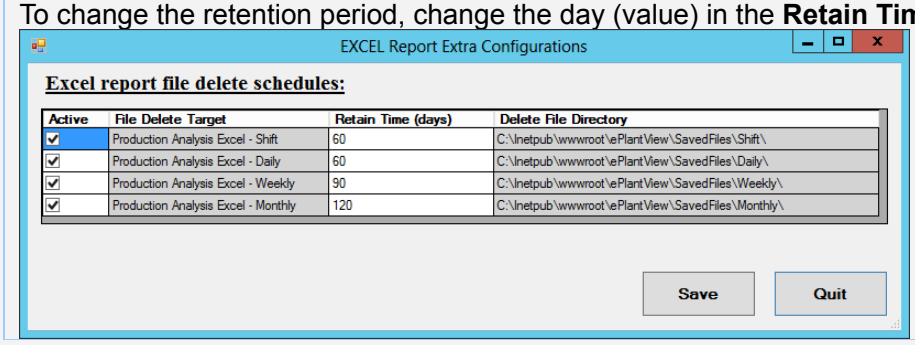
6.2.4 Deleting Excel Export Settings

You can delete the registered Excel export settings.
The procedure for deleting the settings is as follows.

Step	Procedure
1	Launch ePlantView and select **Administration - Task Administration**.
2	Mark the checkbox of a desired data in Schedule Task and click Delete. 

6.2.5 Registering Excel File Deletion Settings

You can register schedule to delete an output excel file.

Step	Procedure
1	Launch Excel Configuration. ・PanaCIM Application Server Execute C:¥ePlantViewDLLs¥EXCELConfigure.exe. 
2	Click **Configuration**.
3	Mark the **Active** checkbox for each file that you want to delete. To change the retention period, change the day (value) in the **Retain Time (days).** 
4	**Click Save** to register.

6.3 Excel File

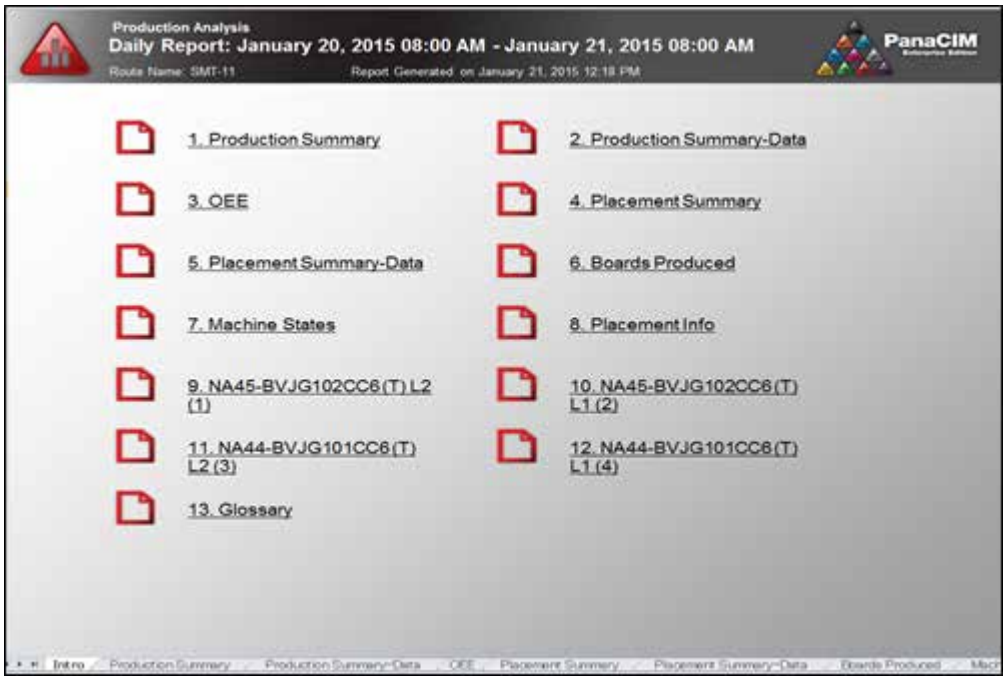
The Excel file can be transmitted to the specified location or send as attachment according to the setting registered in Schedule Task.

6.3.1 Intro

A name of each sheet is displayed in this worksheet as a hyperlink which can take you to each sheet. The following information is included in this worksheet.

- Report type
- Data acquisition start date, and Data acquisition end date
- A route name target for the created Excel report
- Report creation date

Information of these headers is also displayed in each worksheet.



6.3.2 Production Summary

Production summaries such as produced PCB count, cycle time, operation rate, etc., are displayed in this worksheet.
The following charts are included in this worksheet.

- Line performance report
- Line boards produced
- Line cycle time
- Operation rate chart
- Machine status - Hourly chart
- Machine status - Percentage summary chart
- Scrap rate－Hourly (Daily) chart
- Machine status – Chart per machine

■ Line performance report

This chart shows the production, performance and down time of the selected route.
The product name, total, average and each product value are included in the chart.

Total column shows the total value of all products, and Average column shows the average value of all products.
If there is no value shown in the column, the cell is grayed out.

The format of production, performance and down time shown in this table is as follows.

#Productions	− Numeric value
Boards Produced	− Numeric value
Blocks Produced	− Numeric value
Placement Count	− Numeric value
Mounting Count	− Numeric value
Operation Rate	− Percentage value
CPH	− Numeric value
Lot Running Time	− Time in dd, hh:mm:ss format
Pickup Rate	− Percentage value
Placement Rate	− Percentage value
Changeover Time	− Time in dd, hh:mm:ss format
Previous Process Wait Time	− Process time in hh:mm:ss format
Next Process Wait Time	− Process time in hh:mm:ss format
Exhaust Time	− Exhaust time in hh:mm:ss format

■ Line boards produced

This chart shows the total number of boards produced for a route and the number of theoretical boards.
The total number of boards produced for a route is shown in a bar graph and the number of theoretical boards is shown in a line graph.
The aggregation unit in the X-axis is as follows.

Daily report, Shift	Hourly
Weekly report Monthly report	Daily

■ Line cycle time

This chart shows the actual cycle time and logical cycle time of an entire route.
The actual cycle time is shown in a bar graph and the logical cycle time is shown in a line graph.
The aggregation unit in the X-axis is as follows.

Daily report, Shift	Hourly
Weekly report	Daily
Monthly report	

■ Operation rate chart

This pie chart shows the operation rate of the entire line.
The time that the machine is in production (boards are produced), changeover time and down time is displayed as a percentage value.

■ Machine status – Hourly (daily)

This chart shows the machine status as a percentage.
The function of this chart is as follows.

The Y-axis in the stacked bar chart shows the machine status as a percentage.
The aggregation unit in the X-axis is as follows.

Daily report, Shift	Hourly
Weekly report	Daily
Monthly report	

The following machine status is shown as a percentage in a bar graph.

- Actual Operation Time (Time used for producing boards while the machine actually perform production)
- Changeover Time (Time required for setup)
- Machine Error Down Time (Time during which the machine is not working)
- Exhaust Time (Time during which parts are exhausted)
- Previous Process Wait Time (Time during which the machine waits for loading a board before the machine places parts)
- Next Process Wait Time (Time during which the machine waits for unloading a board after the machine placed parts)

Note that the machine status indicates the status of the key machine of the route.
Previous Process Wait Time is a waiting time for the first machine of the route, and Next Process Wait Time is a waiting time for the last machine of the route.

■ Machine status – Percentage summary

This chart shows the machine status as percentages.
The total value of percentage is 100.

■ Scrap rate – Hourly (Daily) chart

This chart shows the scrap rate (PPM) per unit time.

■ Machine status – Chart per machine

This chart shows the status of each machine as a percentage.

6.3.3 Production Summary-Data

This sheet includes the data of charts displayed in the production summary worksheet.
The data of the chart displayed in the production summary worksheet in this sheet is as follows.

- Line boards produced
- Line cycle time
- Machine status – hourly (daily)
- Machine status – percentage summary
- Operation rate
- Scrap rate – Hourly (Daily)
- Machine status – Each machine

6.3.4 OEE

This worksheet shows the line performance and machine total operation rate.
The following charts and table data are included in this worksheet.

- Line performance
- Operation rate
- Hourly (daily) performance
- Quality
- OEE performance
- OEE summary
- Performance for each shift

The aggregation unit is as follows.

Daily report, Shift	Hourly
Weekly report Monthly report	Daily

6.3.5 Placement Summary

Summaries of the pickup rate and placement rate for the entire route are displayed in this worksheet.
Data of the following charts included in this worksheet is as follows.

- Line pickup rate / placement rate
- 10 worst placement rate (nozzle position)
- 10 worst pickup rate (feeder position)

■ Line pickup / placement rate

The function of this chart is as follows.

The Y-axis in the line pickup rate / placement rate bar graph shows the pickup rate /placement rate as a percentage.
The aggregation unit in the X-axis is as follows.

Daily report, Shift	Hourly
Weekly report Monthly report	Daily

The entire route is used to calculate the pickup rate / placement rate.

The pickup rate indicates the percentage of parts that are picked up from the feeder position of the machine.
The placement rate indicates the percentage of picked-up parts that are placed on the board.

■ 10 worst placement rate (nozzle position)

This chart shows the list of worst 10 nozzles with the lowest placement rate, and the placement rate can be compared each other.
The function of this chart is as follows.

This chart shows the 10 records of the nozzle position in the order of the worst placement rate.
The perfect placement rate is 100% that means all picked-up parts are placed.

The X-axis represents the nozzle position and the Y-axis represents the placement rate of the nozzle.
The placement alert limit is shown in a line graph.
If the placement rate falls below this limit, it means that the nozzle is not placed with an adequate level.

The nozzle position shows a combination of a machine name, table number and nozzle number.

■ 10 worst placement rate (nozzle position / feeder serial)

This chart shows the list of worst 10 feeder positions with the lowest pickup rate so that the pickup rate can be compared each other.
The function of this chart is as follows.

This chart shows the 10 records of the feeder positions in the order of the worst pickup rate.
The perfect pickup rate is 100% that means all parts are picked up from the feeder position.
The X-axis represents the feeder position of the machine and the Y-axis represents the pickup rate as a percentage.
The pickup alert limit is shown in a line graph.
If the pickup rate falls below this limit, it means that the parts are not picked up with an adequate level from the feeder position.

The machine feeder position shows a combination of a machine number and feeder position.

6.3.6 Placement Summary-Data

Data of the charts displayed in the placement summary worksheet is included in this worksheet.
Data of the following charts included in this worksheet is as follows.

- Line pickup rate / placement rate
- 10 worst placement rate (nozzle position)
- 10 worst pickup rate (feeder position / feeder serial number)

6.3.7 Boards Produced

The line graph and the data are displayed as a table in the board produced sheet.
The chart data is displayed on the right side table and the chart is displayed on the left side.
The following charts are included in this sheet.

- Boards produced for each shift (actual / planned)
- Boards produced for each product (actual / planned)

Chart for boards produced for shift

Actual boards produced and planned boards produced for each shift are displayed.
The X-axis shows shift and the Y-axis shows the number of boards produced.
The number of boards produced for each shift is added to the total of the boards produced for the route.

Chart for boards produced for each product

Actual boards produced and planned boards produced for each product produced in a route are displayed.
The X-axis shows a product name and the Y-axis shows the number of boards produced (both actual and planned boards).
The total number of boards produced for each product is added to the total of the boards produced for the route.

6.3.8 Machine status

An hourly machine status of a route is displayed as a percentage value in this worksheet
In this sheet, the table including the following machine status value of a route is displayed for each machine.

- Actual Operation Time
- Exhaust Time
- Machine Error Down Time
- Changeover Time
- Previous Process Wait Time
- Next Process Wait Time

The aggregation unit is as follows.

Daily report, Shift	Hourly
Weekly report Monthly report	Daily

In addition, a stacked bar graph that stacks bars representing different elements of the error down-time of each machine on top of each other and a line graph indicating the number of error stops per machine are displayed superimposed on each other.

6.3.9 Placement Info

This worksheet includes the following table and chart.

- Placement information summary (table)
- Pickup rate / placement rate for each machine (chart and table)
- Pickup rate / placement rate for each product (chart and table)
- Hourly (daily) part exhaust (chart and table)

Chart / table name	Description
Placement information summary	Displays the scrap rate (PPM), CPH, total placement count of each machine.
Pickup rate / placement rate for each machine	Displays the pickup rate and placement rate of a machine.
Pickup rate / placement rate for each product	Displays the pickup rate and placement rate of a product
Hourly (daily) part exhaust	Represents the number of parts exhausted and the time used for refill (minutes) during the period.

6.3.10 Sheet for Each Product

The worksheet having a name of the product is created for each product. (*1)
The following sections are included in the worksheet.

- Board information
- Feeder information
- Nozzle information

The period of production information of an acquired product is not 24 hours but a few days.

The production information section shows detailed information such as a board name(*2), the number of boards produced, the number of production blocks, placement count, pickup count, pickup error count, actual operation time and total operation time.

The feeder information section shows detailed information of a feeder used for production such as a machine number, a machine slot with a feeder installed, a feeder barcode of the machine slot, barcode of a reel installed to the feeder (specified by barcode), and parts information such as a part number, a vendor number, a lot number, a part picked up or placed from a feeder installed to the specific machine slot.

The nozzle information section shows a nozzle reference number, a machine associated with the reference number, a position on the machine head, a part picked up on the specific position and a pickup part placed by this nozzle on the specific position.

*1 For daily report or shift, a sheet is created for each product.
For weekly report or monthly report, a sheet is created for each type of board information, feeder information and nozzle information.

*2 For weekly report or monthly report, a board name is not displayed in board information.

6.3.11 Glossary

Formulas and descriptions corresponding to the terms of each item are listed in this worksheet.

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Panasonic Smart Factory Solutions Co., Ltd.

4-1-62 Minoshima, Hakata-ku, Fukuoka City, Fukuoka, Japan
Phone number: +81-92-477-1205
Fax number: +81-92-477-1664

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